IMPROVING SKETCHING ALGORITHMS FOR LOW-RANK MATRIX APPROXIMATION VIA SKETCH-POWER ITERATIONS*

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Abstract. Power iteration can improve the accuracy of randomized SVD, but requires multiple data passes, making it impractical in streaming or memory-constrained settings. We introduce a lightweight yet effective sketch-power iteration, allowing power-like iterations with only a single pass of the data, which can be incorporated into one-pass algorithms for low-rank approximation. As an example, we integrate the sketch-power iteration into a one-pass algorithm proposed by Tropp et al., and introduce strategies to reduce its storage cost. We establish meaningful error bounds: given a fixed storage budget, the sketch sizes derived from the bounds closely match the optimal ones observed in reality. This allows one to preselect reasonable parameters. Numerical experiments on both synthetic and real-world datasets indicate that, under the same storage constraints, applying one or two sketch-power iterations can substantially improve the approximation accuracy of the considered one-pass algorithms. In particular, experiments on real data with flat spectrum show that the method can approximate the dominant singular vectors well.

Key words. Randomized SVD, One-pass, Power iteration, Sketching, Low-rank approximation

MSC codes. 65F55; 15B33; 68W20; 65F35

1. Introduction.

Randomized algorithms have emerged as a popular tool for low-rank matrix approximation. Of particular interest is randomized SVD (RSVD), which was proposed in [24, 22] and later systematically refined and analyzed by Halko, Martinsson, and Tropp [13]. Roughly speaking, given a data matrix $\mathbf{A}$, RSVD first generates a sketch $\mathbf{Y} = \mathbf{A}\mathbf{\Omega}$ approximating the range of $\mathbf{A}$, with $\mathbf{\Omega}$ a random test matrix, orthonormalizes $\mathbf{Y}$, and then performs a QB decomposition, followed by an SVD on a much smaller matrix. RSVD is more efficient than SVD, with detailed theoretical error bounds provided. RSVD requires two passes of the data matrix. In another thread, algorithms using one-pass of the data matrix have been developed [41, 7, 35, 36, 37, 29]. For example, Tropp et al. [35] considered taking two sketches of $\mathbf{A}$, catching its range and co-range, which is able to compute a QB approximation without revisiting $\mathbf{A}$ again. One-pass algorithms are effective for managing data with limited storage, arithmetic, and communication capabilities. Apart from these works, randomized algorithms have been studied extensively in the literature; see, e.g., [11, 41, 7, 23, 40, 8, 4, 29], and the recent monographs [34, 16, 17, 28, 26].

The performance of RSVD is influenced by the decay rate of the singular spectrum. Power iterations $(\mathbf{A}\mathbf{A}^T)^q\mathbf{A}\mathbf{\Omega}$ have been incorporated to amplify the gap between singular values, hence enhancing the decay and improving approximation accuracy [32, 13]. Error bounds have been established, with more refined ones obtained in [12, 34]. The power scheme needs $2q+1$ passes of $\mathbf{A}$, while an alternative approach

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allows for a flexible number of passes [2].

However, power scheme typically requires multiple passes over the data matrix, limiting its applicability in scenarios with limited memory or streaming data. In such settings, the data cannot be fully retained in memory, precluding repeated access. Additionally, in privacy-sensitive contexts where the raw data cannot be exposed [37], power iteration is not feasible. Existing literature on incorporating power iterations into one-pass algorithms to improve accuracy seems sparse. The only relevant instance found is in [42, Remark 4.2] (see also [2, Sect. 6.8]), which briefly suggests a "half" power iteration: $\mathbf{A}\mathbf{A}^T\mathbf{\Omega}$, provided column-wise streaming of $\mathbf{A}$; the details will appear later. These observations motivate the development of alternative methods that approximate the benefits of power iteration without necessitating multiple data passes.

Motivated by sketching, we replace $\mathbf{A}$ in power iterations with a much smaller sketch $\mathbf{Z}$. Specifically, instead of iterating $(\mathbf{A}\mathbf{A}^T)^q\mathbf{A}\mathbf{\Omega} = (\mathbf{A}\mathbf{A}^T)^q\mathbf{Y}$, we perform

$$\hat{\mathbf{Y}} = (\mathbf{Z}\mathbf{Z}^T)^q\mathbf{Y}, \tag{1.1}$$

where $\mathbf{Y} = \mathbf{A}\mathbf{\Omega}$ and $\mathbf{Z} = \mathbf{A}\mathbf{\Phi}$ are both sketches of $\mathbf{A}$ with $\mathbf{\Omega}, \mathbf{\Phi}$ oblivious random test matrices, so $\mathbf{Y}, \mathbf{Z}$ can be formed in a single pass of $\mathbf{A}$. To be reasonable when storage is constrained, $\mathbf{Z}$ should be significantly smaller than $\mathbf{A}$—typically, it is a few times larger than $\mathbf{Y}$, depending on the storage budget. It also allows cheap re-orthonormalization in the process to mitigate round-off errors. Subsequently, $\hat{\mathbf{Y}}$ can be computed in a single pass of $\mathbf{A}$. We refer to (1.1) as Sketch-Power Iteration (SPI).

The intuition behind SPI can be explained by two scenarios. First, consider when $\mathbf{A}$ has insignificant tail energy (e.g. $\mathbf{A}$ has a clear low-rank structure) corresponding to the sketch size of $\mathbf{Z}$. In this case, $\mathbf{Z}$ is a good representation of $\mathbf{A}$'s main subspace, which is especially effective when the tail energy of $\mathbf{A}$ is meaningful corresponding to the sketch size of $\mathbf{Y}$. Alternatively, if the tail energy is non-trivial, $\mathbf{Z}$ provides only a partial approximation. Nevertheless, $\mathbf{Z}$ is constructed with a larger sketch size than $\mathbf{Y}$, ensuring it captures substantially more information than $\mathbf{Y}$. In both scenarios, the operation $(\mathbf{Z}\mathbf{Z}^T)^q\mathbf{Y}$ amplifies the approximate leading directions by concentrating the energy of the captured subspace. In this way, SPI mimics the effect of power iteration while only requires one pass over $\mathbf{A}$.

SPI can be incorporated into various one-pass algorithms for low-rank approximation; here we integrate SPI into the sketching algorithm proposed by Tropp et al. [35] (TYUC17 for the original algorithm and TYUC17-SPI for its SPI enhanced version) to investigate it practically and theoretically. In practice, while a naive implementation of TYUC17-SPI requires additional storage for $\mathbf{Z}$, we propose a mixed-precision and two hardware-dependent strategies to ensure that it maintains comparable storage cost to TYUC17. Numerical evidence demonstrates that TYUC17-SPI achieves a substantial improvement in accuracy over TYUC17, with one or two sketch-power iterations. In particular, experiments on real data exhibiting flat decay spectrum show that enhanced with one iteration, TYUC17 can approximate the dominant singular vectors well. In the supplementary material, we also integrate SPI into the streaming algorithm of Tropp et al. [36].

Two error bounds of TYUC17-SPI have been established, depending on the sketch sizes, q, and tail energy. The first is an expectation bound in the Frobenius norm for the $q = 1$ case, and the second one is a probabilistic deviation bound in the spectral

norm for general q cases. Given a fixed storage budget, we have used the first bound to derive sketch sizes for different singular spectrum decay types and rates. Empirically, the resulting sketch sizes are closely aligned with the optimal choices (oracle ones). This allows us to preselect sketching parameters to achieve higher accuracy under theoretical guidance. The bounds in [35, 36] have also been employed for this purpose, and our work extends their results in this direction.

The remainder is organized as follows. Sect. 2 formally introduces the sketch-power method and its integration with TYUC17. Sect. 3 provides theoretical error bounds and related implications, with proofs deferred to Sect. 5 and 6. Sect. 4 provides numerical experiments to validate its accuracy. Some related approaches are discussed in Sect. 7. Sect. 8 draws conclusions.

Notation. $[\![\mathbf{A}]\!]_r$ means the best rank-r approximation of a matrix $\mathbf{A}$. $\cdot^\dagger$ represents the Moore-Penrose inverse. $\|\cdot\|_p$ standards for the matrix Schatten-p norm; in particular, $\|\cdot\|_F$ and $\|\cdot\|$ respectively means the Frobenius and spectral norms. $\tau_k(\mathbf{A})$ represents the tail energy of $\mathbf{A}$, i.e., $\tau_k^2(\mathbf{A}) = \sum_{i\geq k}\sigma_i^2$, where σ_i is the i-th singular value of $\mathbf{A}$, arranged in a descending order. $\kappa(\cdot)$ stands for the condition number.

Code. The code is available at github.com/Mitchell-Cxyk/SketchPowerIteration.

2. Sketch-Power Iteration.

This section begins with a review of RSVD [13], power iteration, and the TYUC17 algorithm [35]. Next, the sketch-power iteration is proposed and integrated into TYUC17, along with strategies to reduce its storage cost. Finally, we discuss how to choose the sketch sizes for different spectrum decays.

2.1. RSVD and power iteration.

Randomized SVD (RSVD) was proposed in [24, 22] and then systematically refined and analyzed in [13]. Given a data matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$ and the target rank r, RSVD takes the following steps:

1. Draw a random test matrix $\mathbf{\Omega} \in \mathbb{R}^{n\times s}$ ($s > r$ with $p = s - r$ the oversampling parameter) and compute $\mathbf{Y} = \mathbf{A\Omega}$;
2. Compute the orthonormal basis of $\mathbf{Y}$ by thin QR: $[\mathbf{Q}, \sim] = \texttt{QR}(\mathbf{Y}, 0)$;
3. Compute $\mathbf{B} = \mathbf{Q}^T\mathbf{A}$ and its best rank-r approximation $[\![\mathbf{B}]\!]_r$;
4. Output $\hat{\mathbf{A}} := \mathbf{Q}[\![\mathbf{B}]\!]_r$ as the rank-r approximation of $\mathbf{A}$.

RSVD works well when the spectrum of $\mathbf{A}$ has certain decay; otherwise, the small singular values may influence $\mathbf{Y}$ to produce a poor range [13]. To overcome this, the power iteration has been employed to compute $\mathbf{Y}$ [32, 13, 12, 34]:

$$\mathbf{Y} = (\mathbf{A}\mathbf{A}^T)^q\mathbf{A\Omega}; \tag{2.1}$$

usually a few iterations are enough. To avoid numerical issues, instead of (2.1), a re-orthonormalization procedure can be employed:

$$\text{Repeat } q \text{ times}: \ [\mathbf{X}, \sim] = \texttt{QR}(\mathbf{A}^T\mathbf{Y}, 0), \ [\mathbf{Y}, \sim] = \texttt{QR}(\mathbf{AX}, 0). \tag{2.2}$$

(2.2) gives the same basis as that of (2.1) but is more numerically stable. Additional flops for orthonormalizations are $O(q\cdot(ms^2 + ns^2))$.

In practice, there is no need to perform two QRs in each iteration [39]. An option is to replace QR by LU before the last iteration [20]. (2.1) and (2.2) require $2q$ passes of $\mathbf{A}$, while [2] proposed a method that allows any number of passes.

2.2. The TYUC17 algorithm. RSVD needs two passes of $\mathbf{A}$, which is prohibitive in some applications. Several one-pass algorithms have been proposed and studied [41, 7, 40, 8, 4, 36, 29, 37]. We mainly recall that of Tropp et al. [35], which is a numerically stable version of the proposal of [7, Thm. 4.9]. [35] can be regarded as a modification of RSVD to the one-pass scenario. First draw random matrices $\mathbf{\Omega} \in \mathbb{F}^{n\times s}$ and $\mathbf{\Psi} \in \mathbb{F}^{l\times m}$ independently with $r \le s \le d \ll \min\{m, n\}$. Construct two sketches:

$$\mathbf{Y} = \mathbf{A\Omega} \in \mathbb{F}^{m\times s} \quad \text{and} \quad \mathbf{W} = \mathbf{\Psi A} \in \mathbb{F}^{d\times n}. \tag{2.3}$$

Here $\mathbf{W}$ captures the co-range of $\mathbf{A}$. As RSVD, $\mathbf{Q}$ is computed from the thin QR of $\mathbf{Y}$. Then the QB approximation to $\mathbf{A}$ is obtained as

$$\mathbf{A} \approx \mathbf{QB}, \quad \text{with} \quad \mathbf{B} = (\mathbf{\Psi Q})^{\dagger}\,\mathbf{W} \in \mathbb{F}^{s\times n} \Leftrightarrow \mathbf{B} \in \arg\min_{\mathbf{X}} \|\mathbf{\Psi}(\mathbf{A} - \mathbf{QX})\|_F. \tag{2.4}$$

Finally the rank-r approximation is given by $\hat{\mathbf{A}} = \mathbf{Q}[\![\mathbf{B}]\!]_r$. The algorithm only needs a single pass of $\mathbf{A}$, and $\mathbf{Y}, \mathbf{W}$ can be generated in parallel for efficiency. It is a stabilized oblique projection method [29]. In the sequel, we call the algorithm *TYUC17*.

2.3. Sketch-power iteration. TYUC17 exhibits strong numerical performance under rapidly decaying spectra; yet, as RSVD, its performance tends to deteriorate when the spectral decay is flat. However, the power iteration is prohibitive, as it needs extra $2q$ passes of $\mathbf{A}$, violating the nature of one-pass algorithms.

To address this limitation, motivated by sketching algorithms, we consider replacing $\mathbf{A}$ in the power iterations by a sketch, i.e., we perform the following to obtain the rangefinder matrix:

Sketch-Power Iteration

$$\hat{\mathbf{Y}} = (\mathbf{ZZ}^T)^q\mathbf{Y} = \mathbf{Z}(\mathbf{Z}^T\mathbf{Z})^{q-1}\mathbf{Z}^T\mathbf{A\Omega}, \tag{2.5}$$
$$\mathbf{Z} = \mathbf{A\Phi} \in \mathbb{R}^{m\times l};\ \mathbf{\Phi} \in \mathbb{R}^{n\times l} \text{ oblivious random test matrix.}$$

Here $l \ll \min\{m, n\}$. In both practice and theory, $l > s$ is necessary; otherwise $\hat{\mathbf{Y}}$ cannot retain more information than $\mathbf{Y}$. We call (2.5) the Sketch-Power Iteration (SPI). As both $\mathbf{Z}$ and $\mathbf{Y}$ can be simultaneously sketched through a single pass of $\mathbf{A}$, SPI is well-suited for integration into one-pass algorithms.

2.3.1. Intuition. A natural intuition of SPI comes from the subspace embedding perspective. Consider the case $\mathbf{A} = \mathbf{A}_1 + \mathbf{A}_2$, where $\mathbf{A}_1$ is a rank-k matrix carrying the principal information of $\mathbf{A}$, and $\mathbf{A}_2$ is a perturbation with relatively small magnitude. In particular, assume that the spectrum of $\mathbf{A}_1$ decays not so rapidly, while there is a clear spectral gap between $\mathbf{A}_1$ and $\mathbf{A}_2$. This corresponds to the "clear low-rank" scenario in the introduction. The goal is to find a rank-r approximation ($r < k$).

If one simply takes $\mathbf{Y} = \mathbf{A\Omega}$ (with $k > s > r$) as a rangefinder, then $\mathbf{Y}$ may deviate considerably from the subspace in interest. By contrast, if we construct $\mathbf{Z} = \mathbf{A\Phi}$ of

an appropriately large size (e.g., $l > 2k$), we can include enough of $\mathbf{A}_1$'s main spectral directions. Denote $\mathbf{Z}_1 = \mathbf{A}_1\mathbf{\Phi}$; one can see this by noting

$$\mathbf{Z}\,\mathbf{Z}^T\mathbf{Y} = \mathbf{Z}_1\mathbf{Z}_1^T\ \mathbf{A}_1\,\mathbf{\Omega} + f(\mathbf{A}_1, \mathbf{A}_2)\,\mathbf{\Omega},$$

where $f(\mathbf{A}_1, \mathbf{A}_2)$ collects all terms involving $\mathbf{A}_2$, including cross terms with $\mathbf{A}_1$. For the first term, by the Johnson-Lindenstrauss property, the subspace embedding $\mathbf{A}_1\mathbf{\Phi}$ preserves the action of operator $\mathbf{A}_1$ that helps correct the direction of $\mathbf{Y}$, making $\mathbf{Z}_1$ act like $\mathbf{A}_1$, as well as $\mathbf{A}$ (since $\mathbf{A}_1$ well approximates $\mathbf{A}$). For Gaussian embedding, $l > 2k$ is usually enough to ensure this. For the second term, $\mathbf{A}_2$ carries insignificant energy, so $f(\mathbf{A}_1, \mathbf{A}_2)\,\mathbf{\Omega}$ contributes a relatively small perturbation. Thus, by choosing suitable l and s, we preserve the key spectral information of $\mathbf{A}_1$ while limiting overhead. Consequently, SPI approximately replicates the subspace in interest typically revealed by a standard power iteration, yet it does so with only a single pass over $\mathbf{A}$.

2.3.2. Stabilization. Instead of directly performing (2.5), re-orthonormalization can be employed to avoid round-off issues. Additional cost for stabilization of SPI only takes $O(q \cdot ls^2)$ flops, independent of m, n. We shortly introduce it as follows. For $\mathbf{Y}$ at hand, first denote $\hat{\mathbf{Y}} = \mathbf{Y}$; the procedure takes the following form:

Sketch-Power Iteration with Re-orthonormalization
(2.6) Repeat q times : $[\mathbf{X}, \sim] = \mathtt{QR}(\mathbf{Z}^T\hat{\mathbf{Y}}, 0)$, $\hat{\mathbf{Y}} = \mathbf{Z}\mathbf{X}$. $\quad \%\mathbf{Z}^T\hat{\mathbf{Y}} \in \mathbb{R}^{l\times s}$

The range of $\hat{\mathbf{Y}}$ given by (2.6) is equivalent to that of (2.5), provided that both $\mathbf{Z}$ and $\mathbf{Y}$ have full column rank. To see this, just mark $\mathbf{X}$ and $\hat{\mathbf{Y}}$ above with their iteration number (particularly $\hat{\mathbf{Y}}_0 = \mathbf{Y}$), and explicitly write $\mathbf{Z}^T\hat{\mathbf{Y}}_p = \mathbf{X}_p\mathbf{R}_p$; then $\mathbf{Z}^T\hat{\mathbf{Y}}_p \in \mathbb{R}^{l\times s}$ and $\hat{\mathbf{Y}}_p$ all have full column rank and $\mathbf{R}_p$ are invertible. Iteratively express $\hat{\mathbf{Y}}_q$ to see that

$$\hat{\mathbf{Y}}_q = \mathbf{Z}\mathbf{Z}^T\hat{\mathbf{Y}}_{q-1}\mathbf{R}_q^{-1} = \cdots = (\mathbf{Z}\mathbf{Z}^T)^q\hat{\mathbf{Y}}_0\mathbf{R}_q^{-1}\cdots\mathbf{R}_1^{-1}.$$

Comparing (2.6) with (2.5), only additional q thin QR on matrices $\mathbf{Z}^T\hat{\mathbf{Y}}$ sizing $l \times s$ are required. This extra cost is independent of the size of $\mathbf{A}$, such that stabilization in SPI is both computation- and storage-efficient for large-scale data.

2.3.3. Test matrices. Just as sketching algorithms, various oblivious random test matrices are available for SPI. Frequently used examples include Gaussian, Rademacher [7], sparse [40], subsampled randomized Fourier transform (SRFT) [3, 13, 41], and sub-Gaussian [33]. Gaussian embedding is the most widely used in the theoretical analysis. On one hand, it usually provides optimal guarantees and often gives sharper bounds. On the other hand, other randomized embeddings may behave similarly to a Gaussian embedding in practice due to the central limit theorem [26]. Thus, the results from Gaussian theory are often valuable for the general behavior.

2.3.4. Complexity. For dense $\mathbf{A}$ and $\mathbf{\Phi}$ being unstructured distributions, one first forms $\mathbf{Z} \in \mathbb{R}^{m\times l}$, needing mnl flops; an iteration of SPI computes $\mathbf{X} = \mathbf{Z}^T\mathbf{Y} \in \mathbb{R}^{l\times s}$ and $\mathbf{Z}\mathbf{X} \in \mathbb{R}^{m\times s}$, taking $2mls$ flops. Thus SPI with re-orthonormalization takes about $mnl + qmls + O(qls^2)$ flops.

For structured embeddings such as SRFT or sparse test matrices, computing $\mathbf{Z}$ costs much less flops. For SRFT, the flops of computing $\mathbf{Z}$ are $O(mn\log(l))$, and for sparse $\mathbf{\Phi}$, computing $\mathbf{Z}$ requires $m \cdot \mathrm{nnz}(\mathbf{\Phi})$ flops, where $\mathrm{nnz}(\cdot)$ denotes the number of non-zero entries of a matrix. Thus the main cost of SPI with SRFT or sparse $\mathbf{\Phi}$ is $O(mn\log(l)) + 2qmls + O(qls^2)$ and $m \cdot \mathrm{nnz}(\mathbf{\Phi}) + 2qmls + O(qls^2)$, respectively.

2.4. TYUC17 with sketch-power iterations. SPI only acts on the rangefinder, and so it is easily incorporated in various one-pass algorithms. We mainly investigate SPI with the TYUC17 algorithm revisited in Sect. 2.2. This is because 1) TYUC17 is easily implemented; 2) it is numerically stable [29]; 3) it has an accurate error bound that can guide the selection of sketch parameters [35, Sect. 4.5]. In the supplementary material, we also integrate SPI into the streaming algorithm of Tropp et al. [36].

The modified algorithm (TYUC17-SPI for short) is depicted in Algorithm 2.1 and its differences with TYUC17 are highlighted in light blue.

Algorithm 2.1 TYUC17 Algorithm [35] with Sketch-Power Iteration

% **Sketching stage (support linear update):**

Require: Data matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$, random test matrices $\mathbf{\Omega} \in \mathbb{R}^{n\times s}$, $\mathbf{\Psi} \in \mathbb{R}^{d\times n}$, $\mathbf{\Phi} \in \mathbb{R}^{m\times l}$, $l, d, s \ll \min\{m, n\}$, $\min\{l, d\} > s$

Ensure: Sketches $\mathbf{Y} \in \mathbb{R}^{m\times s}, \mathbf{W} \in \mathbb{R}^{d\times n}, \mathbf{Z} \in \mathbb{R}^{m\times l}$

1: $\mathbf{Y} = \mathbf{A\Omega}$, $\mathbf{W} = \mathbf{\Psi A}$, $\mathbf{Z} = \mathbf{A\Phi}$

% **QB approximation stage:**

Require: Test matrix $\mathbf{\Psi} \in \mathbb{R}^{d\times n}$, sketches $\mathbf{Y} \in \mathbb{R}^{m\times s}$, $\mathbf{W} \in \mathbb{R}^{d\times n}$, $\mathbf{Z} \in \mathbb{R}^{m\times l}$, and parameter q

Ensure: Rank-s approximation of $\mathbf{A} \approx \hat{\mathbf{A}} = \mathbf{QB}$ with $\mathbf{Q} \in \mathbb{R}^{m\times s}$ and $\mathbf{B} \in \mathbb{R}^{s\times n}$:

1: Set $\hat{\mathbf{Y}} = \mathbf{Y}$, repeat q times:

$$[\mathbf{X}, \sim] = \mathtt{QR}(\mathbf{Z}^T\hat{\mathbf{Y}}, 0),\ \hat{\mathbf{Y}} = \mathbf{ZX} \qquad \text{\% QR costs } O(ls^2)$$

2: Compute $[\mathbf{Q}, \sim] = \mathtt{QR}(\hat{\mathbf{Y}}, 0)$

3: Compute $\mathbf{B} = (\mathbf{\Psi Q})^\dagger \mathbf{W}$

% **Truncation stage:**

Require: $\mathbf{Q} \in \mathbb{R}^{m\times s}, \mathbf{B} \in \mathbb{R}^{s\times n}$ from **QB stage**, target rank $r < s$

Ensure: Rank-r approximation triple $\mathbf{U} \in \mathbb{R}^{m\times r}$, $\mathbf{V} \in \mathbb{R}^{n\times r}$, and diagonal $\mathbf{\Sigma} \in \mathbb{R}^{r\times r}$, such that $\hat{\mathbf{A}} = \mathbf{U\Sigma V}^T \approx \mathbf{A}$

1: $\tilde{\mathbf{U}}\mathbf{\Sigma V}^T = [\![\mathbf{B}]\!]_r = \mathtt{SVD}(\mathbf{B}, r)$

2: $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$

A naive implementation of TYUC17-SPI requires $\mathbf{Y}, \mathbf{W}, \mathbf{Z}$ simultaneously, which is memory-unfriendly. Note that $\mathbf{Z}$ no longer participates in the computation once $\hat{\mathbf{Y}}$ is obtained. This allows us to design strategies such that TYUC17-SPI has a comparable storage cost as that of TYUC17. We discuss it in Sect. 2.5.

Remark 2.1. Throughout this work, we present SPI in its most transparent form with two sketches $\mathbf{Z}, \mathbf{Y}$, treating $\mathbf{\Phi}, \mathbf{\Omega}$ as independent. This setting highlights the distinct roles of $\mathbf{Y}$ as the initial rangefinder (as in TYUC17) and $\mathbf{Z}$ as an amplifier. In particular, the independence assumption is needed for the proof of our first bound (Theorem 3.1), which is further used to derive a priori sketch sizes under memory constraints (Appendix B). Nevertheless, SPI does not require $\mathbf{Z}, \mathbf{Y}$ be independent

as well: for example, choosing $\mathbf{\Omega} = \mathbf{\Phi}\tilde{\mathbf{\Omega}}$ leads to $\mathbf{Y} = \mathbf{Z}\tilde{\mathbf{\Omega}}$, which can further reduce memory cost in practice (see Sect. 2.7).

2.5. Limiting storage cost of TYUC17-SPI. We propose three strategies to reduce the storage cost of TYUC17-SPI to $dn+ml$, such that it is comparable to that of TYUC17.[1] The first uses mixed-precision, while the latter two are hardware-dependent. For the hardware-dependent approaches, we may consider the storage cost in the sketching stage and the approximation stage (including QB and truncation) separately, since in many architectures, the memory of the former stage is less constrained.

2.5.1. Mixed-precision. Mixed-precision techniques have shown promise in reducing both computational and storage costs in various algorithms. It was recently introduced in randomized algorithms [9, 5] to obtain the sketches in lower-precision, and implement subsequent computations in higher-precision.

This motivates implementing TYUC17-SPI in a mixed-precision fashion. We first sketch $\mathbf{W}, \mathbf{Z}, \mathbf{Y}$ in *single-precision*, and then compute and store $\hat{\mathbf{Y}}$ also in single-precision, overwriting the space of $\mathbf{Y}$. Recent advances in mixed-precision algorithms effectively alleviate the cumulative propagation of rounding errors that typically occurs in single-precision arithmetic [15, 14]. Next, before steps 2 and 3 of Algorithm 2.1, we can convert $\hat{\mathbf{Y}}$ and $\mathbf{W}$ to *double-precision* by reusing the space previously held by $\mathbf{Z}$ if its size equals those of $\hat{\mathbf{Y}}$ and $\mathbf{W}$; see Algorithm SM4.2 in the supplementary material. Higher precision is important for avoiding loss of orthonormality in steps 2 and keeping the accuracy of solving linear systems in step 3. All the subsequent computations are executed in double-precision as well.

We count the storage cost in terms of double-precision words. Storing the single-precision $\mathbf{W}, \mathbf{Z}, \mathbf{Y}$ costs $(d \cdot n + m \cdot s + m \cdot l)/2$ (1/2 represents single-precision); $\hat{\mathbf{Y}}$ takes the space of $\mathbf{Y}$. As the space of $\mathbf{Z}$ can be reused for converting and storing $\mathbf{W}$ and $\hat{\mathbf{Y}}$ into double-precision, we have $ml = ms + dn$. Thus the storage cost is

$$(d \cdot n + m \cdot s + m \cdot l)/2 = d \cdot n + m \cdot s, \tag{2.7}$$

the same as the storage cost of $dn + sm$ of TYUC17 (in double-precision).

The above single-double precision model allows for larger sketch sizes, trading an additional round-off error for better approximation accuracy, which is effective especially when the data has a flat spectrum; see Sect. 4 for example. The single/-double precisions can be replaced by any lower precision $\boldsymbol{\mu}_l$ and higher precision $\boldsymbol{\mu}_h$, respectively; e.g., we can set $\boldsymbol{\mu}_l$ as half-precision to allow larger l.

2.5.2. Multi-level caches system. Consider a setup consisting of a Hard Drive, CPU (with RAM), and GPU (with VRAM). In many GPU-accelerated tasks, VRAM is the real bottleneck due to its limited capacity, whereas RAM is typically larger[2]. The goal is to leverage GPU for speed while limiting its VRAM usage.

[1] Following [36], we disregard the storage cost of test matrices, as they can be structured, sparse, or be generated using random seeds.

[2] For instance, 256GB of RAM is relatively affordable, while a high-end NVIDIA GeForce RTX 5090 graphics card offers only 32GB of VRAM and is much more costly.

Concretely, we first compute the sketches $\mathbf{Y}, \mathbf{Z}, \mathbf{W}$ in a pass of $\mathbf{A}$ using CPU and store them temporarily in the more redundant RAM. Next we transfer $\mathbf{Z}$ and $\mathbf{Y}$ to GPU to compute $\hat{\mathbf{Y}}$. Once finishing, we delete $\mathbf{Z}$ and transfer $\mathbf{W}$ into VRAM, occupying the space previously held by $\mathbf{Z}$ (assume that $d \cdot n = m \cdot l$). Thus the storage needed in VRAM remains $d \cdot n + m \cdot s$, i.e., that for $\mathbf{W}$ and $\hat{\mathbf{Y}}$, matching the storage cost of TYUC17 if also computed by GPU.

Although TYUC17-SPI requires extra $m \cdot l$ storage in RAM (to hold $\mathbf{Z}$), the crucial point is that in the more constrained VRAM, its usage stays the same as TYUC17. Therefore, comparing VRAM usage between the two methods is fair, as both fit within the same hardware limits and maintain the same storage cost in GPU.

2.5.3. Distributed system. Suppose we are in a parallel or distributed system and for simplicity, the system has three computing nodes (labeled as $a_{\mathbf{W}}, b_{\mathbf{Z}}, c_{\mathbf{Y}}$), each with independent RAM. First, $\mathbf{W}, \mathbf{Z}, \mathbf{Y}$ are parallelly computed in node $a_{\mathbf{W}}, b_{\mathbf{Z}}$, and $c_{\mathbf{Y}}$, using a pass of $\mathbf{A}$. Then, $\mathbf{Y}$ is transferred from node $c_{\mathbf{Y}}$ to node $b_{\mathbf{Z}}$ to compute $\hat{\mathbf{Y}}$, and finally, $\hat{\mathbf{Y}}$ is transferred from node $b_{\mathbf{Z}}$ to $a_{\mathbf{W}}$ to execute the rest steps. Thus the storage cost in node $a_{\mathbf{W}}$ is still $d \cdot n + m \cdot s$, and the transferring cost is only $2ms$ ($\mathbf{Y}$ from $c_{\mathbf{Y}}$ to $b_{\mathbf{Z}}$ and $\hat{\mathbf{Y}}$ from $b_{\mathbf{Z}}$ to $a_{\mathbf{W}}$).

2.6. Sketch size guidance. It is necessary to a priori choose sketch sizes for one-pass algorithms. We only consider TYUC17-SPI in the mixed-precision setting. In the sequel, we express storage in terms of double-precision words. For the storage of the three single-precision sketches sizing $(m(l+s)+nd)/2$, it can be parameterized as $T = (c(l+s)+d)/2$ (suppose $m = cn$ for some $c > 0$). Determining l is easy: the discussion in Sect. 2.5.1 shows that $\mathbf{Z}$ takes half of the storage, i.e., $l = T \cdot n/m = T/c = s + d/c$. For s, we present the $c = 1$ case in Table 1, and put the $c \neq 1$ case in Sect. SM1 in the supplementary material.

Table 1: Sketch size s selection in different spectrum decay; $r \leq s \leq T/2$

Flat	Poly ($\sigma_i = i^{-\alpha}$)			Exp ($\sigma_i = e^{-\alpha \cdot i}$)	
	$\alpha < \frac{1}{2}$	$\alpha \approx \frac{1}{2}$	$\alpha > \frac{1}{2}$	$\alpha < \frac{1}{2T}$	$\alpha \geq \frac{1}{2T}$
r	r	$\mathrm{P}_{[r,T/2]}(-\frac{T+1}{2W(-(T+1)/2ne)} - 1)$	$\max\{r, \frac{(2\alpha-1)(T+3)-2}{4\alpha}\}$	r	$\frac{T}{2}$

In the table, "Flat" means flat spectrum, which occurs for example when $\mathbf{A}$ is of low-rank plus noise; "Poly" means polynomial spectrum decay, i.e., singular values $\sigma_i = i^{-\alpha}$ for $\alpha > 0$; "Exp" means exponential decay, i.e., $\sigma_i = e^{-\alpha \cdot i}$ for $\alpha > 0$; $\mathrm{P}_{[a,b]}(\cdot)$ means projection onto $[a, b]$. $W(\cdot)$ refers to the Lambert W function, i.e., $W(a)$ is the solution of $xe^x = a$. In Matlab, one needs to use the command `lambertw`$(-1, a)$ to compute the value for a negative a. Detailed derivations will be provided in Appendix B by minimizing the error bound with respect to the sketch sizes.

2.7. A variant for further reducing the storage cost of TYUC17-SPI. We can obtain a variant of SPI without sketching $\mathbf{Y}$ to further save storage. This can be achieved by using two test matrices $\mathbf{\Phi} \in \mathbb{R}^{n \times l}$ and $\tilde{\mathbf{\Omega}} \in \mathbb{R}^{l \times s}$; then set $\mathbf{Z} = \mathbf{A}\mathbf{\Phi}$ as that in SPI, while compute $\hat{\mathbf{Y}} = (\mathbf{Z}\mathbf{Z}^T)^q \mathbf{Z}\tilde{\mathbf{\Omega}} \in \mathbb{R}^{m \times s}$ instead.

If denoting $\mathbf{\Omega} = \mathbf{\Phi}\tilde{\mathbf{\Omega}}$ and $\mathbf{Y} = \mathbf{A}\mathbf{\Omega}$, then $\mathbf{Z}\tilde{\mathbf{\Omega}} = \mathbf{A}\mathbf{\Phi}\tilde{\mathbf{\Omega}} = \mathbf{A}\mathbf{\Omega} = \mathbf{Y}$, and so $\hat{\mathbf{Y}}$

above can still be written as $\hat{\mathbf{Y}} = (\mathbf{Z}\mathbf{Z}^T)^q\mathbf{Y}$, taking the same form as SPI. Thus this variant can still be regarded as SPI, but without explicitly generating $\mathbf{Y}$.

We count the storage cost. Storing $\mathbf{Z}, \mathbf{W}$ in single-precision requires $(ml + dn)/2$ storage. To compute $\hat{\mathbf{Y}}$, which can be expressed as $\hat{\mathbf{Y}} = \mathbf{Z}(\mathbf{Z}^T\mathbf{Z})^q\tilde{\mathbf{\Omega}}$, we allocate an extra l^2 storage to form $\mathbf{Z}^T\mathbf{Z}$ and reuse this space to compute $(\mathbf{Z}^T\mathbf{Z})^q\tilde{\mathbf{\Omega}}$. Subsequently, we compute $\hat{\mathbf{Y}} \in \mathbb{R}^{m\times s}$ by overwriting the first s columns of $\mathbf{Z}$, using extra s storage for buffering. Thereafter, $\hat{\mathbf{Y}}$ is converted to double-precision (using the space of $\mathbf{Z}$) for maintaining the orthonormal accuracy of $\mathbf{Q}$, while $\mathbf{W}$ remains in single-precision. When $s \le l/2$, the double-precision $\hat{\mathbf{Y}}$ can replace the space previously occupied by $\mathbf{Z}$, so that the overall storage cost is bounded by $(ml + dn)/2 + l^2 + s$.

3. Error Bounds.

This section states two error bounds for TYUC17-SPI: one tailored for $q = 1$ and one valid for general $q \ge 1$. Proofs appear in Sect. 5 and 6, respectively. We include both because the first facilitates a priori sketch parameter selection (Sect. B) and because their proofs rely on different techniques.

3.1. Error bound when $q = 1$.

We begin with the $q = 1$ case, giving an expected Frobenius norm bound. The test matrices $\mathbf{\Phi}, \mathbf{\Omega}, \mathbf{\Psi}$ are assumed independent and standard Gaussian, i.e., each entry is i.i.d. $N(0, 1)$. Recall that l, s, d respectively are the sketch size for $\mathbf{Z}, \mathbf{Y}, \mathbf{W}$.

THEOREM 3.1. *Let $\mathbf{Q} \in \mathbb{R}^{m\times s}, \mathbf{B} \in \mathbb{R}^{s\times n}$ be generated by Algorithm 2.1 with $q = 1$ and $\mathbf{\Phi} \in \mathbb{R}^{n\times l}, \mathbf{\Omega} \in \mathbb{R}^{n\times s}, \mathbf{\Psi} \in \mathbb{R}^{d\times m}$ be independent standard Gaussian. If $l > s \ge \varrho + 4$ and $d > s + 1$, then we have*

$$\mathbb{E}\left[\|\mathbf{A} - \mathbf{Q}\mathbf{B}\|_F^2 | E_F\right] \le \frac{d}{d-s-1}\left[(1+\epsilon\hat{\xi}^2)\tau_{\varrho+1}^2(\mathbf{A}) + \delta\hat{\xi}^2\frac{\sum_{j=\varrho+1}^n \sigma_j^6}{\sigma_\varrho^4} + \mu\hat{\xi}^2\tau_{\varrho+1}^2(\mathbf{A})\frac{\sum_{j=\varrho+1}^n \sigma_j^4}{\sigma_\varrho^4}\right],$$

where $\epsilon = \frac{2\varrho}{(l-\varrho-1)}$,

$$\delta = \frac{e^2(s+\varrho)\varrho(l-1)(l^2+2l)\cdot}{(s-\varrho)^2(l-\varrho)(l-\varrho-1)(l-\varrho-3)},\ \mu = \frac{e^2(s+\varrho)\varrho(l-1)l\cdot}{(s-\varrho)^2(l-\varrho)(l-\varrho-1)(l-\varrho-3)}, \tag{3.1}$$

and $E_F = \left\{\mathbf{\Omega}, \mathbf{\Phi} : \left\|\left(\mathbf{I} + \mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{\Sigma}_2^2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\mathbf{\Sigma}_1^{-2}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\right)^{-1}\right\| < \xi\right\}$ for some given $\xi > 1$, and $\hat{\xi} = \xi/\sqrt{\mathbb{P}(E_F)} \approx \xi$ (see the discussion below).

The matrices involved in E_F will be detailed in Sect. 5.2. ϱ is a parameter that varies between r and s; one can minimize the bound with respect to ϱ to get a tighter bound. The following gives some discussions on the bound.

On the event E_F. E_F is introduced to characterize the event that the matrix $\mathbf{I} + \mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{\Sigma}_2^2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\mathbf{\Sigma}_1^{-2}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}$ stays close to the identity, with spectral norm of its inverse bounded by a moderate constant ξ. A relatively larger SPI parameter l reduces its spectral norm and increases the likelihood of E_F. Specifically, if l is relatively large such that the term $(e_1e_4 + e_2e_3 + e_2e_4u)e_5^2\sigma_\varrho^{-2} < 1 - \xi^{-1}$, then $\mathbb{P}(E_F) > 1 - p_f$, where e_i's are defined in (5.17), failure probability $p_f = 5e^{-u^2/2} + 2t^{-(l-\varrho)} + 2t^{-(s-\varrho)}$, and $u > 0, t > 1$. See Theorem 5.12 and its discussions in Sect. 5.3 for more details. The underlying reason is the favorable embedding properties of $\mathbf{\Phi}_1$. p_f can be small such that $\hat{\xi} \approx \xi$.

Simplification under spectral gap. Assume a large gap between σ_ϱ and $\sigma_{\varrho+1}$. Then the last two terms in the bound can be viewed as higher-order terms of $\tau^2_{\varrho+1}(\mathbf{A})$. Meanwile, the coefficient of the first term can be simplified: $1+\epsilon\xi^2 \le 2\xi^2 \cdot \frac{l}{l-\varrho-1} = O(\frac{l}{l-\varrho-1})$. Up to constants, the bound reduces to the following:

$$\mathbb{E}\left[\|\mathbf{A}-\mathbf{QB}\|_F^2 \,|\, E_F\right] \le O\left(\frac{d}{d-s-1}\right)\cdot O\left(\frac{l}{l-\varrho-1}\right)\tau^2_{\varrho+1}(\mathbf{A}) + o(\tau^2_{\varrho+1}(\mathbf{A})). \tag{3.2}$$

While heuristic, this simplification provides effective sketch sizes guidance (Sect. B).

The higher-order terms arise from one iteration of SPI. The coefficient $d/(d-s-1)$ matches that in the TYUC17 bound (see (3.3)), due to both sketching the same corange $\mathbf{W} = \mathbf{\Psi A}$. We also see that a larger SPI parameter l yields smaller bound (also observe from (5.16) that a larger l does not increase δ, μ, the coefficients of $o(\tau^2_{\varrho+1}(\mathbf{A}))$). This aligns with the intuition that a larger $\mathbf{Z}$ captures more information of $\mathbf{A}$, and so does the new rangefinder $\hat{\mathbf{Y}} = \mathbf{ZZ}^T\mathbf{Y}$ produces a better approximation.

Comparison with TYUC17 bound. Comparing (3.2) with [35, Thm. 4.3][3]

$$\mathbb{E}\|\mathbf{A}-\mathbf{QB}\|_F^2 \le \frac{d}{d-s-1}\frac{s}{s-\varrho-1}\tau^2_{\varrho+1}(\mathbf{A}), \tag{3.3}$$

we see that SPI replaces the relatively small s in $s/(s-\varrho-1)$ with a larger l, which lowers the TYUC17 bound when higher-order terms are negligible. This helps explain why TYUC17, when equipped with SPI, achieves better performance in practice.

On orthogonal projection. Bounding the orthogonal projection error $\left\|\mathbf{A}-P_{\hat{\mathbf{Y}}}\mathbf{A}\right\|_F$ is the key to establish the main bound, where $P_{\hat{\mathbf{Y}}}$ is the projection onto the column space of $\hat{\mathbf{Y}}$. With $\left\|\mathbf{A}-P_{\hat{\mathbf{Y}}}\mathbf{A}\right\|_F$ obtained, the relation between oblique and orthogonal projection error[4] follows by the argument of [35]: conditioned on $\mathbf{\Omega}, \mathbf{\Phi}$,

$$\mathbb{E}_{\mathbf{\Psi}}\|\mathbf{A}-\mathbf{QB}\|_F^2 = \frac{d}{d-s-1}\left\|\mathbf{A}-P_{\hat{\mathbf{Y}}}\mathbf{A}\right\|_F^2.$$

The orthogonal projection is the main difference between TYUC17 and TYUC17-SPI.

We do not state the final error $\|\mathbf{A}-\mathbf{Q}\tilde{\mathbf{U}}\mathbf{\Sigma}\mathbf{V}^T\|_F$; it can be derived directly from Theorem 3.1 and [35, Prop. 6.1], and is therefore omitted.

3.2. Error bound for general $q \ge 1$. We now present a probabilistic bound in spectral norm. Here $\mathbf{\Psi} \in \mathbb{R}^{d\times n}, \mathbf{\Phi} \in \mathbb{R}^{m\times l}$ are independent standard Gaussian, while $\mathbf{\Omega} = \mathbf{\Phi}\tilde{\mathbf{\Omega}} \in \mathbb{R}^{m\times s}$, where $\tilde{\mathbf{\Omega}} \in \mathbb{R}^{l\times s}$ is standard Gaussian independent of $\mathbf{\Psi}, \mathbf{\Phi}$. This setting connects to Sect. 2.7.

THEOREM 3.2. *Let $\mathbf{Q}, \mathbf{B}$ be generated by Algorithm 2.1, with $\mathbf{\Phi}, \mathbf{\Psi}$ being independent standard Gaussian, and $\mathbf{\Omega} = \mathbf{\Phi}\tilde{\mathbf{\Omega}}$ where $\tilde{\mathbf{\Omega}}$ is standard Gaussian independent of*

[3]There was a typo in the bound of [35, Thm. 4.3]: the factor $k/(l-k-1)$ should be $l/(l-k-1)$, as $1+k/(l-k-1) \le l/(l-k-1)$. Note that l, k in [35] correspond to d, s in our notation.

[4]"Orthogonal" is emphasized to distinguish it from the oblique projection error $\|\mathbf{A}-\mathbf{QB}\|_F$. Unless stated otherwise, "projection" hereafter refers to orthogonal projection.

$\mathbf{\Phi}, \mathbf{\Psi}$. *Assume that* $d \geq s+4$, *and for any natural numbers* ϱ, k *with* $\varrho < k \leq l-4$ *and* $\varrho \leq s-4$. *Then* $\|\mathbf{A} - \mathbf{QB}\|$ *is upper bounded by*

$$
\begin{aligned}
&((\eta_1 + u \cdot \eta_2)\eta_3 + \eta_2\eta_4)\, \tau_{k+1}(\mathbf{A}) + ((\eta_1 + u \cdot \eta_2)(\eta_3 + \eta_4) + u\eta_3)\, \sigma_{k+1}(\mathbf{A}) \\
&+ \gamma_1\gamma_2\gamma_3 \left(\left(1 + \sqrt{\frac{k}{l}} + \frac{\beta}{\sqrt{l}} \right) \sigma_{\varrho+1}(\mathbf{A}) + \left(1 + \frac{\beta}{\sqrt{l}} \right) \sigma_{k+1}(\mathbf{A}) + \tau_{k+1}(\mathbf{A}) / \sqrt{l} \right),
\end{aligned}
$$

where

$$
\begin{aligned}
\eta_1 &= 1 + t\sqrt{\frac{3s}{d-s+1}};\ \eta_2 = t\frac{e\sqrt{d}}{d-s+1}; \\
\eta_3 &= \frac{e\sqrt{l}}{l-k+1}t;\ \eta_4 = 1 + t\sqrt{\frac{3k}{l-k+1}}; \\
\gamma_1 &= 1 + t\sqrt{\frac{3s}{d-s+1}} + t\frac{e\sqrt{dl}}{d-s+1} + ut\frac{e\sqrt{d}}{d-s+1};\ \gamma_2 = \frac{el}{l-k+1}; \\
\gamma_3 &= \left(1 + t \cdot \sqrt{\frac{3\varrho}{s-\varrho+1}} + t \cdot \frac{e \cdot \left(\sqrt{sl} + u\sqrt{s} \right)}{s-\varrho+1} \right)^{\frac{1}{2q+1}},
\end{aligned}
$$

with failure probability at most $p_e = 3e^{-u^2/2} + 2t^{-(d-s)} + 2t^{-(l-k)} + 2t^{-(s-\varrho)} + e^{-\beta^2/2}$, *where* $u, \beta > 0$, *and* $t \geq 1$.

As ϱ, k are relatively flexible, the bound can be improved by suitable choices.

Simplification under parameter choices. For simplicity, let $d = 2s$, $l = 2k$, which are reasonable. We also assume $l = O(d)$ (c.f. Sect. 2.6), In this case, $\eta_1, \eta_4, \gamma_1, \gamma_2 = O(1)$, $\eta_2 = O(1/\sqrt{d})$, and $\eta_3 = O(1/\sqrt{l})$. The bound then simplifies to

$$
\begin{aligned}
&O\left(\frac{1}{\sqrt{d}} + \frac{1}{\sqrt{l}} \right) \tau_{k+1}(\mathbf{A}) + O\left(1 + O(\frac{1}{\sqrt{d}}) + O(\frac{1}{\sqrt{l}}) \right) \sigma_{k+1}(\mathbf{A}) \\
&+ O(1)\gamma_3 \left(\left(O(1) + O\left(\frac{1}{\sqrt{l}} \right) \right) \sigma_{\varrho+1}(\mathbf{A}) + (1 + O(1/\sqrt{l}))\sigma_{k+1}(\mathbf{A}) + \frac{\tau_{k+1}(\mathbf{A})}{\sqrt{l}} \right) \\
\approx &\gamma_3 O\left(1 + \frac{1}{\sqrt{l}} \right) \sigma_{\varrho+1}(\mathbf{A}) \\
&+ (O(1) + O(1)\gamma_3)\, O\left(1 + \frac{1}{\sqrt{l}} \right) \sigma_{k+1}(\mathbf{A}) + O\left(\frac{\gamma_3}{\sqrt{l}} + \frac{1}{\sqrt{l}} \right) \tau_{k+1}(\mathbf{A}).
\end{aligned}
$$

Two observations follow. First, γ_3, which depends on the number of SPI steps, decreases exponentially with q, thereby reducing the bound, acting a similar role as the standard power iterations [13]. Second, the bound decreases further as l increases.

Tail energy improvement. The tail energy is the price for obtaining a spectral norm bound [13, 34, 12]. A key distinction of our result is that the tail does not start from a small $\varrho + 1$ [13] nor $s + 1$ [34, 12], but from a considerably larger index $k + 1$ (as k scales with l, it can be several times larger than s). Consequently, $\tau_{k+1}(\mathbf{A})$ may be far smaller than $\tau_{\varrho+1}(\mathbf{A})$, especially when the spectrum decays slowly before the

first k singular values but rapidly thereafter. A special case is $\operatorname{rank}(\mathbf{A}) = k$, where $\sigma_{k+1}(\mathbf{A}) = \tau_{k+1}(\mathbf{A}) = 0$, so that for large q, $\|\mathbf{A} - \mathbf{QB}\| \lesssim O(1)\sigma_{\varrho+1}(\mathbf{A})$ precisely. This illustrates the benefit of SPI: whereas a small rangefinder $\mathbf{Y} = \mathbf{A\Omega}$ may fail to capture sufficient range information due to storage limits, SPI drives the sketch size from s to a considerably large l to capture much more information (with feasible storage), powers it to enhance the rangefinder, yielding improved results.

Comparison with TYUC17 bound. To compare Theorem 3.2 with the bound of TYUC17, we also need a spectral norm bound for it (as [35] only provides a Frobenius norm one in expectation). Similar to the $q = 1$ case, the key is to convert oblique to orthogonal projection errors. By Lemma 6.8, with high probability,

$$\|\mathbf{A} - \mathbf{QB}\| \leq \beta_1 \|\mathbf{A} - P_{\hat{\mathbf{Y}}}\mathbf{A}\|_F + \beta_2 \|\mathbf{A} - P_{\hat{\mathbf{Y}}}\mathbf{A}\|.$$

where $\beta_1 = \frac{e\sqrt{d}t}{d-s+1}, \beta_2 = 1 + t\sqrt{\frac{3s}{d-s+1}} + u\frac{e\sqrt{d}t}{d-s+1}$. For TYUC17, similarly,

$$\|\mathbf{A} - \mathbf{QB}\| \leq \beta_1 \|\mathbf{A} - P_{\mathbf{A\Omega}}\mathbf{A}\|_F + \beta_2 \|\mathbf{A} - P_{\mathbf{A\Omega}}\mathbf{A}\|.$$

Since both decompositions are analogous, it suffices to compare their orthogonal projection errors. For clarity, we only consider the spectral norm case. Theorem 6.7 shows that with high probability,

$$\begin{aligned}\|\mathbf{A} - P_{\hat{\mathbf{Y}}}\mathbf{A}\| \leq\ & 2\left(1 + t \cdot \sqrt{\frac{3k}{l-k+1}} + \frac{ut \cdot e\sqrt{l}}{l-k+1}\right)\sigma_{k+1}(\mathbf{A}) + \frac{2t \cdot e\sqrt{l}}{l-k+1}\tau_{k+1}(\mathbf{A}) \\ & + \frac{te\sqrt{l}}{l-k+1} \cdot \gamma_3 \cdot \left(\sigma_{\varrho+1}(\mathbf{A})(\sqrt{l} + \sqrt{k} + \beta) + \sigma_{k+1}(\mathbf{A})(\sqrt{l} + \beta) + \tau_{k+1}(\mathbf{A})\right).\end{aligned}$$

On the other hand, [13] shows that with high probability,

$$\|\mathbf{A} - P_{\mathbf{A\Omega}}\mathbf{A}\| \leq \left(1 + t \cdot \sqrt{\frac{3\varrho}{s-\varrho+1}} + \frac{ut \cdot e\sqrt{s}}{s-\varrho+1}\right)\sigma_{\varrho+1}(\mathbf{A}) + t \cdot \frac{e\sqrt{s}}{s-\varrho+1}\tau_{\varrho+1}(\mathbf{A}).$$

The first-line bound for $\left\|\mathbf{A} - P_{\hat{\mathbf{Y}}}\mathbf{A}\right\|$ essentially reflects the error of $\|\mathbf{A} - P_{\mathbf{Z}}\mathbf{A}\|$. Comparing with $\|\mathbf{A} - P_{\mathbf{A\Omega}}\mathbf{A}\|$, this term is typically far smaller, especially when $\tau_{k+1}(\mathbf{A})$ is substantially smaller than $\tau_{\varrho+1}(\mathbf{A})$. This improvement is accompanied by an additional term $\|\mathbf{Z} - P_{(\mathbf{ZZ}^T)^q\mathbf{A\Omega}}\mathbf{Z}\|$ that corresponds to the second line. When l is considerably large, the coefficient of $\sigma_{\varrho+1}(\mathbf{A})$ is approximately a constant and moreover, γ_3 is driven down to 1 as q increases. As a result, the sketch size l and the number of SPI iterations q are the primary mechanisms that help improve the bound.

4. Numerical Experiments.

This section presents numerical experiments to validate the effectiveness of SPI. The following are observed: 1) TYUC17-SPI in mixed-precision can improve the accuracy of TYUC17 with the same storage budget[5]; 2) The sketch sizes guidance in Sect. 2.6 is reliable; 3) Only a few iterations of SPI are generally sufficient in our tests. All the experiments were conducted on a computer with an Intel Core i9-12700 CPU and 64GB of RAM. The Matlab version is 2023b.

4.1. Experimental setup.

[5]TYUC17-SPI with the other two strategies in Sect. 2.5 can clearly improve the accuracy, though it is hardware-dependence. To better illustrate the broad applicability of SPI, we thus employ the mixed-precision strategy here that runs on any computer.

4.1.1. Synthetic data. Data matrices similar to those in [35, 36] are used.

1. Low-rank + noise: The matrices are generated by $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T + \frac{\gamma R}{n^2}\mathbf{E}$, where $\mathbf{U}, \mathbf{V}$ are randomly orthonormal, $\mathbf{\Sigma} = \begin{bmatrix} I_R & 0 \\ 0 & 0 \end{bmatrix}$ is diagonal, γ is signal-to-noise ratio (SNR), and $\mathbf{E}$ is a Gaussian noise matrix. We consider the case where $R = 10$ and $m = n = 1000$ and
   (a) LowNoise: $\gamma = 10^{-4}$; (b) MediumNoise: $\gamma = 0.01$; (c) HighNoise: $\gamma = 0.1$.

2. Polynomially (power law) decaying spectrum: The matrices are generated by $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$, where $\mathbf{\Sigma} = \operatorname{diag}(1, \ldots, 1, 2^{-\alpha}, \ldots, n^{-\alpha})$ having R ones, and $\mathbf{U}, \mathbf{V}$ are randomly orthonormal. $m = n = 1000$, and the decay rate satisfies
   (a) SlowDecay: $\alpha = 0.5$; (b) MediumDecay: $\alpha = 1$; (c) FastDecay: $\alpha = 2$.

3. Exponentially decaying spectrum: The matrices are generated by $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$, where $\mathbf{U}, \mathbf{V}$ are randomly orthonormal, $\mathbf{\Sigma} = \operatorname{diag}(1, \ldots, 1, e^{-\alpha}, \ldots, e^{-n\alpha})$ which has R ones, $m = n = 1000$, and the decay rate satisfies
   (a) SlowDecay: $\alpha = 0.01$; (b) MediumDecay: $\alpha = 0.1$; (c) FastDecay: $\alpha = 0.5$.

4.1.2. Real data. Our real data contains the following two datasets:

1. NIST Special Database 19: the dataset is a key resource for handprinted document and character recognition research. It contains handprinted sample forms from 3600 writers, over 810000 segmented handprinted character images covering digits, uppercase and lowercase letters, and ground truth classifications for each image. This large-scale, diverse dataset is invaluable for training robust recognition models and advancing related AI applications.
   We have obtained a 16384×20000 matrix from a subset containing all 20000 images of the digit 0 from different writers, each being a 128×128 image. Each image is reshaped into a column vector and stacked to form the matrix.
2. Climate dataset: The climate datasets are obtained from the NCEP/NCAR Reanalysis 1 project. The NCEP-NCAR Reanalysis 1 (R1) is a comprehensive atmospheric dataset providing historical atmospheric data from 1948 to the present. It is generated using a global weather model that combines observational data (from weather stations, satellites, etc.) with numerical weather prediction models to produce a consistent and high-resolution record of atmospheric variables (temperature, pressure, winds, etc.) at multiple pressure levels. The dataset has a spatial resolution of 2.5° latitude by 2.5° longitude and a temporal resolution of 6 hours, making it invaluable for studying both short-term weather patterns and long-term climate trends.
   We have obtained a 10512×28144 matrix from the dataset, each column being a vector reshaped by the pressure data of the 73×144 meshgrid on the surface. All daily data from 1948 to 2025 forms the matrix.

4.1.3. Relative Error. The relative error metric follows that in [35, 36]:

$$\text{Relative error: } S_F = \frac{\|\mathbf{A} - \hat{\mathbf{A}}\|_F}{\|\mathbf{A} - [\![\mathbf{A}]\!]_r\|_F} - 1 \text{ and } S_\infty = \frac{\|\mathbf{A} - \hat{\mathbf{A}}\|}{\|\mathbf{A} - [\![\mathbf{A}]\!]_r\|} - 1, \tag{4.1}$$

where $\mathbf{A}$ is the original data and $\hat{\mathbf{A}}$ is a rank-r approximation computed by algorithms.

4.1.4. Oracle error. The sketch sizes affect the results; thus it is possible to report the best possible performance for a fair comparison. This asks to enumerate over all possible parameters and evaluate them to select the most appropriate ones. The associated error is called the oracle error [35, 36]. Due to computer limit, it is only reachable for small test data in our experiments, e.g., the synthetic data.

4.1.5. Test matrix. Although the parameter guidance is based on the Gaussian test matrix assumption, to save storage in practice, we use the sparse Rademacher test matrix generated as follows: first draw a Rademacher matrix and then randomly set 99% of the entries to be zero. It is unclear whether this distribution enjoys the Johnson-Lindenstrauss (JL) property in theory; our use here is purely empirical, motivated by its implementation and storage advantage. In practice, we observe that it empirically exhibits JL-like behavior, and it yields results similar to those obtained with standard test matrices; see Sect. SM2 and SM3 in the supplementary material for numerical evidence.

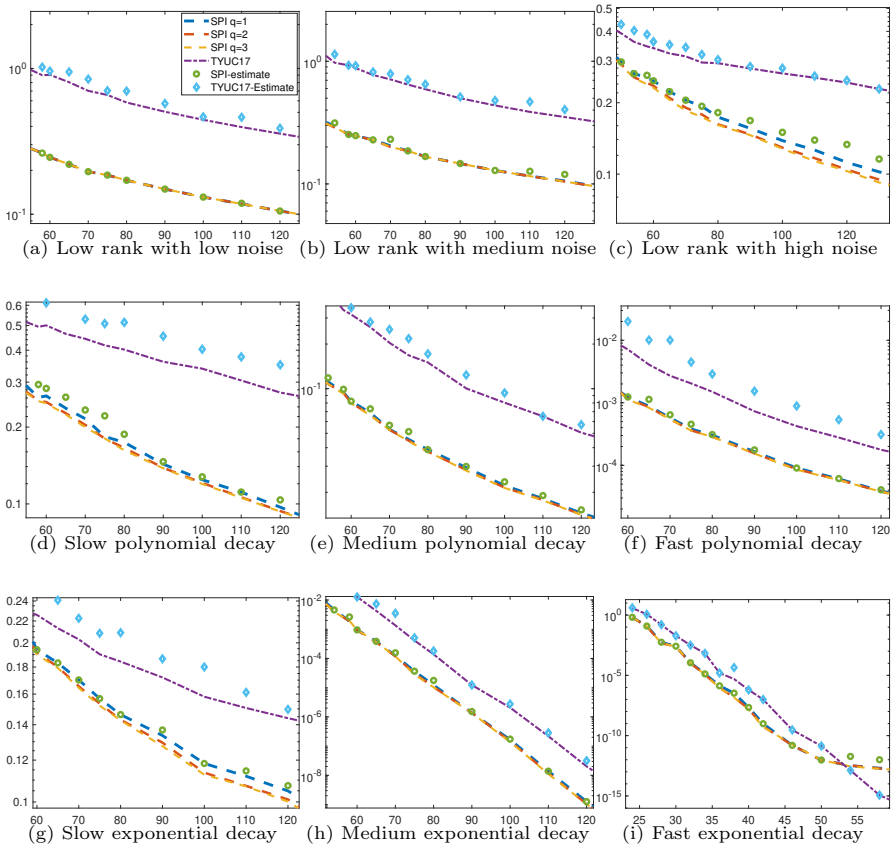
4.2. Numerical results for synthetic data.

4.2.1. Reconstruction error.

EXAMPLE 4.1. *This example compares the performance between TYUC17-SPI (Algorithm 2.1) and its TYUC17 ([35, Alg. 7]) on synectic data with the same storage budget. All oracle errors both in Frobenius norm and spectral norm were computed by averaging* 20 *independently repeated experiments (the errors are represented by lines) in Fig. 1 and 2, respectively. We also plot the relative errors of the algorithms, whose parameters were computed by theoretical guidance (see Sect. B for TYUC17-SPI and [35, Sect. 4.5] for TYUC17) to show that the guided parameters are reliable (the errors are represented by markers). In the figure, x-axis means the storage budget $\hat{T}n$ (parameterized by $\hat{T}$) and y-axis means the relative errors.* Dashed lines *represent oracle errors, while the* markers *indicate errors calculated with parameter guidance.*

Figures 1 and 2 show three points:

- In most cases, the errors of TYUC17-SPI are smaller than those of TYUC17, demonstrating the effectiveness of SPI with the same storage budget. The improvement is more significant when the data matrix has a slow to moderate spectrum decay, such as the low rank matrix plus noise and the polynomial decay spectrum cases. The only exception is that when the spectrum exhibits a fast exponential decay and a larger storage budget is available (in the bottom-right subfigure), TYUC17-SPI is outperformed by TYUC17. This performance drop is due to TYUC17-SPI reaching the precision limit of its mixed-precision strategy, with round-off errors becoming dominant. Nevertheless, its accuracy remains acceptable for most applications, and this exception does not affect the overall trend, which still favors TYUC17-SPI.
- The Frobenius errors (indicated by markers) in Fig. 1 computed by TYUC17-SPI using theoretically guided sketch sizes are nearly identical to the best performance achieved (oracle errors, shown by lines). For spectral errors in Fig. 2, most of the errors are also well aligned with the oracle errors, except for the slow polynomial decay case. Nevertheless, the estimation becomes more accurate when the storage budget is increased. This suggests that



(a) Low rank with low noise (b) Low rank with medium noise (c) Low rank with high noise

(d) Slow polynomial decay (e) Medium polynomial decay (f) Fast polynomial decay

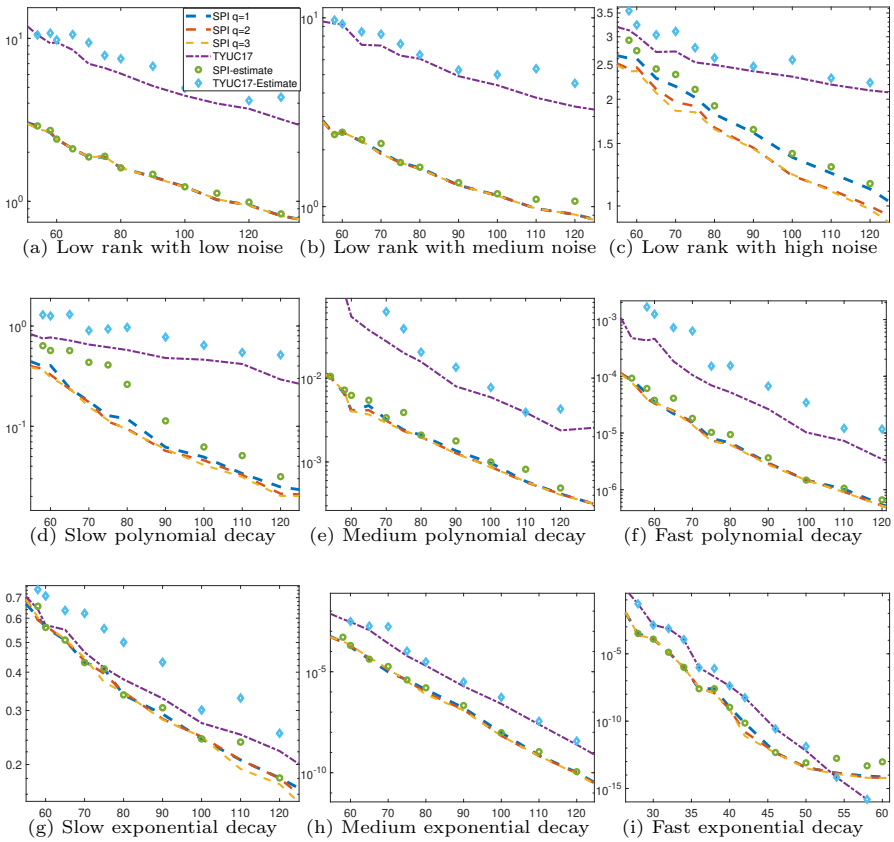
(g) Slow exponential decay (h) Medium exponential decay (i) Fast exponential decay

Fig. 1: Figures of synthetic data. x-axis means the storage budget $\hat{T}n$ (parameterized by $\hat{T}$). y-axis means the relative Frobenius error S_F. All *dashed lines* represent oracle errors, while the *markers* indicate errors calculated with parameter guidance. All results are averaged by 20 independent repeated experiments.

our parameter guidance is reliable. Meanwhile, the errors of TYUC17 using parameter guidance are also close to its oracle errors. This demonstrates that we can set sketch sizes with theoretical guidance a priori to improve the performance of the algorithm without resorting to an exhaustive search.

- We also observe that in several cases, the errors of TYUC17-SPI with $q = 1$ are close to those with more iterations ($q = 2, 3$). However, when the data matrix has a slower decay, (e.g, highly noisy low-rank, slow polynomial or slow exponential; see the subfigures (c), (d), (g) in both Fig. 1 and 2), larger q improves the accuracy, especially in spectral norm. This indicates that a single iteration usually suffices when the spectrum decays rapidly, whereas more iterations are beneficial for slow decay.

  A possible reason why only a few sketch-power iterations suffice is that the sketch $\mathbf{Z} = \mathbf{A}\mathbf{\Phi}$ has a relatively faster decay spectrum, and especially faster than that of the original data matrix, as visualized in Fig. 3. This implies that the energy of the sketched matrix can be effectively captured by a few sketch-power iterations.

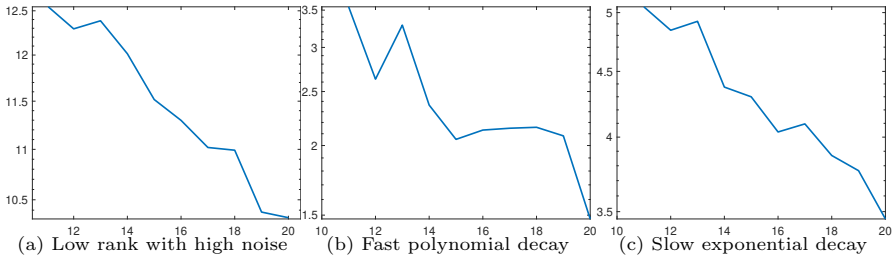


(a) Low rank with low noise (b) Low rank with medium noise (c) Low rank with high noise

(d) Slow polynomial decay (e) Medium polynomial decay (f) Fast polynomial decay

(g) Slow exponential decay (h) Medium exponential decay (i) Fast exponential decay

Fig. 2: Figures of synectic data. y-axis means the relative Spectral error S_∞.

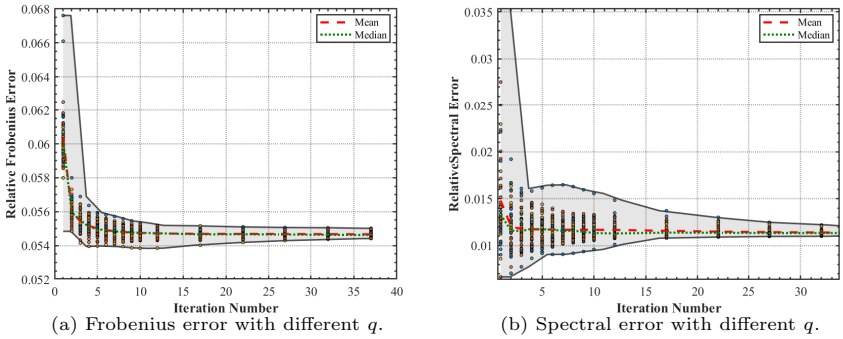


(a) Low rank with high noise (b) Fast polynomial decay (c) Slow exponential decay

Fig. 3: Distortion ratio $\sigma_i(\mathbf{A\Phi})/\sigma_i(\mathbf{A})$ for $r = 10$ and $s = 20$. The x-axis is the singular value index.

4.2.2. Effect of iterations in SPI. For slow decay spectrum, we observed that TYUC17-SPI with more iterations ($q = 2, 3$) has improved accuracy compared to $q = 1$. In this section we further investigate the long-term effect of increasing q, so as to understand the general trend and the potential benefits of additional iterations.

EXAMPLE 4.2. *We test TYUC17-SPI with different q on slow polynomial data in Example 4.1, using* 50 *independent runs for each q with $l = 60$, $s = 20$, $d = 40$. Results are shown in Fig. 4, where the x-axis is q and the y-axis means the error S_F (left) and S_∞ (right).*



(a) Frobenius error with different q. (b) Spectral error with different q.

Fig. 4: Error comparison of different q. The scatter points represent the results of individual experiments and the shaded areas represent the regions between the minimum and maximum errors. The dashed lines represent the average result across all 50 experiments.

Fig. 4 shows the following observations:

- In the average sense, more iterations q can bring better accuracy. Particularly, the spectral error S_∞ has more significant improvement than the Frobenius error S_F as q increases.
- The stability of the algorithm is significantly improved with more iterations q, as the error range (the shaded area) becomes smaller.
- From the density of the scatter points, we can see that most of the results are concentrated in a small range around the mean (the dashed line), especially for larger q and the outliers are rare.
- By computing the coefficient of variation (CV, the ratio of the standard deviation to the mean) of the errors, we have two trends: 1) the CV decreases as q increases, indicating that the results become more stable; 2) the CV of the spectral error is larger than that of the Frobenius error, indicating that the spectral error is more sensitive to randomness.

4.2.3. Wall-clock time comparisons. Comparing with TYUC17, the additional cost of TYUC17-SPI arises primarily from the SPI stage, which involves generating $\mathbf{\Phi}$, constructing $\mathbf{Z} = \mathbf{A\Phi}$, and performing power iterations. When sparse random matrices are employed, the cost of forming $\mathbf{\Phi}$ is negligible and computing $\mathbf{Z}$ scales with the number of nonzero entries in $\mathbf{\Phi}$.

The iteration phase consists of two main components. The first involves matrix multiplications of the form $\mathbf{Z}^T\mathbf{Y}$ and $\mathbf{ZX}$, requiring $O(mls)$ flops. The second involves orthogonalization of $\mathbf{Z}^T\mathbf{Y} \in \mathbb{R}^{l\times s}$, which can be efficiently accomplished using a QR factorization in $O(ls^2)$ operations. Since $l, s \ll \min\{m, n\}$, this cost is negligible. Therefore, the dominant operations within the SPI procedure are the BLAS-3 matrix multiplications, and the incremental cost introduced by TYUC17-SPI remains modest.

Example 4.3. *In this experiment, we compare the wall-clock time of TYUC17-SPI with $q = 1, 2, 3$ against TYUC17, using the synthetic data with medium polynomial decay as defined in Section 4.1.1. The dimensions are set to $m = 20000$ and $n = 30000$, with the parameters fixed at $l = 1000$, $s = 500$, and $d = 1000$. The total wall-clock time, along with that of each step, is reported in Table 2.*

Table 2: Test additional time cost of TYUC17-SPI

Step	Time Complexity	TYUC17	TYUC17-SPI		
			$q=1$	$q=2$	$q=3$
$\mathbf{\Phi}$ $(n \times l)$ generation	$O(nnz)$	-	0.0646	0.0646	0.0646
$\mathbf{\Omega}$ $(n \times s)$ generation	$O(nnz)$	0.0645	0.0640	0.0640	0.0640
$\mathbf{Z} = \mathbf{A\Phi}$	$O(m \times nnz(\mathbf{\Phi}))$	-	0.1591	0.1591	0.1591
$\mathbf{Y} = \mathbf{A\Omega}$	$O(m \times nnz(\mathbf{\Omega}))$	0.1373	0.1370	0.1370	0.1370
$\mathbf{Y}_2 = \mathbf{Z}^T\mathbf{Y}$ $(or\ \mathbf{Y}_3)$	$O(mls)$	-	0.0497	0.1116	0.1528
$[\mathbf{X}, \sim] = \texttt{QR}(\mathbf{Y}_2, 0)$	$O(ls^2)$	-	0.0042	0.0112	0.0150
$\mathbf{Y}_3 = \mathbf{ZX}$	$O(mls)$	-	0.0652	0.1332	0.1998
QR on $\mathbf{Y}$ or $\mathbf{Y}_3$	$O(ms^2)$	0.1146	0.1190	0.1190	0.1190
$\mathbf{\Psi}$ $(d \times m)$ generation	$O(nnz)$	0.0431	0.0429	0.0429	0.0429
$\mathbf{W} = \mathbf{\Psi A}$	$O(n \times nnz(\mathbf{\Psi}))$	0.1626	0.1613	0.1613	0.1613
Solve $\mathbf{B} = (\mathbf{\Psi Q})\backslash\mathbf{W}$	$O(nls)$	0.2419	0.2419	0.2419	0.2419
$[\mathbf{U}, \mathbf{\Sigma}, \mathbf{V}] = \texttt{trun_svd}(\mathbf{B}, r)$	$O(ns^2)$	0.2672	0.2705	0.2705	0.2705
$\mathbf{U} \leftarrow \mathbf{QU}$	$O(msr)$	0.0260	0.0256	0.0256	0.0256
TOTAL	-	1.0571	1.4049	1.5415	1.6625

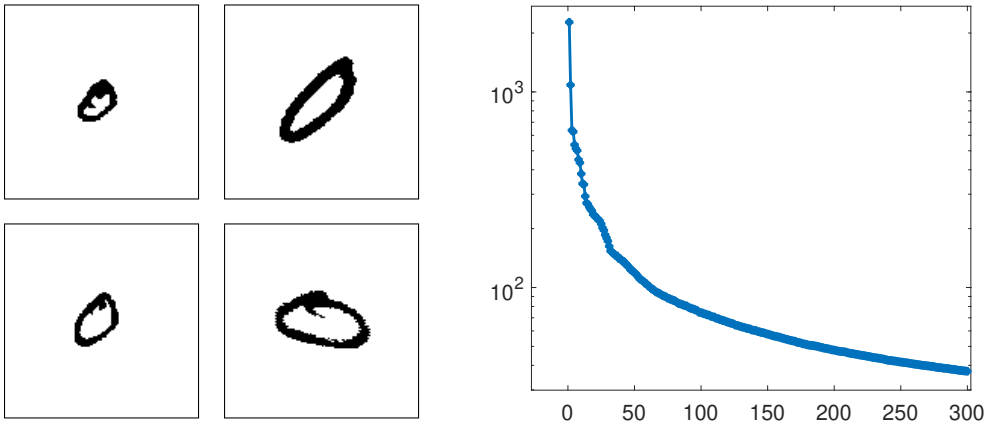


Fig. 5: Left images are four samples of the digit "0" from different writers in NIST dataset. Right figure shows the singular values of the matrix generated in Sect. 4.1.2.

Table 2 shows the wall-clock time of TYUC17 and TYUC17-SPI with $q = 1, 2, 3$. We can see that the additional cost is from 32% when $q = 1$ to 56% when $q = 3$ compared to TYUC17, which mainly comes from constructing $\mathbf{Z}$. The cost of one iteration is smaller compared to the cost of constructing $\mathbf{Z}$. Particularly, orthogonalizing $\mathbf{Z}^T\mathbf{Y}$ is negligible. In summary, the computational overhead of TYUC17-SPI is small, which is consistent with our previous analysis.

4.3. Numerical results for real data. This section evaluates different algorithms on NIST and climate datasets, which are commonly used for machine leraning and scientific computing, respectively. The former exhibits a relatively smooth polynomial spectral decay, whereas for the latter, the leading five singular values drop by two orders of magnitude, with the rest decaying at a slow exponential rate. The visualization of these datasets and their singular values is shown in Fig. 5 and 6.

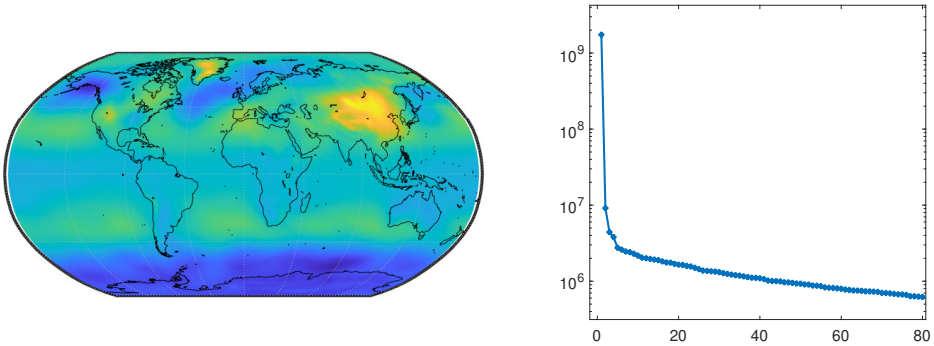


Fig. 6: Left figure visualizes the original data for a single day. Right figure shows the singular values.

4.3.1. Metrics. As discussed at the end of Sect. 3.1 and 3.2, the key difference between TYUC17 and TYUC17-SPI lies in the orthogonal projection. Therefore, in addition to the relative error (4.1), we further consider the following two errors that will be used in the remainder of this subsection:

1) **RangeError**: $\|\mathbf{A} - \mathbf{U}\mathbf{U}^T\mathbf{A}\|_p / \|\mathbf{A} - \lfloor \mathbf{A} \rfloor_r\|_p - 1$;

2) **ExtraError**: $\|\mathbf{U}\tilde{\mathbf{U}}^T(\mathbf{Q}^T\mathbf{A} - (\mathbf{\Psi}\mathbf{Q})^\dagger \mathbf{\Psi}\mathbf{A})\|_p / \|\mathbf{A} - \lfloor \mathbf{A} \rfloor_r\|_p$;

"RangeError" refers to orthogonal projection error (with $\mathbf{U}$ in place of $\mathbf{Q}$, consistent with $\|\mathbf{A} - \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T\|_p$). In "ExtraError", the subtraction of one is omitted as it may be less than one. The term in its numerator comes because the error of TYUC17 and TYUC17-SPI have two error sources:

$$\|\mathbf{A} - \hat{\mathbf{A}}\|_p = \left\|\mathbf{A} - \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T\right\|_p = \|(\mathbf{A} - \mathbf{U}\mathbf{U}^T\mathbf{A}) + (\mathbf{U}\tilde{\mathbf{U}}^T(\mathbf{Q}^T\mathbf{A} - (\mathbf{\Psi}\mathbf{Q})^\dagger \mathbf{\Psi}\mathbf{A}))\|_p,$$

where the second term is caused by the sketch-and-solve stage (the decomposition is in the same spirit to that in [35, Lemma A.3]; see Sect. SM6 in the supplementary material for a proof). Note that it is perpendicular to the first term.

4.3.2. A RSVD-Onepass implementation under row-wise format. Before the experiment, we include another algorithm: a one-pass implementation of the original RSVD [42, 2] (RSVD-Onepass for short). First, it computes two sketches:

$$\mathbf{Y} = \mathbf{A}\mathbf{\Omega}, \mathbf{W} = \mathbf{A}^T\mathbf{A}\mathbf{\Omega};$$

here $\mathbf{A}$ must be in row-wise format such that $\mathbf{W}$ can be constructed in a single pass of $\mathbf{A}$. Then, it computes the thin QR of $\mathbf{Y}$ and computes $\mathbf{B}$ as follows:

$$[\mathbf{Q}, \mathbf{R}] = \mathtt{qr}(\mathbf{Y}, 0), \mathbf{B} = \mathbf{R}^{-T}\mathbf{W}^T.$$

Finally, the algorithm computes the truncated SVD of $\mathbf{B} \approx \tilde{\mathbf{U}}\mathbf{\Sigma}\mathbf{V}^T$ and the left approximate singular vectors $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$, and the approximation is $\hat{\mathbf{A}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$. It is not hard to see that RSVD-Onepass is algebraically equivalent to RSVD of [13]. The algorithm requires two sketches $\mathbf{Y}, \mathbf{W}$, taking storage $ms + ns$.

A key difference is that RSVD-Onepass is a orthogonal projection method, while TYUC17 and TYUC17-SPI are oblique projection methods. Thus RSVD-Onepass has

only one error source:

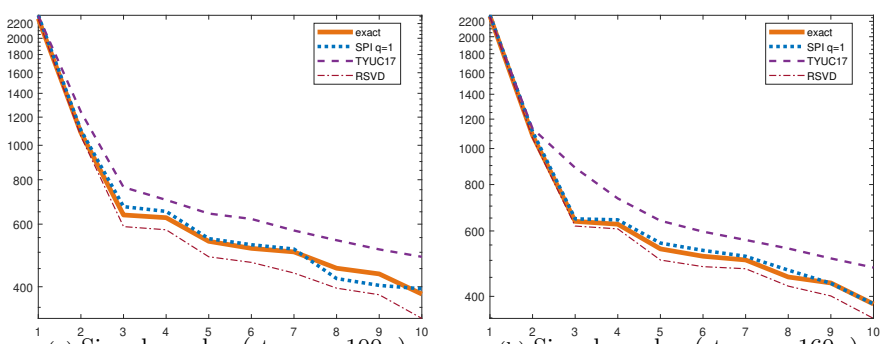
$$\|\mathbf{A} - \hat{\mathbf{A}}\|_p = \|\mathbf{A} - \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T\|_p = \|\mathbf{A} - \mathbf{U}\mathbf{U}^T\mathbf{A}\|_p,$$

because $\mathbf{U}^T\mathbf{A} = \tilde{\mathbf{U}}^T\mathbf{Q}^T\mathbf{A} = \tilde{\mathbf{U}}^T\mathbf{B} = \mathbf{\Sigma}\mathbf{V}^T$ due to $\mathbf{B} = \mathbf{R}^{-T}\mathbf{W}^T = \mathbf{Q}^T\mathbf{A}$.

As RSVD-Onepass only supports row-wise streaming data, it is not directly comparable to TYUC17 or TYUC17-SPI that allow general linear updates of the data matrix. Nevertheless, we still report its results for two reasons. First, as discussed, RSVD only involves the orthogonal projection error, making it a strong baseline under storage-limited settings. Second, the comparison helps illustrate how SPI affects the two error sources. On the other hand, it is also conceivable that RSVD-Onepass itself could be enhanced by SPI, which we leave for future investigation.

4.3.3. NIST dataset performance. The main task in the NIST dataset is to extract the principal components of the data matrix, i.e., its left singular vectors in the context. Hence, we mainly focus on approximating the dominant 10 singular values and visualizing the approximated left singular vectors. Reconstruction errors are also reported for completeness. In the sequel, TYUC17-SPI and TYUC17 employ their theoretically guided sketch sizes, while RSVD-Onepass uses an exhaustive search to find the best parameter s.

EXAMPLE 4.4. *This example shows the feature extraction quality of the tested algorithms in NIST dataset. We used the* 16384×20000 *matrix generated in Sect. 4.1.2. To test the performance, we mainly focusing on the dominant singular values and left singular vectors. All results are averaged over 10 independent repeated experiments. We set the same storage budget* $100n$ *and* $160n$ *for all the algorithms.*

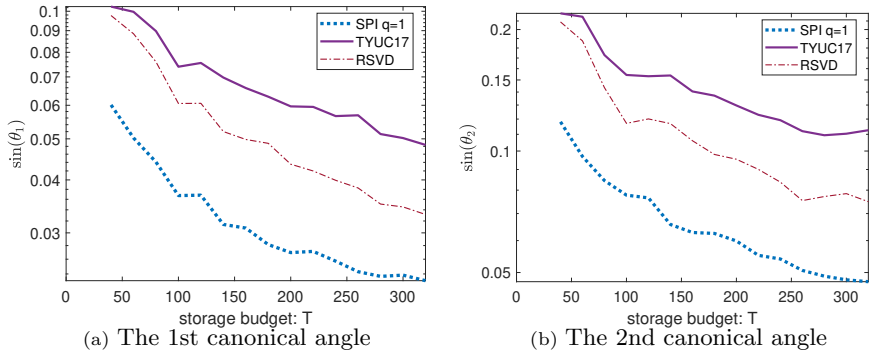


(a) Singular value (storage: $100n$) (b) Singular value (storage: $160n$)

Fig. 7: Singular value estimation. The x-axis means the singular value index and the y-axis shows the magnitude of the singular values. TYUC17-SPI uses the parameter guided for polynomial decay and TYUC17 uses the parameters suggested in [35, Sect. 4.5.2]. The exact singular values are computed by Matlab's `svds` function.

Fig. 7 illustrates the accuracy of approximating the first ten singular values under two storage budgets: $100n$ and $160n$. The x-axis corresponds to the singular value index, while the y-axis shows the singular values. The red curve represents the true singular values. We observe that TYUC17-SPI ($q = 1$) closely follows the exact singular values, indicating its high accuracy. By contrast, RSVD-Onepass tends to underestimate the singular values, and TYUC17 overestimates them. When the

storage budget increases from $100n$ to $160n$, all methods improve their accuracy, but TYUC17-SPI remains the closest to the exact values.



(a) The 1st canonical angle (b) The 2nd canonical angle

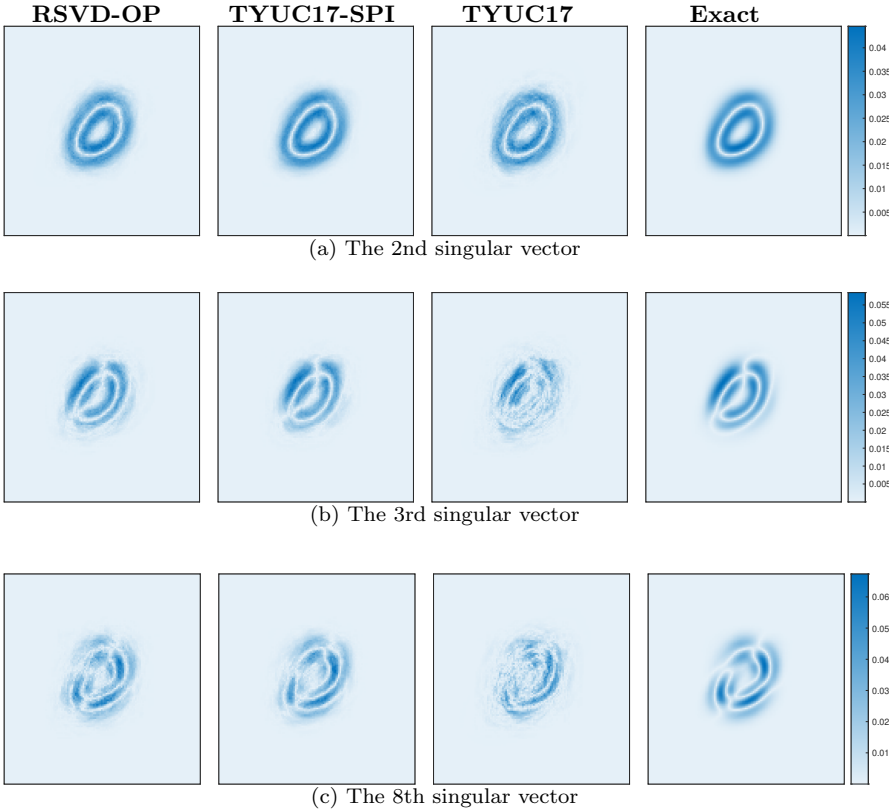
Fig. 8: Sines of the first two canonical angles between the true and computed left dominant singular subspaces, with increasing storage budget $\hat{T}n$ (x-axis: $\hat{T}$).

Fig. 8 displays the first two canonical angles between true and computed left dominant singular subspaces by TYUC17-SPI, TYUC17 and RSVD-Onepass algorithms, respectively, with increasing storage budget $\hat{T}n$. x-axis represents $\hat{T}$, while the y-axis shows the sines of the canonical angles. We observe that TYUC17-SPI consistently achieves the highest accuracy among the three. RSVD-Onepass perform better than TYUC17, because its storage budget $\hat{T}n = (m+n)s$ leads to a larger effective sketch size than TYUC17's $ms + dn$ (since $d > s$, $\mathbf{Y}$ of the former can be larger). TYUC-SPI outperforms both by leveraging storage for $\mathbf{Z}$ via the mixed-precision strategy, resulting in the strongest $\mathbf{Y}$ and overall best performance.

Fig. 9 visualizes the 2nd, 3rd, and 8th singular vectors produced by the algorithms with storage $320n$. Each vector is reshaped into a 128×128 matrix. We observe that TYUC17-SPI (second column) yields the most accurate approximations, closely matching the exact vectors (fourth column). In contrast, TYUC17 (third column) shows noticeable deviations starting from the 3rd singular vector. This illustrates SPI's effectiveness in improving the accuracy of singular vector approximations. While RSVD-Onepass (first column) also produces relatively high-quality results, the TYUC17-SPI solutions are sharper and exhibit less noise, demonstrating the accuracy using SPI.

Table 3 summarizes the errors of the algorithms under three storage budgets: $200n$, $260n$, and $320n$. The table includes both Frobenius and spectral norm errors. We observe that TYUC17-SPI consistently outperforms TYUC17 across all metrics, demonstrating the effectiveness of the SPI strategy. Specifically, for the RangeError, TYUC17-SPI shows over $5\times$ improvement for the Frobenius norm and $10\times$ improvement for the spectral norm compared to TYUC17. Furthermore, for the ExtraError caused by the sketch-and-solve stage, SPI also has considerable improvements.

We also see that the RangeError of TYUC17-SPI is lower than that of RSVD (whose RangError=Error according to Sect. 4.3.2), particularly under the Frobenius norm. However, RSVD performs best on the Error metric, as its orthogonal projection avoids the corange $\mathbf{V}$ bias inherent in oblique methods. This does not contradict Fig. 7, 8, and 9: while these figures only reflect singular value and range $\mathbf{U}$ estimation, the Error metric here is additionally influenced by the corange $\mathbf{V}$. This suggests that SPI



(a) The 2nd singular vector

(b) The 3rd singular vector

(c) The 8th singular vector

Fig. 9: Three singular vectors computed by the algorithms. RSVD-OP means RSVD-Onepass.

Table 3: Reconstruction errors of different algorithms on NIST. "Error" refers to the relative error (4.1). RSVD-Onepass's Error=RangeError and it has no ExtraError.

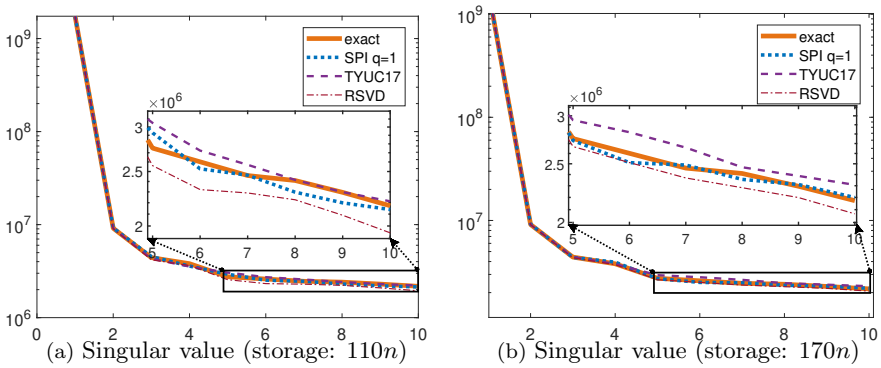
Method	Type	Frobenius Error			Spectral Error		
		200	260	320	200	260	320
TYUC17 -SPI	Error	0.0423	0.0335	0.0221	7.648e-3	4.177e-3	1.742e-3
	RangeError	0.0178	0.0123	0.0098	5.099e-3	3.062e-3	1.309e-3
	ExtraError	0.2244	0.2082	0.1586	0.5436	0.4268	0.3703
TYUC17	Error	0.1386	0.1846	0.0773	1.562e-1	5.195e-2	3.155e-2
	RangeError	0.0958	0.0748	0.0545	1.277e-1	2.878e-2	2.379e-2
	ExtraError	0.3093	0.2450	0.2205	0.7085	0.6708	0.4612
RSVD-OP	Error	0.0326	0.0222	0.0165	5.133e-3	3.447e-3	1.371e-3

may be applied to also enhance the corange estimation in the future study.

4.3.4. Climate dataset performance.

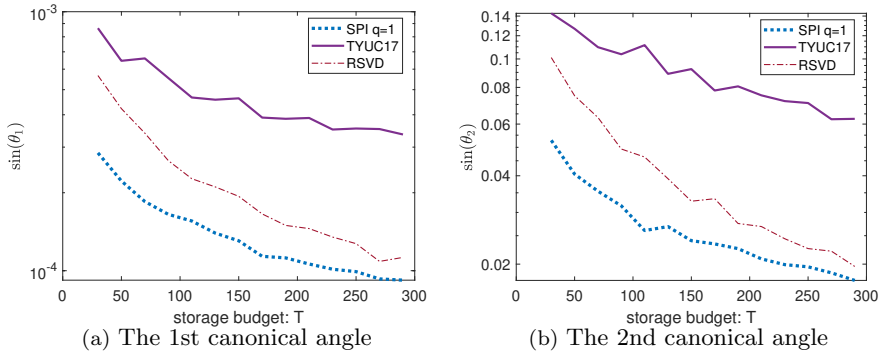
EXAMPLE 4.5. *We tested the algorithms in scientific computation, using the* 10512×28144 *climate data introduced in Sect. 4.1.2. Feature vectors derived from*

climate data are paramountly important in both scientific research and practical applications. These vectors encapsulate critical information extracted from the datasets. For the pressure data of the 73×144 *meshgrid on the surface, the left singular vectors reveal the spatial structure of the data, for example, the distribution of pressure belts. We test rank-*10 *approximation and all results are averaged over* 10 *independent trials.*



(a) Singular value (storage: $110n$) (b) Singular value (storage: $170n$)

Fig. 10: Singular value estimation in climate dataset.



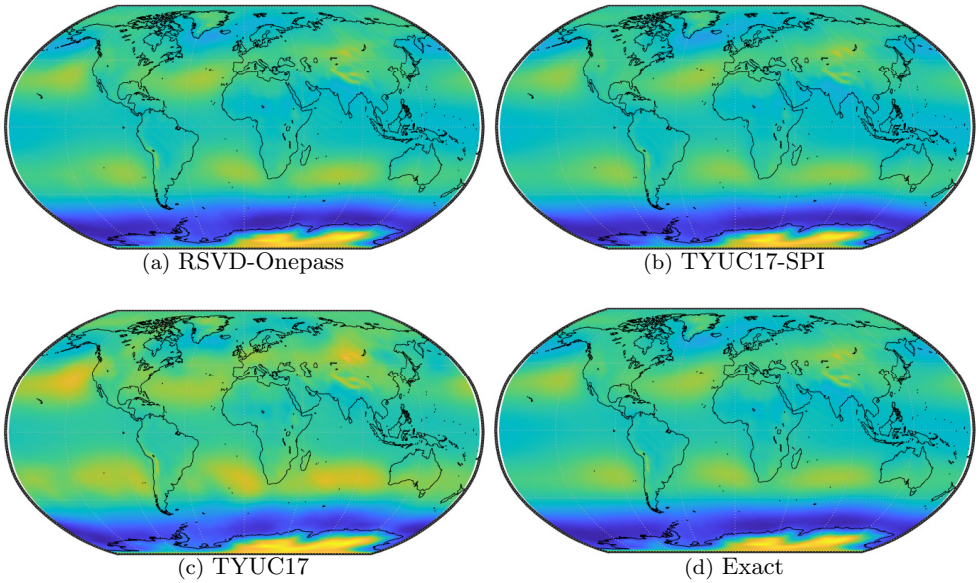
(a) The 1st canonical angle (b) The 2nd canonical angle

Fig. 11: Sines of the first two canonical angles between the true and computed left dominant singular subspace, with increasing storage budget $\hat{T}n$ (x-axis: $\hat{T}$).

The singular value estimation and canonical angles are respectively plotted in Fig. 10 and 11. The behavior on the climate dataset is similar to that on NIST, but all algorithms perform better owing to the faster spectral decay of the climate matrix (c.f. Fig. 6). In particular, Fig. 11 shows that the curves of RSVD-Onepass are closer to those of TYUC17-SPI, as the rapid spectral decay enables RSVD to perform well.

Fig. 12 and 13 visualize the features extracted from the climate dataset. They reveal some long-term distinct climatological patterns and can be explained respectively. We set storage $60n$. The leading singular vector in Fig. 12 captures persistent pressure belt structures with remarkable fidelity. While both SPI and RSVD-Onepass generate physically consistent approximations, the TYUC17 solution exhibits notable deviations.

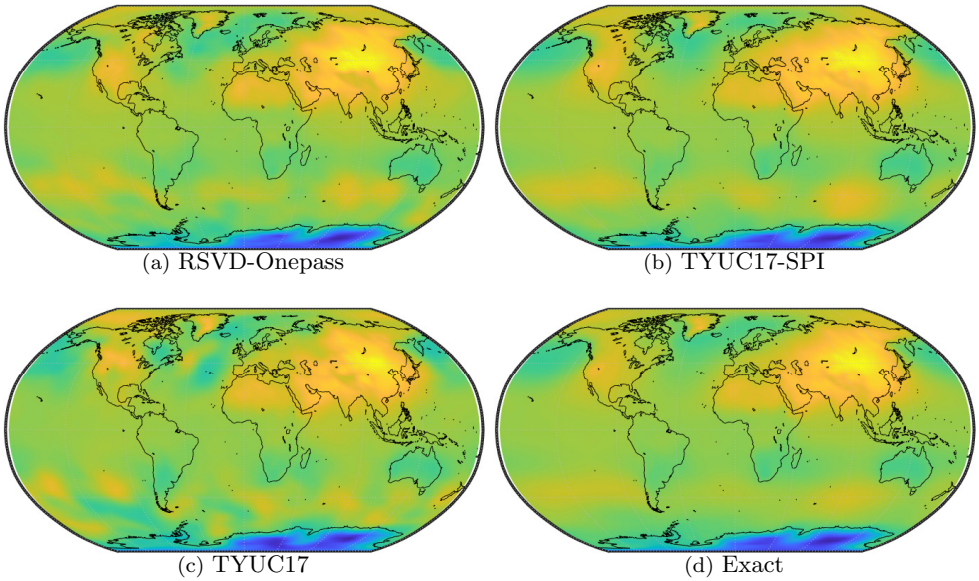
Fig. 13 further illustrates the second singular vector's sensitivity to Eurasian landmass dynamics and Southern Hemisphere westerlies. The SPI reconstruction



(a) RSVD-Onepass (b) TYUC17-SPI

(c) TYUC17 (d) Exact

Fig. 12: The 1st singular vector of climate data. All heat maps use the same color bar (omitted here for brevity).



(a) RSVD-Onepass (b) TYUC17-SPI

(c) TYUC17 (d) Exact

Fig. 13: The 2nd singular vector of climate data.

closely matches the reference solution, particularly in resolving the pressure gradient over prevailing westerlies. By contrast, TYUC17 and RSVD-Onepass show systematic errors south of 45°S latitude. This differential performance suggests that TYUC17 enhanced by SPI effectively preserves critical synoptic-scale features that are attenuated by prior methods.

Table 4: Reconstruction errors of different algorithms on climate dataset. "Error" refers to the relative error (4.1). RSVD-Onepass's Error=RangeError.

Method	Type	Frobenius Error			Spectral Error		
		110	140	170	110	140	170
TYUC17 -SPI	Error	0.0281	0.0175	0.0099	2.316e-3	1.282e-3	6.402e-4
	RangeError	0.0079	0.0043	0.0031	1.445e-3	9.454e-4	3.219e-4
	ExtraError	0.2023	0.1635	0.1170	0.5928	0.4072	0.2961
TYUC17	Error	0.0933	0.0659	0.0452	1.825e-1	4.766e-2	1.333e-2
	RangeError	0.0590	0.0440	0.0298	1.124e-1	2.887e-2	8.560e-3
	ExtraError	0.2715	0.2148	0.1789	0.7737	0.6137	0.5224
RSVD-OP	Error	0.0126	0.0078	0.0050	2.511e-3	1.190e-3	4.467e-4

Table 4 reports the errors on the climate dataset. Overall errors are smaller than those on NIST, owing to the faster spectral decay of the climate data, such that algorithms can obtain lower errors with fewer storage budgets (compared with NIST). Meanwhile, the performance is consistent with that in Table 3: TYUC17-SPI again achieves the smallest RangeError, while RSVD performs best on the Error metric.

4.3.5. Performance with different q. We conduct quantitative experiments to examine multiple SPI iterations, as follows.

EXAMPLE 4.6. *We test the performance of TYUC17-SPI with $q = 1, 2, 3$ on both datasets and set the storage $190n$. All results are averaged over 10 independent trials.*

Table 5: Performance of TYUC-SPI with $q = 1, 2, 3$ on both real datasets.

Data	Type	Frobenius Error			Spectral Error		
		1	2	3	1	2	3
NIST	Error	0.0496	0.0489	0.0487	0.0159	0.0138	0.0124
	RangeError	0.0222	0.0217	0.0215	0.0093	0.0080	0.0077
	ExtraError	0.2383	0.2373	0.2373	0.5771	0.5691	0.5652
Climate	Error	0.0087	0.0084	0.0083	2.124e-4	2.006e-4	1.965e-4
	RangeError	2.661e-3	2.626e-3	2.615e-3	1.550e-4	1.463e-4	1.379e-4
	ExtraError	0.1102	0.1078	0.1069	0.4315	0.4312	0.4310

Table 5 reports the performance of TYUC17-SPI with $q = 1, 2, 3$ on the NIST and climate datasets. Errors generally decrease as q increases, especially for the spectral norm, similar to Fig. 4, confirming the effectiveness of multiple SPI iterations.

5. Analysis for the $q = 1$ Bound. This section is focused on proving Theorem 3.1. The key is to establish the projection error $\|(\mathbf{I} - P_{\hat{\mathbf{Y}}})\mathbf{A}\|_F$. The difficulty lies in the interaction between $\mathbf{A}$ and $\mathbf{Z}$. Roughly speaking, in the standard power iteration $\mathbf{Y} = (\mathbf{A}\mathbf{A}^T)^q\mathbf{A}\boldsymbol{\Omega}$, one can analyze $\boldsymbol{\Sigma}^{(2q+1)}\mathbf{V}^T\boldsymbol{\Omega}$, with $\mathbf{A} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T$ its SVD, so that the analysis of RSVD applies similarly. However, SPI $\hat{\mathbf{Y}} = \mathbf{Z}\mathbf{Z}^T\mathbf{A}\boldsymbol{\Omega}$ does not have such a representation. We address this using similar ideas as [13], but with some more

involved decompositions. Fortunately, when $q = 1$ these decompositions still work. In the following, Sect. 5.1 introduces necessary lemmas, Sect. 5.2 provides deterministic projection error, while Sect. 5.3 and 5.4 establish the averaged error bounds.

5.1. Technical lemmas. Some useful lemmas are recalled first.

LEMMA 5.1 (Parallel sums [34]). *For positive semidefinite (PSD) matrices* $\mathbf{A}, \mathbf{B}$ *with the same dimension, define the parallel sum* $\mathbf{A} : \mathbf{B}$ *to be the PSD matrix*

$$\mathbf{A} : \mathbf{B} = \mathbf{A} - \mathbf{A}(\mathbf{A} + \mathbf{B})^{\dagger}\mathbf{A},$$

where † *denotes the Moore–Penrose inverse. The following hold:*

1. *The parallel sum is symmetric; that is,* $\mathbf{A} : \mathbf{B} = \mathbf{B} : \mathbf{A}$.
2. *If* $\mathbf{A}$ *and* $\mathbf{B}$ *are strictly positive definite, then* $\mathbf{A} : \mathbf{B} = (\mathbf{A}^{-1} + \mathbf{B}^{-1})^{-1}$.
3. *The mapping* $\mathbf{B} \mapsto \mathbf{A} : \mathbf{B}$ *is monotone and concave with respect to the PSD ordering, and it is bounded above as* $\mathbf{A} : \mathbf{B} \preceq \mathbf{A}$.
4. $\|\mathbf{A} : \mathbf{B}\|_p \leq \|\mathbf{A}\|_p : \|\mathbf{B}\|_p$ *for any matrix Schatten p-norm with* $1 \leq p \leq \infty$.

Here $\|\mathbf{A}\|_p = (\sum_{i=1} \sigma_i(\mathbf{A})^p)^{1/p}$. In particular $\|\mathbf{A}\|_2 = \|\mathbf{A}\|_F$ and $\|\mathbf{A}\|_\infty = \|\mathbf{A}\|$.

LEMMA 5.2 (Moment bounds [34]). *Consider fixed matrices* $\mathbf{S} \in \mathbb{R}^{l\times n}$ *and* $\mathbf{T} \in \mathbb{R}^{p\times q}$ *and a standard Gaussian matrix* $\mathbf{G} \in \mathbb{R}^{n\times p}$. *Then*

$$\mathbb{E}\|\mathbf{SGT}\|_F^2 = \|\mathbf{S}\|_F^2 \|\mathbf{T}\|_F^2, \ \mathbb{E}\|\mathbf{SGT}\|^2 \leq (\|\mathbf{S}\|_F \|\mathbf{T}\| + \|\mathbf{S}\| \|\mathbf{T}\|_F)^2.$$

LEMMA 5.3 (Inverse moment bounds, Frobenius norm [34, 13]). *Let* $\mathbf{G} \in \mathbb{R}^{r\times k}$ *with* $r \leq k$ *be standard Gaussian. Then*

$$\mathbb{E}(\mathbf{GG}^T)^{-1} = \frac{1}{k-r-1}\mathbf{I}_r, \qquad r \leq k-2,$$

$$\mathbb{E}\|(\mathbf{GG}^T)^{-1}\|_F^2 = \frac{r(k-1)}{(k-r)(k-r-1)(k-r-3)}, \qquad r \leq k-4.$$

For any $t > 1$ *and* $r \leq k-4$,

$$\mathbb{P}\left\{\|\mathbf{G}^{\dagger}\|_F > t \cdot \sqrt{\frac{3r}{k-r+1}}\right\} \leq t^{-(k-r)}.$$

LEMMA 5.4 (Inverse moment bounds, spectral norm [34, 13]). *For* $0 < p \leq 18$ *and* $p \leq \frac{k-r}{2}$, *the standard Gaussian matrix* $\mathbf{G} \in \mathbb{R}^{r\times k}$ *satisfies*

$$\left(\mathbb{E}\|(\mathbf{GG}^T)^{-1}\|^p\right)^{1/p} \leq \frac{e^2(k+r)}{2(k-r)^2}, \ \mathbb{P}\left(\|\mathbf{G}^{\dagger}\| \geq \frac{e\sqrt{k}}{k-r+1} \cdot t\right) \leq t^{-(k-r+1)}, \quad (t > 1).$$

COROLLARY 5.5. *Let* $\mathbf{A} \in \mathbb{R}^{m\times n}$ *be fixed, and* $\mathbf{G}_1 \in \mathbb{R}^{s\times k}$ $(k \geq s+2), \mathbf{G}_2 \in \mathbb{R}^{n\times k}$ *be independent standard Gaussian matrices. Then*

$$\mathbb{E}\|\mathbf{AG}_2\mathbf{G}_1^{\dagger}\|_F^2 = \frac{s\|\mathbf{A}\|_F^2}{k-s-1}; \ \mathbb{E}\|\mathbf{AG}_2\mathbf{G}_1^{\dagger}\|^2 \leq \left(\|\mathbf{A}\|\sqrt{\frac{s}{k-s-1}} + \|\mathbf{A}\|_F \frac{e\sqrt{k+s}}{\sqrt{2}(k-s)}\right)^2.$$

LEMMA 5.6 ([38]). *Let* h *be a Lipschitz function on matrices:* $|h(\mathbf{X}) - h(\mathbf{Y})| \leq L\|\mathbf{X} - \mathbf{Y}\|_F$, $\forall \mathbf{X}, \mathbf{Y}$. *Draw a standard Gaussian matrix* $\mathbf{G}$. *Then for* $u > 0$,

$$\mathbb{P}\left(h(\mathbf{G}) - \mathbb{E}h(\mathbf{G}) > uL\right) \leq e^{-u^2/2}.$$

Corollary 5.7. *For standard Gaussian matrix* $\mathbf{G}$ *and two random matrices* $\mathbf{S}$ *and* $\mathbf{T}$ *independent from* $\mathbf{G}$*, for* $u > 0$ *we have:*

$$\mathbb{P}\left(\|\mathbf{SGT}\| > e_1e_4 + e_2e_3 + e_2e_4u\right) < e^{-u^2/2} + \mathbb{P}\left(E^C\right), \tag{5.1}$$

where $E = \{\mathbf{S}, \mathbf{T} \,|\, \|\mathbf{S}\|_F \le e_1, \|\mathbf{S}\| \le e_2, \|\mathbf{T}\|_F \le e_3, \|\mathbf{T}\| \le e_4\}$ *and* $e_i > 0,\ i = 1, \ldots, 4.$

The proofs of Corollaries 5.5 and 5.5 are given in Appendix A. Using the above results, the following bounds can be derived. They have appeared in the proofs of [13, Thm. 10.7 and Thm. 10.8].

Corollary 5.8. *Under the setting of Corollary 5.5, for any* $t > 1$*,* $u > 0$*,* $k - s \ge 4$*, we have*

$$\mathbb{P}\left(\|\mathbf{AG}_2\mathbf{G}_1^\dagger\| > (\|\mathbf{A}\|_F + u\|\mathbf{A}\|)\frac{e\sqrt{k}}{k-s+1}t + \|\mathbf{A}\|\sqrt{\frac{3s}{k-s+1}}t\right) < 2t^{-(k-s)} + e^{-u^2/2};$$

$$\mathbb{P}\left(\left\|\mathbf{AG}_2\mathbf{G}_1^\dagger\right\|_F > \|\mathbf{A}\|_F\sqrt{\frac{3s}{k-s+1}}t + \|\mathbf{A}\|\frac{e\sqrt{k}}{k-s+1}tu\right) < 2t^{-(k-s)} + e^{-u^2/2}.$$

5.2. Deterministic projection error. For $\mathbf{A} \in \mathbb{R}^{m\times n}$, without loss of generality assume that $m \ge n$. For any $\varrho \ge r$ with r the target rank, partition $\mathbf{A}$ as

$$\mathbf{A} = \mathbf{U\Sigma V}^T = \mathbf{U}\begin{array}{c} \begin{array}{cc} \varrho & n-\varrho \end{array} \\ \begin{bmatrix} \mathbf{\Sigma}_1 & \\ & \mathbf{\Sigma}_2 \end{bmatrix} \end{array} \begin{array}{c} \begin{array}{c} n \end{array} \\ \begin{bmatrix} \mathbf{V}_1^T \\ \mathbf{V}_2^T \end{bmatrix} \end{array} \begin{array}{c} \varrho \\ n-\varrho \end{array}, \tag{5.2}$$

where $\mathbf{U} \in \mathbb{R}^{m\times n}$, $\mathbf{V} \in \mathbb{R}^{n\times n}$ are orthonormal, $\mathbf{\Sigma} \in \mathbb{R}^{n\times n}$, $\mathbf{\Sigma}_1 \in \mathbb{R}^{\varrho\times\varrho}$, $\mathbf{\Sigma}_2 \in \mathbb{R}^{(n-\varrho)\times(n-\varrho)}$ are diagonal matrices, and $\mathbf{V}_1 \in \mathbb{R}^{n\times\varrho}, \mathbf{V}_2 \in \mathbb{R}^{n\times(n-\varrho)}$. The singular values in $\mathbf{\Sigma}$ are arranged in a descending order.

Let $\mathbf{\Omega} \in \mathbb{R}^{n\times s}, \mathbf{\Phi} \in \mathbb{R}^{n\times l}$ be two test matrices with $l > s > \varrho$. We decompose $\mathbf{\Omega}$ and $\mathbf{\Phi}$ determined by the partition of $\mathbf{V}$ as follows:

$$\begin{aligned} &\mathbf{\Omega}_1 = \mathbf{V}_1^T\mathbf{\Omega} \in \mathbb{R}^{\varrho\times s},\ \mathbf{\Omega}_2 = \mathbf{V}_2^T\mathbf{\Omega} \in \mathbb{R}^{(n-\varrho)\times s}, \\ &\mathbf{\Phi}_1 = \mathbf{V}_1^T\mathbf{\Phi} \in \mathbb{R}^{\varrho\times l},\ \mathbf{\Phi}_2 = \mathbf{V}_2^T\mathbf{\Phi} \in \mathbb{R}^{(n-\varrho)\times l}. \end{aligned} \tag{5.3}$$

In addition, we introduce the auxiliary notation

$$\mathbf{H} = \mathbf{\Sigma}_2^2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\mathbf{\Sigma}_1^{-2}. \tag{5.4}$$

Let $\mathbf{Y} = \mathbf{A\Omega} \in \mathbb{R}^{m\times s}$, $\mathbf{Z} = \mathbf{A\Phi} \in \mathbb{R}^{m\times l}$, and $\hat{\mathbf{Y}} = \mathbf{ZZ}^T\mathbf{Y} \in \mathbb{R}^{m\times s}$. The projection error with respect to $\mathbf{I} - P_{\hat{\mathbf{Y}}} = \mathbf{I} - \hat{\mathbf{Y}}\hat{\mathbf{Y}}^\dagger = \mathbf{I} - \hat{\mathbf{Y}}(\hat{\mathbf{Y}}^T\hat{\mathbf{Y}})^\dagger\hat{\mathbf{Y}}^T$ is given as follows.

Theorem 5.9. *Let* $\mathbf{A}$ *be partitioned as in* (5.2) *and recall the notations introduced in* (5.3) *and* (5.4)*. Assume that* $\mathbf{\Omega}_1, \mathbf{\Phi}_1$ *have full row rank. For* $\hat{\mathbf{Y}} = \mathbf{ZZ}^T\mathbf{Y} = \mathbf{A\Phi}(\mathbf{A\Phi})^T\mathbf{A\Omega}$*, we have the following estimation:*

$$\left\|\mathbf{A} - \hat{\mathbf{Y}}\hat{\mathbf{Y}}^\dagger\mathbf{A}\right\|_p^2 \le \|\mathbf{\Sigma}_2\|_p^2 + \left\|\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_1^\dagger + \mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\right\|_p^2 \tag{5.5}$$

$$\cdot\left\|\left(\mathbf{I} + \mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\right)^{-1}\right\|^2 \tag{5.6}$$

by assuming the invertibility of $\mathbf{I} + \mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}$.

When $\mathbf{\Omega}$ and $\mathbf{\Phi}$ are standard Gaussian, $\mathbf{\Omega}_1, \mathbf{\Phi}_1$ are also standard Gaussian due to the rotational invariance, which together with their sizes imply that they have full row rank with probability 1. Proposition 5.10 shows that $\mathbf{I} + \mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}$ is invertible with probability 1 in the same situation.

Proof. The proof can be divided into four steps. The first step is to consider a trivial case. The second is to remove $\mathbf{U}$ from the projection error. The third step is to find an acceptable replacement for $\hat{\mathbf{Y}}$ to further simplify the analysis, and the last one is to decompose the terms.

Step 1: Trivial case. First we consider the case that $\mathbf{\Sigma}_1$ is singular. Thus it must hold that $\mathbf{\Sigma}_2 = 0$, and $\mathbf{\Sigma}_1$ can be further written as $\mathbf{\Sigma}_1 = \begin{bmatrix} \mathbf{\Sigma}_0 & \\ & 0 \end{bmatrix}$ with $\mathbf{\Sigma}_0 \in \mathbb{R}^{r_0 \times r_0}$ invertible and r_0 the rank of $\mathbf{A}$. In this case, the SVD of $\mathbf{A}$ can be expressed as $\mathbf{A} = \mathbf{U}_1\mathbf{\Sigma}_1\mathbf{V}_1^T = \mathbf{U}_0\mathbf{\Sigma}_0\mathbf{V}_0^T$, where $\mathbf{U}_0$ and $\mathbf{V}_0$ are composed of the columns in $\mathbf{U}_1, \mathbf{V}_1$ corresponding to $\mathbf{\Sigma}_0$. Therefore, $\mathbf{V}_0^T\mathbf{\Phi}$ and $\mathbf{V}_0^T\mathbf{\Omega}$, as sub-matrices of $\mathbf{\Phi}_1 = \mathbf{V}_1^T\mathbf{\Phi}$ and $\mathbf{\Omega}_1 = \mathbf{V}_1^T\mathbf{\Omega}$, have full row rank due to the assumption. Thus

$$\begin{aligned}\operatorname{range}(\hat{\mathbf{Y}}) &= \operatorname{range}\left(\mathbf{A}\mathbf{\Phi}(\mathbf{A}\mathbf{\Phi})^T\mathbf{A}\mathbf{\Omega}\right) = \operatorname{range}\left(\mathbf{U}_0\mathbf{\Sigma}_0\mathbf{V}_0^T\mathbf{\Phi}\mathbf{\Phi}^T\mathbf{V}_0\mathbf{\Sigma}_0^2\mathbf{V}_0^T\mathbf{\Omega}\right)\\ &= \operatorname{range}(\mathbf{U}_0) = \operatorname{range}(\mathbf{A}),\end{aligned}$$

where the third equality holds because $\mathbf{\Sigma}_0\mathbf{V}_0^T\mathbf{\Phi}\mathbf{\Phi}^T\mathbf{V}_0\mathbf{\Sigma}_0^2\mathbf{V}_0^T\mathbf{\Omega}$ has full row rank. Therefore, $\left\|(\mathbf{I} - P_{\hat{\mathbf{Y}}})\mathbf{A}\right\|_p = 0$ and the estimation naturally holds. Thus in the remaining proof, we only consider that $\mathbf{\Sigma}_1$ is invertible.

Step 2: Reducing U. We then remove $\mathbf{U}$ from the projection error. Note that

$$\hat{\mathbf{Y}} = \mathbf{A}\mathbf{\Phi}(\mathbf{A}\mathbf{\Phi})^T\mathbf{Y} = \mathbf{A}\mathbf{\Phi}(\mathbf{A}\mathbf{\Phi})^T\mathbf{A}\mathbf{\Omega} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T\mathbf{\Phi}\mathbf{\Phi}^T\mathbf{V}\mathbf{\Sigma}^2\mathbf{V}^T\mathbf{\Omega} = \mathbf{U}\mathbf{D}\mathbf{V}^T\mathbf{\Omega} \in \mathbb{R}^{m\times s},$$

where the randomized embedding properties are defined by the following notation:

$$\mathbf{D} := \mathbf{\Sigma}\mathbf{V}^T\mathbf{\Phi}\mathbf{\Phi}^T\mathbf{V}\mathbf{\Sigma}^2 \in \mathbb{R}^{n\times n}. \tag{5.7}$$

Here $\mathbf{\Phi}$ generates a well approximation to the invariant subspace corresponding to $\mathbf{\Sigma}_1$, such that the dominant part of $\mathbf{D}$ is close to $\mathbf{\Sigma}_1^3$ (note that in the standard power iteration, $\mathbf{D} = \mathbf{\Sigma}\mathbf{V}^T\mathbf{V}\mathbf{\Sigma}^2 = \mathbf{\Sigma}^3$). As $P_{\hat{\mathbf{Y}}} = \hat{\mathbf{Y}}(\hat{\mathbf{Y}}^T\hat{\mathbf{Y}})^\dagger\hat{\mathbf{Y}}^T$, we have

$$\begin{aligned}\left\|(\mathbf{I} - P_{\hat{\mathbf{Y}}})\mathbf{A}\right\|_p &= \left\|\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T - \mathbf{U}\mathbf{D}\mathbf{V}^T\mathbf{\Omega}(\mathbf{\Omega}^T\mathbf{V}\mathbf{D}^T\mathbf{D}\mathbf{V}^T\mathbf{\Omega})^\dagger\mathbf{\Omega}^T\mathbf{V}\mathbf{D}^T\mathbf{U}^T\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T\right\|_p\\ &= \left\|\mathbf{\Sigma} - \mathbf{D}\mathbf{V}^T\mathbf{\Omega}(\mathbf{\Omega}^T\mathbf{V}\mathbf{D}^T\mathbf{D}\mathbf{V}^T\mathbf{\Omega})^\dagger\mathbf{\Omega}^T\mathbf{V}\mathbf{D}^T\mathbf{\Sigma}\right\|_p\\ &= \left\|(\mathbf{I} - P_{\mathbf{D}\mathbf{V}^T\mathbf{\Omega}})\mathbf{\Sigma}\right\|_p, \end{aligned} \tag{5.8}$$

where the third equality is due to $\mathbf{U}^T\mathbf{U} = \mathbf{I}$ and unitary invariance of the Schattern-p norm. Thus it suffices to bound $\left\|(\mathbf{I} - P_{\mathbf{D}\mathbf{V}^T\mathbf{\Omega}})\mathbf{\Sigma}\right\|_p$ instead.

Step 3: Further simplification. Recalling the notations in (5.3) and (5.7), we may express $\mathbf{D}\mathbf{V}^T\mathbf{\Omega}$ as

$$\begin{aligned}\mathbf{D}\mathbf{V}^T\mathbf{\Omega} &= \mathbf{\Sigma}(\mathbf{V}^T\mathbf{\Phi})(\mathbf{\Phi}^T\mathbf{V})\mathbf{\Sigma}^2(\mathbf{V}^T\mathbf{\Omega})\\ &= \begin{bmatrix}\mathbf{\Sigma}_1\mathbf{\Phi}_1\\ \mathbf{\Sigma}_2\mathbf{\Phi}_2\end{bmatrix}\begin{bmatrix}\mathbf{\Phi}_1^T\mathbf{\Sigma}_1^2 & \mathbf{\Phi}_2^T\mathbf{\Sigma}_2^2\end{bmatrix}\begin{bmatrix}\mathbf{\Omega}_1\\ \mathbf{\Omega}_2\end{bmatrix} \in \mathbb{R}^{n\times s}.\end{aligned} \tag{5.9}$$

Expression (5.9) shows that directly dealing with $\mathbf{D}\mathbf{V}^T\mathbf{\Omega}$ is too complicated due to the coupling of $\mathbf{\Phi}_1, \mathbf{\Phi}_2, \mathbf{\Omega}_1, \mathbf{\Omega}_2$. Thus we find an acceptable replacement of $\mathbf{D}\mathbf{V}^T\mathbf{\Omega}$, preparing for decoupling. To this end, first denote

$$\begin{bmatrix}\mathbf{F}_1\\ \mathbf{F}_2\end{bmatrix} := \mathbf{D}\mathbf{V}^T\mathbf{\Omega}(\mathbf{\Omega}_1^\dagger\mathbf{\Sigma}_1^{-2}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\mathbf{\Sigma}_1^{-1}) \in \mathbb{R}^{n\times\varrho}, \tag{5.10}$$

where the invertibility of $\mathbf{\Phi}_1\mathbf{\Phi}_1^T$ follows from the assumption. Here $\mathbf{F}_1 \in \mathbb{R}^{\varrho\times\varrho}$ and $\mathbf{F}_2 \in \mathbb{R}^{(n-\varrho)\times\varrho}$. It follows from basic linear algebra that the range spanned by $\mathbf{D}\mathbf{V}^T\mathbf{\Omega}$ and $\begin{bmatrix}\mathbf{F}_1\\ \mathbf{F}_2\end{bmatrix}$ satisfy:

$$\operatorname{range}\left(\begin{bmatrix}\mathbf{F}_1\\ \mathbf{F}_2\end{bmatrix}\right) \subseteq \operatorname{range}\left(\mathbf{D}\mathbf{V}^T\mathbf{\Omega}\right). \tag{5.11}$$

On the other hand, from (5.9) we have

$$\begin{aligned}
\begin{bmatrix}\mathbf{F}_1\\ \mathbf{F}_2\end{bmatrix} &= \begin{bmatrix}\mathbf{\Sigma}_1\mathbf{\Phi}_1\mathbf{\Phi}_1^T\mathbf{\Sigma}_1^2\mathbf{\Omega}_1 + \mathbf{\Sigma}_1\mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{\Sigma}_2^2\mathbf{\Omega}_2\\ \mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_1^T\mathbf{\Sigma}_1^2\mathbf{\Omega}_1 + \mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_2^T\mathbf{\Sigma}_2^2\mathbf{\Omega}_2\end{bmatrix}\mathbf{\Omega}_1^\dagger\mathbf{\Sigma}_1^{-2}\left(\mathbf{\Phi}_1\mathbf{\Phi}_1^T\right)^{-1}\mathbf{\Sigma}_1^{-1}\\
&= \begin{bmatrix}\mathbf{I}_\varrho + \mathbf{\Sigma}_1\mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{\Sigma}_2^2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\mathbf{\Sigma}_1^{-2}\left(\mathbf{\Phi}_1\mathbf{\Phi}_1^T\right)^{-1}\mathbf{\Sigma}_1^{-1}\\ \mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_1^\dagger\mathbf{\Sigma}_1^{-1} + \mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_2^T\mathbf{\Sigma}_2^2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\mathbf{\Sigma}_1^{-2}\left(\mathbf{\Phi}_1\mathbf{\Phi}_1^T\right)^{-1}\mathbf{\Sigma}_1^{-1}\end{bmatrix}\\
&= \begin{bmatrix}\mathbf{I}_\varrho + \mathbf{\Sigma}_1\mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{H}\left(\mathbf{\Phi}_1\mathbf{\Phi}_1^T\right)^{-1}\mathbf{\Sigma}_1^{-1}\\ \mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_1^\dagger\mathbf{\Sigma}_1^{-1} + \mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_2^T\mathbf{H}\left(\mathbf{\Phi}_1\mathbf{\Phi}_1^T\right)^{-1}\mathbf{\Sigma}_1^{-1}\end{bmatrix},
\end{aligned}$$

where the second equality holds because $\mathbf{\Omega}_1$ has full row rank. Observe that $\mathbb{E}\left[\mathbf{\Phi}_2^T\right] = 0$ and $\mathbf{\Sigma}_2$ is small, indicating that $\mathbf{F}_1$ is close to $\mathbf{I}_\varrho$. The assumption and the discussion in step 1 shows that $\mathbf{F}_1$ is invertible. Right multiply $\mathbf{F}_1^{-1}$ to $\begin{bmatrix}\mathbf{F}_1\\ \mathbf{F}_2\end{bmatrix}$ to obtain

$$\mathbf{N} := \begin{bmatrix}\mathbf{F}_1\\ \mathbf{F}_2\end{bmatrix}\mathbf{F}_1^{-1} = \begin{bmatrix}\mathbf{I}_\varrho\\ \mathbf{F}\end{bmatrix} \in \mathbb{R}^{n\times\varrho}, \tag{5.12}$$

where $\mathbf{F} := \mathbf{F}_2\mathbf{F}_1^{-1}$. (5.11) gives $\operatorname{range}(\mathbf{N}) = \operatorname{range}\left(\begin{bmatrix}\mathbf{F}_1\\ \mathbf{F}_2\end{bmatrix}\right) \subseteq \operatorname{range}\left(\mathbf{D}\mathbf{V}^T\mathbf{\Omega}\right)$, and so

$$\|(\mathbf{I} - P_{\mathbf{D}\mathbf{V}^T\mathbf{\Omega}})\mathbf{\Sigma}\|_p \le \|(\mathbf{I} - P_{\mathbf{N}})\mathbf{\Sigma}\|_p. \tag{5.13}$$

Thus it suffices to estimate $\|(\mathbf{I} - P_{\mathbf{N}})\mathbf{\Sigma}\|_p$ to obtain the assertion (5.5).

Step 4: Decoupling. We first provide some intuition regarding $\mathbf{N}$ in (5.12). We have previously argued that $\mathbf{F}_1$ is close to $\mathbf{I}_\varrho$, suggesting that $\mathbf{F} \approx \mathbf{F}_2$. In expectation, the first term in $\mathbf{F}_2$ vanishes, while the second term is of order $O(\mathbf{\Sigma}_2^3)$, implying that $\mathbf{F}$ is also small. Consequently, $\mathbf{N}$ is close to $\begin{bmatrix}\mathbf{I}_\varrho\\ 0\end{bmatrix}$. Note that $\operatorname{range}\left(\begin{bmatrix}\mathbf{I}_\varrho\\ 0\end{bmatrix}\right)$ primarily reflects the information of $\mathbf{\Sigma}_1$. Therefore, we expect that $\operatorname{range}(\mathbf{N})$ can do it similarly.

Considering $\|(\mathbf{I}-P_{\mathbf{N}})\mathbf{\Sigma}\|_p$, the following inequalities hold:

$$\begin{aligned}\|(\mathbf{I}-P_{\mathbf{N}})\mathbf{\Sigma}\|_p &= \left\|(\mathbf{I}-P_{\mathbf{N}})\mathbf{\Sigma}^2(\mathbf{I}-P_{\mathbf{N}})\right\|_{p/2}\\ &\le \left\|(\mathbf{I}-P_{\mathbf{N}})\begin{bmatrix}0 & \\ & \mathbf{\Sigma}_2^2\end{bmatrix}(\mathbf{I}-P_{\mathbf{N}})\right\|_{p/2} + \left\|(\mathbf{I}-P_{\mathbf{N}})\begin{bmatrix}\mathbf{\Sigma}_1^2 & \\ & 0\end{bmatrix}(\mathbf{I}-P_{\mathbf{N}})\right\|_{p/2}\\ &\le \|\mathbf{\Sigma}_2\|_p^2 + \left\|\begin{bmatrix}\mathbf{\Sigma}_1 & \\ & 0\end{bmatrix}(\mathbf{I}-P_{\mathbf{N}})\begin{bmatrix}\mathbf{\Sigma}_1 & \\ & 0\end{bmatrix}\right\|_{p/2}, \qquad (5.14)\end{aligned}$$

where the last inequality comes from $\|\mathbf{I}-P_{\mathbf{N}}\|\le 1$ and the relation between $\|\cdot\|_{p/2}$ and $\|\cdot\|_p$ as $\|\mathbf{M}\|_p^2 = \left\|\mathbf{M}^T\mathbf{M}\right\|_{p/2} = \left\|\mathbf{M}\mathbf{M}^T\right\|_{p/2}$. The second term of the last line above can be written as a parallel sum:

$$\begin{aligned}&\left\|\begin{bmatrix}\mathbf{\Sigma}_1 & \\ & 0\end{bmatrix}(\mathbf{I}-P_{\mathbf{N}})\begin{bmatrix}\mathbf{\Sigma}_1 & \\ & 0\end{bmatrix}\right\|_{p/2} = \left\|\begin{bmatrix}\mathbf{\Sigma}_1 & \\ & 0\end{bmatrix}(\mathbf{I}-P_{\mathbf{N}\mathbf{\Sigma}_1})\begin{bmatrix}\mathbf{\Sigma}_1 & \\ & 0\end{bmatrix}\right\|_{p/2}\\ &= \left\|\mathbf{\Sigma}_1^2 - \mathbf{\Sigma}_1^2(\mathbf{\Sigma}_1^2+\mathbf{\Sigma}_1\mathbf{F}^T\mathbf{F}\mathbf{\Sigma}_1)^\dagger\mathbf{\Sigma}_1^2\right\|_{p/2} = \left\|\mathbf{\Sigma}_1^2 : \mathbf{\Sigma}_1\mathbf{F}^T\mathbf{F}\mathbf{\Sigma}_1\right\|_{p/2}\\ &\le \left\|\mathbf{\Sigma}_1^2\right\|_{p/2} : \left\|\mathbf{\Sigma}_1\mathbf{F}^T\mathbf{F}\mathbf{\Sigma}_1\right\|_{p/2} = \left\|\mathbf{\Sigma}_1\mathbf{F}^T\mathbf{F}\mathbf{\Sigma}_1\right\|_{p/2} : \left\|\mathbf{\Sigma}_1^2\right\|_{p/2} \le \|\mathbf{F}\mathbf{\Sigma}_1\|_p^2, \qquad (5.15)\end{aligned}$$

where the first equality is due to the invertibility of $\mathbf{\Sigma}_1$; the two inequalities are from Lemma 5.1. Then

$$\begin{aligned}&\|\mathbf{F}\mathbf{\Sigma}_1\|_p^2 = \left\|\mathbf{F}_2\mathbf{F}_1^{-1}\mathbf{\Sigma}_1\right\|_p^2\\ &= \left\|\left(\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_1^\dagger\mathbf{\Sigma}_1^{-1}+\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\mathbf{\Sigma}_1^{-1}\right)\left(\mathbf{I}+\mathbf{\Sigma}_1\mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\mathbf{\Sigma}_1^{-1}\right)^{-1}\mathbf{\Sigma}_1\right\|_p^2\\ &= \left\|\left(\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_1^\dagger + \mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\right)\left(\mathbf{I}+\mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\right)^{-1}\right\|_p^2\\ &\le \left\|\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_1^\dagger + \mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\right\|_p^2 \left\|(\mathbf{I}+\mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1})^{-1}\right\|^2.\end{aligned}$$

This together with (5.8), (5.13), (5.14), and (5.15) gives the assertion. ∎

PROPOSITION 5.10. *For $\mathbf{\Omega}$ and $\mathbf{\Phi}$ being standard Gaussian matrices and recall the notations of $\mathbf{\Omega}_1, \mathbf{\Omega}_2, \mathbf{\Phi}_1, \mathbf{\Phi}_2$ in* (5.3), $\mathbf{F}_1 = \mathbf{I}+\mathbf{\Sigma}_1\mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{\Sigma}_2^2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\mathbf{\Sigma}_1^{-2}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\mathbf{\Sigma}_1^{-1} \in \mathbb{R}^{\varrho\times\varrho}$ *is invertible with probability* 1.

Proof. It suffices to prove that each eigenvalue of $\mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{\Sigma}_2^2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\mathbf{\Sigma}_1^{-2}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}$ equals -1 with probability 0. Let $\lambda_i(\cdot)$ denote an eigenvalue of a matrix. For a function

$$f(\mathbf{X}) = \lambda_i(\mathbf{S}\mathbf{X}\mathbf{T}),$$

it is continuous with respect to the entries of $\mathbf{X}$, where $\mathbf{S}, \mathbf{T}, \mathbf{X}$ are matrices of proper sizes. For a matrix, the zeros of the characteristic polynomial exhibit continuous dependence on the coefficients, and their quantitative description is provided in [19]. Thus $f(\mathbf{X})$ is continuous. Therefore, either $f(\mathbf{X})$ is a constant, or its image contains uncountably many points in $\mathbb{C}$.

Now take $\mathbf{X} = \mathbf{\Omega}_2$, $\mathbf{S} = \mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{\Sigma}_2^2$, and $\mathbf{T} = \mathbf{\Omega}_1^\dagger\mathbf{\Sigma}_1^{-2}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}$. It follows from that $\mathbf{\Omega}$ and $\mathbf{\Phi}$ are standard Gaussian and the unitary invariance property that $\mathbf{\Omega}_1, \mathbf{\Omega}_2, \mathbf{\Phi}_1, \mathbf{\Phi}_2$ are independent standard Gaussian matrices; in particular, $\mathbf{X} = \mathbf{\Omega}_2$ is

independent of $\mathbf{S}$ and $\mathbf{T}$. Now consider $\mathbf{\Omega}_1, \mathbf{\Phi}_1, \mathbf{\Phi}_2$ as events in their related probability spaces and $\mathbf{X}$ a variable. Clearly, $f(\mathbf{X})$ can take value 0 provided $\mathbf{X} = 0$; as a result, if $f(\mathbf{X})$ takes value -1, then $f(\mathbf{X})$ is not a constant, and the argument above shows that the image of $f(\cdot)$ with respect to the probability space of $\mathbf{\Omega}_2$ must contain uncountable many points in $\mathbb{C}$. Therefore, with respect to the product probability space of $(\mathbf{\Omega}_1, \mathbf{\Omega}_2, \mathbf{\Phi}_1, \mathbf{\Phi}_2)$, the image of $f(\cdot)$ also has uncountable many points in $\mathbb{C}$, which means that $f(\mathbf{X}) = \lambda_i(\mathbf{\Sigma}_1\mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{\Sigma}_2^2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\mathbf{\Sigma}_1^{-2}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\mathbf{\Sigma}_1^{-1}) = -1$ with probability 0. This completes the proof. ∎

5.3. Averaged projection error. We derive the bound of Theorem 5.9 in expectation. We first estimate $\mathbb{E}\|\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_1^\dagger + \mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\|_p^2$, which can be upper bounded by $2\mathbb{E}\|\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_1^\dagger\|_p^2 + 2\mathbb{E}\|\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\|_p^2$. The first can be bounded by Corollary 5.5. For the second one we have:

THEOREM 5.11. *Let $\mathbf{\Omega}, \mathbf{\Phi}$ be standard Gaussian. If $l > \varrho + 4$ and $s \geq \varrho + 2$, then*

$$\mathbb{E}\left\|\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\right\|_p^2 \leq \frac{\delta}{2}\frac{\sum_{j=\varrho+1}\sigma_j^6}{\sigma_\varrho^4} + \frac{\mu}{2}\tau_{\varrho+1}^2(\mathbf{A})\frac{\sum_{j=\varrho+1}\sigma_j^4}{\sigma_\varrho^4},$$

where

$$\delta = \frac{e^2(s+\varrho)\varrho(l-1)(l^2+2l)\cdot}{(s-\varrho)^2(l-\varrho)(l-\varrho-1)(l-\varrho-3)},\ \mu = \frac{e^2(s+\varrho)\varrho(l-1)l\cdot}{(s-\varrho)^2(l-\varrho)(l-\varrho-1)(l-\varrho-3)}. \tag{5.16}$$

Proof. The Schatten p-norm is decreasing with respect to p, so we can bound it by that with $p = 2$, i.e, the Frobenius norm. Recall that $\mathbf{H} = \mathbf{\Sigma}_2^2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\mathbf{\Sigma}_1^{-2}$; by the independence of $\mathbf{\Omega}_1, \mathbf{\Omega}_2, \mathbf{\Phi}_1, \mathbf{\Phi}_2$ and using Lemma 5.2,

$$\begin{aligned}
\mathbb{E}_{\mathbf{\Phi},\mathbf{\Omega}}\left[\left\|\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\right\|_F^2\right] &= \mathbb{E}_{\mathbf{\Phi},\mathbf{\Omega}_1}\left[\mathbb{E}_{\mathbf{\Omega}_2}\left[\left\|\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\right\|_F^2 |\mathbf{\Omega}_1, \mathbf{\Phi}\right]\right] \\
&= \mathbb{E}_{\mathbf{\Phi},\mathbf{\Omega}_1}\left[\left\|\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_2^T\mathbf{\Sigma}_2^2\right\|_F^2 \cdot \left\|\mathbf{\Omega}_1^\dagger\mathbf{\Sigma}_1^{-2}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\right\|_F^2\right] \\
&\leq \mathbb{E}_{\mathbf{\Phi},\mathbf{\Omega}_1}\left[\left\|\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_2^T\mathbf{\Sigma}_2^2\right\|_F^2 \cdot \left\|\mathbf{\Omega}_1^\dagger\right\|^2 \left\|\mathbf{\Sigma}_1^{-2}\right\|^2 \left\|(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\right\|_F^2\right] \\
&= \sigma_\varrho^{-4}\mathbb{E}\left\|\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_2^T\mathbf{\Sigma}_2^2\right\|_F^2 \cdot \mathbb{E}\left\|\mathbf{\Omega}_1^\dagger\right\|^2 \cdot \mathbb{E}\left\|(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\right\|_F^2.
\end{aligned}$$

The three expectations above are computed in the sequel.

Let $\mathbf{W} = \mathbf{\Phi}_2\mathbf{\Phi}_2^T$; then $\mathbf{W}_{i,j} = \sum_{k=1}^l \phi_{i,k}\phi_{j,k}$ where $\phi_{i,j}$ are independent Gaussian variables. We have:

$$\begin{aligned}
\mathbb{E}\mathbf{W}_{i,i}^2 &= \mathbb{E}(\sum_{k=1}^l \phi_{i,k}\phi_{i,k})^2 = \mathbb{E}\sum_{k=1}^l(\phi_{i,k}\phi_{i,k})^2 + \mathbb{E}\sum_{k\neq p}(\phi_{i,k}\phi_{i,k})(\phi_{i,p}\phi_{i,p}) \\
&= \sum_{k=1}^l \mathbb{E}\phi_{i,k}^4 + \sum_{k\neq p}\mathbb{E}\phi_{i,k}^2\mathbb{E}\phi_{i,p}^2 = 3l + l^2 - l = l^2 + 2l; \\
\mathbb{E}\mathbf{W}_{i,j}^2 &= \mathbb{E}(\sum_{k=1}^l \phi_{i,k}\phi_{j,k})^2 = l,\ i \neq j.
\end{aligned}$$

Thus $\mathbb{E}\left\|\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_2^T\mathbf{\Sigma}_2^2\right\|_F^2 = \sum_{i,j}\sigma_i^2\mathbb{E}\mathbf{W}_{i,j}^2\sigma_j^4 = (l^2+2l)\left\|\mathbf{\Sigma}_2^3\right\|_F^2 + l\left\|\mathbf{\Sigma}_2\right\|_F^2\left\|\mathbf{\Sigma}_2^2\right\|_F^2$.

Lemma 5.4 gives $\mathbb{E}\left\|\mathbf{\Omega}_1^\dagger\right\|^2 \leq \frac{e^2(s+\varrho)}{2(s-\varrho)^2}$. Lemma 5.3 indicates that

$$\mathbb{E}\left\|(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\right\|_F^2 \leq \frac{\varrho(l-1)}{(l-\varrho)(l-\varrho-1)(l-\varrho-3)}$$

when $l-\varrho \geq 4$. Combining the pieces yields the assertion. ∎

The term $\|(\mathbf{I}+\mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1})^{-1}\|$ in (5.6) cannot be bounded in expectation because there exists a positive Lebesgue measure around singularity, facing a similar trouble as that in the analysis of [30, Thm. 3.1 and Remark 3.3]. As an alternative, we show that it is not large with high probability when $\mathbf{A}$ does not have too flattened spectrum. The bound is related to classical results in [13, 34, 26]. Define the events:

$$E_1 = \left\{\mathbf{\Phi} : \left\|(\mathbf{\Phi}_1^T)^\dagger\mathbf{\Phi}_2^T\mathbf{\Sigma}_2^2\right\|_F < e_1 \text{ and } \left\|(\mathbf{\Phi}_1^T)^\dagger\mathbf{\Phi}_2^T\mathbf{\Sigma}_2^2\right\| < e_2\right\},$$
$$E_2 = \left\{\mathbf{\Omega} : \left\|\mathbf{\Omega}_1^\dagger\right\|_F < e_3 \text{ and } \left\|\mathbf{\Omega}_1^\dagger\right\| < e_4\right\},$$
$$E_3 = \left\{\mathbf{\Phi} : \kappa(\mathbf{\Phi}_1\mathbf{\Phi}_1^T) < e_5^2\right\} \quad (\kappa(\cdot) \text{ stands for the condition number}),$$

where

$$e_1 = t\cdot\sqrt{\frac{3\varrho}{l-\varrho+1}}\left\|\mathbf{\Sigma}_2^2\right\|_F + ut\cdot\frac{e\sqrt{l}}{l-\varrho+1}\cdot\sigma_{\varrho+1}^2,$$
$$e_2 = \left[t\cdot\sqrt{\frac{3\varrho}{l-\varrho+1}}\sigma_{\varrho+1}^2 + t\cdot\frac{e\sqrt{l}}{l-\varrho+1}\left\|\mathbf{\Sigma}_2^2\right\|_F\right] + ut\cdot\frac{e\sqrt{l}}{l-\varrho+1}\sigma_{\varrho+1}^2, \tag{5.17}$$
$$e_3 \leq \sqrt{\frac{3\varrho}{s-\varrho+1}}t, \quad e_4 \leq \frac{e\sqrt{s}}{s-\varrho+1}t, \quad e_5 \leq \frac{\sqrt{l}+\sqrt{\varrho}+u}{\sqrt{l}-\sqrt{\varrho}-u},$$

with $u>0, t>1$. The first event E_1 refines from Corollary 5.8. The second one E_2 is from Lemmas 5.3 and 5.4, and the third one E_3 is from [38]. The following theorem provides a detailed deviation error bound with specific coefficients related to these classical results.

THEOREM 5.12. *Let $\mathbf{\Omega}, \mathbf{\Phi}$ be standard Gaussian. If $l > s \geq \varrho+4$ and $(e_1e_4 + e_2e_3 + e_2e_4u)e_5\sigma_\varrho^{-2} < 1$, then for any $\mu > 0, t > 1$,*

$$\mathbb{P}\left(\left\|\left(\mathbf{I}+\mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\right)^{-1}\right\| > \frac{1}{1-(e_1e_4+e_2e_3+e_2e_4u)e_5^2\sigma_\varrho^{-2}}\right)$$
$$< p_f := 5e^{-u^2/2} + 2t^{-(l-\varrho)} + 2t^{-(s-\varrho)}. \tag{5.18}$$

Proof. The main idea is from the Neumann series: if $\|\mathbf{X}\| < \alpha < 1$, then $\left\|(\mathbf{I}-\mathbf{X})^{-1}\right\| \leq (1-\alpha)^{-1}$. Thus we estimate the spectral norm of $\mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}$ and recall $\mathbf{H} = \mathbf{\Sigma}_2^2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\mathbf{\Sigma}_1^{-2}$:

$$\begin{aligned}\left\|\mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\right\| &= \left\|(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)(\mathbf{\Phi}_1^\dagger)^T\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\right\| \\ &\leq \left\|(\mathbf{\Phi}_1^\dagger)^T\mathbf{\Phi}_2^T\mathbf{\Sigma}_2^2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\right\|\left\|\mathbf{\Sigma}_1^{-2}\right\|\left\|\mathbf{\Phi}_1\mathbf{\Phi}_1^T\right\|\|(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\| \\ &= \left\|(\mathbf{\Phi}_1^\dagger)^T\mathbf{\Phi}_2^T\mathbf{\Sigma}_2^2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\right\|\left\|\mathbf{\Sigma}_1^{-2}\right\|\kappa(\mathbf{\Phi}_1\mathbf{\Phi}_1^T).\end{aligned}$$

By Corollary 5.7 with $\mathbf{S} = (\mathbf{\Phi}_1^\dagger)^T \mathbf{\Phi}_2^T \mathbf{\Sigma}_2^2, \mathbf{G} = \mathbf{\Omega}_2, \mathbf{T} = \mathbf{\Omega}_1^\dagger$ and conditioned on E_3:

$$\mathbb{P}\left(\left\|\mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\right\| > (e_1e_4 + e_2e_3 + e_2e_4u)\sigma_\varrho^{-2}e_5^2 \mid E_3\right) \le e^{-u^2/2} + \mathbb{P}\left(E_1^c\right) + \mathbb{P}\left(E_2^c\right),$$

where e_1, e_2, e_3, e_4 were defined in (5.17). On the other hand, we have

$$\mathbb{P}\left(E_1^c\right) \le 2t^{-(l-\varrho)} + 2e^{-u^2/2}, \quad \mathbb{P}\left(E_2^c\right) \le 2t^{-(s-\varrho)}, \quad \mathbb{P}\left(E_3^c\right) \le 2e^{-u^2/2}, \tag{5.19}$$

for $l - \varrho \ge 4$ and $u > 0, t > 1$, where the first one can be derived from Corollary 5.8, the second from Lemmas 5.3 and 5.4, and the last follows the spectral estimation of Gaussian matrices [38]. We finally remove the condition on E_3 by (5.19) and obtain the assertion. ∎

$(e_1e_4 + e_2e_3 + e_2e_4u)e_5^2\sigma_\varrho^{-2}$ can be smaller than a constant $\alpha < 1$ with proper choice of u, t and l, ϱ, s. To simplify, assume that $s = 2\varrho = 2r$, $e_5 \le 2$, which is not a strong assumption in practice. It follows $e_3 \le \sqrt{3}t$ and $e_4 \le \frac{e\sqrt{2}}{\sqrt{\varrho}}t$. Then $(e_1e_4 + e_2e_3 + e_2e_4u)e_5^2\sigma_\varrho^{-2}$ is simplified as:

$$2et^2\left(\sqrt{\frac{6}{l-\varrho+1}} + \frac{\sqrt{3l}}{l-\varrho+1}\right)\frac{\left\|\mathbf{\Sigma}_2^2\right\|_F}{\sigma_\varrho^2} + 2t^2\left(\frac{3\sqrt{\varrho}}{l-\varrho+1} + u\cdot\frac{e\sqrt{3l}}{l-\varrho+1} + u\cdot\frac{e^2\sqrt{2l/\varrho}}{l-\varrho+1}\right)\frac{\sigma_{\varrho+1}^2}{\sigma_\varrho^2}.$$

Clearly, for a reasonably large l, the above can be smaller than a constant $\alpha < 1$. These discussions together with Theorem 5.12 show that the event

$$E_F := \left\{\mathbf{\Omega}, \mathbf{\Phi} : \left\|\left(\mathbf{I} + \mathbf{\Phi}_1\mathbf{\Phi}_2^T\mathbf{\Sigma}_2^2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\mathbf{\Sigma}_1^{-2}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\right)^{-1}\right\| < \xi\right\}. \tag{5.20}$$

holds with probability at least $1 - p_f$ for a constant $\xi = 1/(1-\alpha)$.

As a consequence, the averaged projection error is upper bounded as follows.

THEOREM 5.13. *Let $\mathbf{Q}$ be computed by Algorithm 2.1 with $q = 1$ and $\mathbf{\Phi}, \mathbf{\Omega}$ be independent standard Gaussian. If $l > s \ge \varrho + 4$, then*

$$\mathbb{E}\left[\|\mathbf{A} - P_{\mathbf{Q}}\mathbf{A}\|_F^2 \mid E_F\right] \le (1 + \epsilon\hat{\xi}^2)\tau_{\varrho+1}^2(\mathbf{A}) + \delta\hat{\xi}^2\frac{\sum_{j=\varrho+1}^n \sigma_j^6}{\sigma_\varrho^4} + \mu\hat{\xi}^2\tau_{\varrho+1}^2(\mathbf{A})\frac{\sum_{j=\varrho+1}^n \sigma_j^4}{\sigma_\varrho^4},$$

where $\hat{\xi} := \xi/\sqrt{\mathbb{P}(E_F)}$ with E_F defined in (5.20) and $\xi > 1$, δ, μ are as in (5.16), and $\epsilon = \frac{2\varrho}{(l-\varrho-1)}$.

In particular, for given $\xi > 1, u > 0, t > 1$, if l is large enough so that $(e_1e_4 + e_2e_3 + e_2e_4u)e_5\sigma_\varrho^{-2} < 1 - \xi^{-1}$ (e_i's are defined in (5.17)), then $\mathbb{P}(E_F) \ge (1 - p_f)$ with p_f as in (5.18), hence $\hat{\xi} \le \xi/\sqrt{1-p_f} \approx \xi$.

Proof. Note that $P_{\mathbf{Q}} = P_{\hat{\mathbf{Y}}}$. Conditioned on E_F, Theorem 5.9 shows that the expectation is upper bounded by $\tau_{\varrho+1}^2(\mathbf{A}) + \xi^2\mathbb{E}[Z|E_F]$, where $Z := \|\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_1^\dagger + \mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\|_F^2$. By the definition of conditional expectation for nonnegative random variables, $\mathbb{E}[Z|E_F] \le \mathbb{E}[Z]/\mathbb{P}(E_F)$. Hence the upper bound becomes $\tau_{\varrho+1}^2(\mathbf{A}) + \hat{\xi}^2\mathbb{E}[Z]$ with $\hat{\xi} = \xi/\sqrt{\mathbb{P}(E_F)}$. Then, decompose $Z \le 2\|\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_1^\dagger\|_F^2 + 2\|\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_2^T\mathbf{H}(\mathbf{\Phi}_1\mathbf{\Phi}_1^T)^{-1}\|_F^2$ and apply Corollary 5.5 and Theorem 5.11 to bound each term, completing the proof of the main statement.

If, in addition, l is large enough such that $(e_1e_4 + e_2e_3 + e_2e_4u)e_5\sigma_\varrho^{-2} \le 1 - \xi^{-1}$, then Theorem 5.12 ensures that $\mathbb{P}(E_F) > 1 - p_f$. Consequently, $\hat{\xi} \le \xi/\sqrt{1-p_f}$. ∎

5.4. Averaged oblique projection error. Previous subsections have established the orthogonal projection error. To obtain the main bound stated in Sect. 3.1, we still need a results established in [35] that characteristizes the connection between oblique and orthogonal projections. The proof is given in Appendix A.

LEMMA 5.14. *Let* $\mathbf{Q}, \mathbf{B}$ *be generated by Algorithm 2.1 and* $\mathbf{\Psi}$ *be standard Gaussian and independent from arbitrarily random matrices* $\mathbf{\Omega}, \mathbf{\Phi}$*. If* $d > s + 1$*, then*

$$\mathbb{E}_{\mathbf{\Psi},\mathbf{\Omega},\mathbf{\Phi}}\|\mathbf{A} - \mathbf{QB}\|_F^2 \leq \frac{d}{d-s-1}\mathbb{E}_{\mathbf{\Omega},\mathbf{\Phi}}\|\mathbf{A} - P_{\mathbf{Q}}\mathbf{A}\|_F^2;$$

$$\mathbb{E}_{\mathbf{\Psi},\mathbf{\Omega},\mathbf{\Phi}}\|\mathbf{A} - \mathbf{QB}\| \leq \left(1 + \sqrt{\frac{s}{d-s-1}}\right)\mathbb{E}_{\mathbf{\Omega},\mathbf{\Phi}}\|\mathbf{A} - P_{\mathbf{Q}}\mathbf{A}\| + \frac{e\sqrt{d+s}}{\sqrt{2}(d-s)}\mathbb{E}_{\mathbf{\Omega},\mathbf{\Phi}}\|\mathbf{A} - P_{\mathbf{Q}}\mathbf{A}\|_F.$$

Proof of Theorem 3.1. Since Lemma 5.14 holds for any random $\mathbf{\Phi}, \mathbf{\Omega}$, it also applies when they are Gaussian and further restricted to E_F. Therefore, the assertion follows by combining Theorem 5.13 and Lemma 5.14. ∎

Remark 5.15. We only present the bound in Frobenius norm. That in spectral norm can be derived similarly, which is omitted for simplicity.

6. Analysis for the $q \geq 1$ Bound. To prove the bound in Theorem 3.2, the analysis in the previous section might not yield reasonable bounds. As an alternative, we adapt the decomposition idea in [25, 32, 41], with several refinements to keep the bound independent of the ambient matrix size. Throughout, we assume $\mathbf{\Psi} \in \mathbb{R}^{d\times n}$, $\mathbf{\Phi} \in \mathbb{R}^{m\times l}$ are still independent standard Gaussian, while $\mathbf{\Omega} = \mathbf{\Phi}\tilde{\mathbf{\Omega}}$ with $\tilde{\mathbf{\Omega}} \in \mathbb{R}^{l\times s}$ standard Gaussian independent of $\mathbf{\Psi}, \mathbf{\Phi}$. The analysis still works when $\mathbf{\Omega}$ is standard Gaussian, but the bound will be worse. Thus we only present the case $\mathbf{\Omega} = \mathbf{\Phi}\tilde{\mathbf{\Omega}}$. This section is organized as follows: Sect. 6.1 presents technical lemmas; Sect. 6.2 derives the deterministic error, with probabilistic error bounds established in Sect. 6.3 and 6.4.

6.1. Technical lemmas. For some positive integer k satisfying $k < l \ll \min\{m, n\}$, partition $\mathbf{A}$ via SVD as

$$\mathbf{A} = \mathbf{U\Sigma V}^T = \mathbf{U}\begin{array}{c} \begin{array}{cc} k & n-k \end{array} \\ \begin{bmatrix} \mathbf{\Sigma}_1 & \\ & \mathbf{\Sigma}_2 \end{bmatrix} \end{array} \begin{array}{c} n \\ \begin{bmatrix} \mathbf{V}_1^T \\ \mathbf{V}_2^T \end{bmatrix} \end{array} \begin{array}{c} k \\ n-k \end{array}, \tag{6.1}$$

where $\mathbf{\Sigma}_1 \in \mathbb{R}^{k\times k}$, $\mathbf{\Sigma}_2 \in \mathbb{R}^{(n-k)\times(n-k)}$, $\mathbf{U} \in \mathbb{R}^{m\times n}$, $\mathbf{V}_1 \in \mathbb{R}^{n\times k}$, and $\mathbf{V}_2 \in \mathbb{R}^{n\times(n-k)}$. The singular values are arranged in a descending order. Then define the block partition

$$\mathbf{V}^T\mathbf{\Phi} = \begin{bmatrix} \mathbf{V}_1^T\mathbf{\Phi} \\ \mathbf{V}_2^T\mathbf{\Phi} \end{bmatrix} = \begin{array}{c} l \\ \begin{bmatrix} \mathbf{\Phi}_1 \\ \mathbf{\Phi}_2 \end{bmatrix} \end{array} \begin{array}{c} k \\ n-k \end{array}. \tag{6.2}$$

The following estimates the upper bound of $\sigma_{\varrho+1}(\mathbf{Z})$. Martinsson et al. derived $\sigma_{\varrho+1}(\mathbf{Z}) \leq O(\sqrt{\max\{\varrho+j, l\}})\sigma_{\varrho+1}(\mathbf{A}) + O(\sqrt{\max\{m-\varrho-j, l\}})\sigma_{\varrho+j+1}(\mathbf{A})$ [25]. Similar to their analysis but credited to Lemma 5.2, we obtain a bound depending on sketch sizes and tail energy, which is crucial for our final results.

LEMMA 6.1. *With the notations in* (6.1) *and* (6.2) *and* $\mathbf{Z} = \mathbf{A\Phi}$ *with* $\mathbf{\Phi}$ *standard*

Gaussian, for any ϱ with $0 \le \varrho \le k-1 < l-1$, we have

$$\sigma_{\varrho+1}(\mathbf{Z}) \le \sigma_{\varrho+1}(\mathbf{A})\left(\sqrt{l}+\sqrt{k}+\beta\right) + \sigma_{k+1}(\mathbf{A})\left(\sqrt{l}+\beta\right) + \tau_{k+1}(\mathbf{A});$$

$$\tau_{\varrho+1}(\mathbf{Z}) \le \tau_{\varrho+1}(\mathbf{A})\left(\sqrt{l}+\sqrt{k}+\beta\right) + \sigma_{k+1}(\mathbf{A})\beta + \tau_{k+1}(\mathbf{A})\sqrt{l}$$

with exception probability at most $2e^{-\beta^2/2}$ perspectively, $\beta > 0$.

Proof. With the block representation of (6.1) and (6.2), we have:

$$\sigma_{\varrho+1}(\mathbf{Z}) \le \sigma_{\varrho+1}\left(\mathbf{\Sigma}\begin{bmatrix}\mathbf{\Phi}_1\\0\end{bmatrix}\right) + \left\|\mathbf{\Sigma}\begin{bmatrix}0\\\mathbf{\Phi}_2\end{bmatrix}\right\| = \sigma_{\varrho+1}(\mathbf{\Sigma}_1\mathbf{\Phi}_1) + \|\mathbf{\Sigma}_2\mathbf{\Phi}_2\|; \tag{6.3}$$

$$\tau^2_{\varrho+1}(\mathbf{Z}) \le \tau^2_{\varrho+1}\left(\mathbf{\Sigma}\begin{bmatrix}\mathbf{\Phi}_1\\0\end{bmatrix}\right) + \left\|\mathbf{\Sigma}\begin{bmatrix}0\\\mathbf{\Phi}_2\end{bmatrix}\right\|_F^2 = \tau^2_{\varrho+1}(\mathbf{\Sigma}_1\mathbf{\Phi}_1) + \|\mathbf{\Sigma}_2\mathbf{\Phi}_2\|_F^2, \tag{6.4}$$

where the inequality in the first line uses that $\mathbf{Z}$ and $\mathbf{\Sigma}\mathbf{V}^T\mathbf{\Phi}$ share the same singular values and uses Weyl's inequality. For the second line, observe $\|\mathbf{Z}\|_F^2 = \|\mathbf{\Sigma}\mathbf{V}^T\mathbf{\Phi}\|_F^2 = \|\mathbf{\Sigma}_1\mathbf{\Phi}_1\|_F^2 + \|\mathbf{\Sigma}_2\mathbf{\Phi}_2\|_F^2$, and $\sigma_i(\mathbf{\Sigma}\mathbf{V}^T\mathbf{\Phi}) \ge \sigma_i(\mathbf{\Sigma}\begin{bmatrix}\mathbf{\Phi}_1\\0\end{bmatrix})$ for any i (see e.g., [12, Remark 3.1]). Thus,

$$\begin{aligned}\|\mathbf{\Sigma}_2\mathbf{\Phi}_2\|_F^2 &= \sum_{i=1}^{\varrho}\sigma_i(\mathbf{\Sigma}\mathbf{V}^T\mathbf{\Phi}) + \tau^2_{\varrho+1}(\mathbf{\Sigma}\mathbf{V}^T\mathbf{\Phi}) - \sum_{i=1}^{\varrho}\sigma_i^2\left(\mathbf{\Sigma}\begin{bmatrix}\mathbf{\Phi}_1\\0\end{bmatrix}\right) - \tau^2_{\varrho+1}\left(\mathbf{\Sigma}\begin{bmatrix}\mathbf{\Phi}_1\\0\end{bmatrix}\right)\\ &\ge \tau^2_{\varrho+1}(\mathbf{\Sigma}\mathbf{V}^T\mathbf{\Phi}) - \tau^2_{\varrho+1}\left(\mathbf{\Sigma}\begin{bmatrix}\mathbf{\Phi}_1\\0\end{bmatrix}\right).\end{aligned}$$

Next, the terms on the right hand-side of (6.3) and (6.4) satisfy:

$$\sigma_{\varrho+1}(\mathbf{\Sigma}_1\mathbf{\Phi}_1) \le \sigma_{\varrho+1}(\mathbf{\Sigma}_1)\|\mathbf{\Phi}_1\| \le \sigma_{\varrho+1}(\mathbf{\Sigma}_1)\left(\sqrt{l}+\sqrt{k}+\beta\right);$$

$$\tau_{\varrho+1}(\mathbf{\Sigma}_1\mathbf{\Phi}_1) \le \tau_{\varrho+1}(\mathbf{\Sigma}_1)\|\mathbf{\Phi}_1\| \le \tau_{\varrho+1}(\mathbf{\Sigma}_1)\left(\sqrt{l}+\sqrt{k}+\beta\right);$$

$$\|\mathbf{\Sigma}_2\mathbf{\Phi}_2\| = \|\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{I}_l\| \le \|\mathbf{\Sigma}_2\|\left(\sqrt{l}+\beta\right) + \|\mathbf{\Sigma}_2\|_F; \quad \|\mathbf{\Sigma}_2\mathbf{\Phi}_2\|_F \le \|\mathbf{\Sigma}_2\|_F\sqrt{l} + \|\mathbf{\Sigma}_2\|\beta,$$

where the first two lines uses the spectral estimation of Gaussian matrices [38], while the last line can be obtained by combining Lemmas 5.2 and 5.6, all with exception probability at most $e^{-\beta^2/2}$. By definitions of $\mathbf{\Sigma}_1$ and $\mathbf{\Sigma}_2$, $\sigma_{\varrho+1}(\mathbf{\Sigma}_1) = \sigma_{\varrho+1}(\mathbf{A})$, $\|\mathbf{\Sigma}_1\| = \sigma_{k+1}(\mathbf{A})$, and $\|\mathbf{\Sigma}_2\|_F = \tau_{k+1}(\mathbf{A})$. Combining the above, we have the assertion. ∎

LEMMA 6.2 (Proposition 8.6 of [13]). *Let P be an orthogonal projector, and let $\mathbf{M}$ be a matrix. For each positive number q,*

$$\|P\mathbf{M}\| \le \left\|P\left(\mathbf{M}\mathbf{M}^T\right)^q\mathbf{M}\right\|^{1/(2q+1)}.$$

LEMMA 6.3 (c.f. [13]). *For any matrix $\mathbf{M} \in \mathbb{R}^{m\times n}$ and standard Gaussian matrix $\mathbf{\Omega} \in \mathbb{R}^{n\times s}$, for any $\varrho \le s-4$ and $t \ge 1$, $u > 0$,*

$$\|(\mathbf{I}-P_{\mathbf{M}\mathbf{\Omega}})\mathbf{M}\|_F \le \left(1 + t\cdot\sqrt{\frac{3\varrho}{s-\varrho+1}}\right)\tau_{\varrho+1}(\mathbf{M}) + ut\cdot\frac{e\sqrt{s}}{s-\varrho+1}\cdot\sigma_{\varrho+1}(\mathbf{M});$$

$$\|(\mathbf{I}-P_{\mathbf{M}\mathbf{\Omega}})\mathbf{M}\| \le \left(1 + t\cdot\sqrt{\frac{3\varrho}{s-\varrho+1}} + ut\cdot\frac{e\sqrt{s}}{s-\varrho+1}\right)\sigma_{\varrho+1}(\mathbf{M}) + t\cdot\frac{e\sqrt{s}}{s-\varrho+1}\tau_{\varrho+1}(\mathbf{M}),$$

with failure probability at most $2t^{-(s-\varrho)} + e^{-u^2/2}$.

6.2. Deterministic projection error. We first analyze the behavior of the projection error without randomization. Denote

$$\mathbf{F} := \begin{bmatrix} \mathbf{\Phi}_1^\dagger & 0 \end{bmatrix} \in \mathbb{R}^{l\times n}; \quad (\text{recall } \mathbf{\Phi}_1^\dagger \in \mathbb{R}^{l\times k},\ l \geq k).$$

We have the following result:

THEOREM 6.4. *With the notations in* (6.1) *and* (6.2)*; assume that* $\mathbf{\Phi}_1$ *has full row rank. Then the following estimation holds:*

$$\left\|\mathbf{A} - P_{(\mathbf{Z}\mathbf{Z}^T)^q\mathbf{A}\mathbf{\Omega}}\mathbf{A}\right\|_p \leq 2\left(\|\mathbf{\Sigma}_2\|_p + \|\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_1^\dagger\|_p\right) + \left\|\left(\mathbf{I} - P_{(\mathbf{Z}\mathbf{Z}^T)^q\mathbf{A}\mathbf{\Omega}}\right)\mathbf{Z}\right\|_p \|\mathbf{F}\|.$$

Proof. We have the following estimation by triangle inequality:

$$\begin{aligned}
\left\|\mathbf{A} - P_{(\mathbf{Z}\mathbf{Z}^T)^q\mathbf{A}\mathbf{\Omega}}\mathbf{A}\right\|_p \leq & \|\mathbf{A} - \mathbf{Z}\mathbf{F}\|_p + \left\|\mathbf{Z}\mathbf{F} - P_{(\mathbf{Z}\mathbf{Z}^T)^q\mathbf{A}\mathbf{\Omega}}\mathbf{Z}\mathbf{F}\right\|_p \\
& + \left\|P_{(\mathbf{Z}\mathbf{Z}^T)^q\mathbf{A}\mathbf{\Omega}}\mathbf{Z}\mathbf{F} - P_{(\mathbf{Z}\mathbf{Z}^T)^q\mathbf{A}\mathbf{\Omega}}\mathbf{A}\right\|_p \\
= & \|\mathbf{A} - \mathbf{Z}\mathbf{F}\|_p + \left\|\left(\mathbf{I} - P_{(\mathbf{Z}\mathbf{Z}^T)^q\mathbf{A}\mathbf{\Omega}}\right)\mathbf{Z}\mathbf{F}\right\|_p + \left\|P_{(\mathbf{Z}\mathbf{Z}^T)^q\mathbf{A}\mathbf{\Omega}}\left(\mathbf{Z}\mathbf{F} - \mathbf{A}\right)\right\|_p \\
\leq & 2\|\mathbf{A} - \mathbf{Z}\mathbf{F}\|_p + \left\|\left(\mathbf{I} - P_{(\mathbf{Z}\mathbf{Z}^T)^q\mathbf{A}\mathbf{\Omega}}\right)\mathbf{Z}\right\|_p \|\mathbf{F}\|,
\end{aligned}$$

where the last line uses the nonexpansiveness of the projection operator. We then have

$$\|\mathbf{A} - \mathbf{Z}\mathbf{F}\|_p = \|\mathbf{A} - \mathbf{A}\mathbf{\Phi}\mathbf{F}\|_p = \left\|\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T - \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T\mathbf{\Phi}\mathbf{F}\right\|_p = \left\|\mathbf{\Sigma} - \mathbf{\Sigma}\mathbf{V}^T\mathbf{\Phi}\mathbf{F}\right\|_p$$

by the rotational invariance of any Schatten-p norm. This yields

$$\begin{aligned}
\|\mathbf{A} - \mathbf{Z}\mathbf{F}\|_p &= \left\|\mathbf{\Sigma}\left(\mathbf{I} - \begin{bmatrix}\mathbf{\Phi}_1 \\ \mathbf{\Phi}_2\end{bmatrix}[\mathbf{\Phi}_1^\dagger, 0]\right)\right\|_p = \left\|\mathbf{\Sigma}\left(\mathbf{I} - \begin{bmatrix}\mathbf{I} & 0 \\ \mathbf{\Phi}_2\mathbf{\Phi}_1^\dagger & 0\end{bmatrix}\right)\right\|_p \\
&= \left\|\mathbf{\Sigma}\begin{bmatrix}0 & 0 \\ -\mathbf{\Phi}_2\mathbf{\Phi}_1^\dagger & \mathbf{I}\end{bmatrix}\right\|_p \leq \|\mathbf{\Sigma}_2\|_p + \left\|\mathbf{\Sigma}_2\mathbf{\Phi}_2\mathbf{\Phi}_1^\dagger\right\|_p.
\end{aligned}$$

Combining the above yields the assertion. ∎

The above framework, adapted from [25, 32, 41], uses $\mathbf{Z}$ as a bridge to decompose the error into two terms $\|\mathbf{A} - \mathbf{Z}\mathbf{F}\|_p$ and $\|(\mathbf{I} - P_{\mathbf{Z}\mathbf{Z}^T)^q\mathbf{A}\mathbf{\Omega}}\mathbf{Z}\|$, which are relatively easier to analyze. We will estimate the second term in the next subsection. Here for the Frobenius norm error we further have

$$\begin{aligned}
\left\|\mathbf{A} - P_{(\mathbf{Z}\mathbf{Z}^T)^q\mathbf{A}\mathbf{\Omega}}\mathbf{A}\right\|_F \leq & 2\|\mathbf{A} - \mathbf{Z}\mathbf{F}\|_F + \left\|\left(\mathbf{I} - P_{(\mathbf{Z}\mathbf{Z}^T)^q\mathbf{A}\mathbf{\Omega}}\right)\mathbf{Z}\right\|_F \|\mathbf{F}\| \\
\leq & 2\|\mathbf{A} - \mathbf{Z}\mathbf{F}\|_F + \sqrt{l}\left\|\left(\mathbf{I} - P_{(\mathbf{Z}\mathbf{Z}^T)^q\mathbf{A}\mathbf{\Omega}}\right)\mathbf{Z}\right\| \|\mathbf{F}\|, \qquad (6.5)
\end{aligned}$$

where the inequality is due to the relation between Frobenius and spectral norm, and that $\mathbf{Z} \in \mathbb{R}^{n\times l}$.

6.3. Probabilistic projection error. We complete the randomized projection error analysis. We first give a lemma.

LEMMA 6.5. *Let* $\mathbf{Z} = \mathbf{A}\mathbf{\Phi}$ *and* $\mathbf{\Omega} = \mathbf{\Phi}\tilde{\mathbf{\Omega}}$ *with* $\mathbf{\Phi} \in \mathbb{R}^{n\times l}$ *and* $\tilde{\mathbf{\Omega}} \in \mathbb{R}^{l\times s}$ *independent standard Gaussian matrices. For any* ϱ, k *with* $\varrho < k < l$ *and* $\varrho < s - 4$*, we have*

$$\left\|\left(\mathbf{I} - P_{(\mathbf{Z}\mathbf{Z}^T)^q\mathbf{A}\mathbf{\Omega}}\right)\mathbf{Z}\right\| \leq \alpha \cdot \left(\sigma_{\varrho+1}(\mathbf{A})\left(\sqrt{l} + \sqrt{k} + \beta\right) + \sigma_{k+1}(\mathbf{A})\left(\sqrt{l} + \beta\right) + \tau_{k+1}(\mathbf{A})\right)$$

with exception probability at most $2t^{-(s-\varrho)} + e^{-u^2/2} + e^{-\beta^2/2}$*, where the parameter*

$$\alpha = \left(1 + t \cdot \sqrt{\frac{3\varrho}{s-\varrho+1}} + t \cdot \frac{e \cdot \left(\sqrt{sl} + u\sqrt{s}\right)}{s-\varrho+1}\right)^{\frac{1}{2q+1}}. \tag{6.6}$$

Proof. For notational convenience, we denote $\mathbf{D} := (\mathbf{Z}\mathbf{Z}^T)^q \mathbf{Z} \in \mathbb{R}^{m\times l}$. It follows from $\mathbf{\Omega} = \mathbf{\Phi}\tilde{\mathbf{\Omega}}$ that $P_{(\mathbf{Z}\mathbf{Z}^T)^q \mathbf{A}\mathbf{\Omega}} = P_{\mathbf{D}\tilde{\mathbf{\Omega}}}$. Consider the following fact from Lemma 6.2:

$$\|(\mathbf{I} - P_{\mathbf{D}\tilde{\mathbf{\Omega}}})\,\mathbf{Z}\| \le \left\|(\mathbf{I} - P_{\mathbf{D}\tilde{\mathbf{\Omega}}})\left(\mathbf{Z}\mathbf{Z}^T\right)^q \mathbf{Z}\right\|^{\frac{1}{2q+1}} = \|(\mathbf{I} - P_{\mathbf{D}\tilde{\mathbf{\Omega}}})\,\mathbf{D}\|^{\frac{1}{2q+1}}; \tag{6.7}$$

note that $\tilde{\mathbf{\Omega}} \in \mathbb{R}^{l\times s}$ is Gaussian; applying Lemma 6.3 we have for any $\varrho < s-4$:

$$\|(\mathbf{I} - P_{\mathbf{D}\tilde{\mathbf{\Omega}}})\,\mathbf{D}\| \le \left(1 + t \cdot \sqrt{\frac{3\varrho}{s-\varrho+1}} + ut \cdot \frac{e\sqrt{s}}{s-\varrho+1}\right)\sigma_{\varrho+1}(\mathbf{D}) + t \cdot \frac{e\sqrt{s}}{s-\varrho+1}\tau_{\varrho+1}(\mathbf{D})$$

with failure probability at most $2t^{-(s-\varrho)} + e^{-u^2/2}$. Next, $\sigma_{\varrho+1}(\mathbf{D})$ and $\tau_{\varrho+1}(\mathbf{D})$ satisfy

$$\tau_{\varrho+1}(\mathbf{D}) \le \sqrt{l-\varrho}\,\sigma_{\varrho+1}(\mathbf{D}),\ \sigma_{\varrho+1}(\mathbf{D}) = \sigma_{\varrho+1}\left(\left(\mathbf{Z}\mathbf{Z}^T\right)^q \mathbf{Z}\right) = \sigma_{\varrho+1}^{2q+1}(\mathbf{Z}); \tag{6.8}$$

combining the above pieces yields:

$$\begin{aligned}
\|(\mathbf{I} - P_{\mathbf{D}\tilde{\mathbf{\Omega}}})\,\mathbf{Z}\| &\le \left(1 + t \cdot \sqrt{\frac{3\varrho}{s-\varrho+1}} + t \cdot \frac{e \cdot \left(\sqrt{sl} + u\sqrt{s}\right)}{s-\varrho+1}\right)^{\frac{1}{2q+1}} \sigma_{\varrho+1}^{\frac{1}{2q+1}}(\mathbf{D}) \\
&= \alpha\sigma_{\varrho+1}(\mathbf{Z}) \\
&\le \alpha \cdot \left(\sigma_{\varrho+1}(\mathbf{A})(\sqrt{l} + \sqrt{k} + \beta) + \sigma_{k+1}(\mathbf{A})(\sqrt{l} + \beta) + \tau_{k+1}(\mathbf{A})\right),
\end{aligned}$$

where the second inequality follows from Lemma 6.1. This completes the proof. ∎

Remark 6.6. For $\tau_{\varrho+1}(\mathbf{D})$ in (6.8), we could have used $\tau_{\varrho+1}(\mathbf{D}) \le \tau_{\varrho+1}^{2q+1}(\mathbf{Z})$ and the second relation of Lemma 6.1 to obtain a slightly better bound. However, we prefer to keep the current form for clarity and simplicity.

With the preparations, we have the following theorem.

THEOREM 6.7. *Let* $\mathbf{Z} = \mathbf{A}\mathbf{\Phi}$ *and* $\mathbf{\Omega} = \mathbf{\Phi}\tilde{\mathbf{\Omega}}$ *with* $\mathbf{\Phi} \in \mathbb{R}^{n\times l}, \tilde{\mathbf{\Omega}} \in \mathbb{R}^{l\times s}$ *independent standard Gaussian matrices. For any* ϱ, k *with* $\varrho < k \le l-4$ *and* $\varrho \le s-4$*, we have*

$$\begin{aligned}
\left\|\mathbf{A} - P_{(\mathbf{Z}\mathbf{Z}^T)^q \mathbf{A}\mathbf{\Omega}}\mathbf{A}\right\| &\le 2\left(1 + t \cdot \sqrt{\frac{3k}{l-k+1}} + \frac{ut \cdot e\sqrt{l}}{l-k+1}\right)\sigma_{k+1}(\mathbf{A}) + \frac{2t \cdot e\sqrt{l}}{l-k+1}\tau_{k+1}(\mathbf{A}) \\
&\quad + \frac{t\alpha e\sqrt{l}}{l-k+1} \cdot \left(\sigma_{\varrho+1}(\mathbf{A})(\sqrt{l} + \sqrt{k} + \beta) + \sigma_{k+1}(\mathbf{A})(\sqrt{l} + \beta) + \tau_{k+1}(\mathbf{A})\right) \\
\left\|\mathbf{A} - P_{(\mathbf{Z}\mathbf{Z}^T)^q \mathbf{A}\mathbf{\Omega}}\mathbf{A}\right\|_F &\le 2\left(1 + t\sqrt{\frac{3k}{l-k+1}}\right)\tau_{k+1}(\mathbf{A}) + ut\frac{e\sqrt{l}}{l-k+1}\sigma_{k+1}(\mathbf{A}) \\
&\quad + \frac{t\alpha el}{l-k+1} \cdot \left(\sigma_{\varrho+1}(\mathbf{A})(\sqrt{l} + \sqrt{k} + \beta) + \sigma_{k+1}(\mathbf{A})(\sqrt{l} + \beta) + \tau_{k+1}(\mathbf{A})\right),
\end{aligned}$$

where $\alpha = \left(1 + t \cdot \sqrt{\frac{3\varrho}{s-\varrho+1}} + t \cdot \frac{e\cdot\left(\sqrt{sl}+u\sqrt{s}\right)}{s-\varrho+1}\right)^{\frac{1}{2q+1}}$ *with failure probability at most*

$$p_e = 2e^{-u^2/2} + 2t^{-(l-k)} + 2t^{-(s-\varrho)} + e^{-\beta^2/2}.$$

Proof. By Theorem 6.4, we have:

$$\left\|\mathbf{A}-P_{(\mathbf{Z}\mathbf{Z}^T)^q\mathbf{A}\boldsymbol{\Omega}}\mathbf{A}\right\|_p \leq 2\left(\|\boldsymbol{\Sigma}_2\|_p+\|\boldsymbol{\Sigma}_2\boldsymbol{\Phi}_2\boldsymbol{\Phi}_1^\dagger\|_p\right)+\left\|\left(\mathbf{I}-P_{(\mathbf{Z}\mathbf{Z}^T)^q\mathbf{A}\boldsymbol{\Omega}}\right)\mathbf{Z}\right\|_p\|\mathbf{F}\|.$$

When $p=2,\infty$, the norms of $\boldsymbol{\Sigma}_2\boldsymbol{\Phi}_2\boldsymbol{\Phi}_1^\dagger$ are given by Corollary 5.8:

$$\left\|\boldsymbol{\Sigma}_2\boldsymbol{\Phi}_2\boldsymbol{\Phi}_1^\dagger\right\| \leq \left(t\cdot\sqrt{\frac{3k}{l-k+1}}+ut\cdot\frac{e\sqrt{l}}{l-k+1}\right)\sigma_{k+1}(\mathbf{A})+t\cdot\frac{e\sqrt{l}}{l-k+1}\tau_{k+1}(\mathbf{A});$$

$$\left\|\boldsymbol{\Sigma}_2\boldsymbol{\Phi}_2\boldsymbol{\Phi}_1^\dagger\right\|_F \leq t\cdot\sqrt{\frac{3k}{l-k+1}}\tau_{k+1}(\mathbf{A})+ut\frac{e\sqrt{l}}{l-k+1}\sigma_{k+1}(\mathbf{A});$$

on the other hand, Lemma 5.4 shows that $\|\boldsymbol{\Phi}_1^\dagger\| \leq \frac{e\sqrt{l}}{l-k+1}t$; these three inequalities simultaneously hold with failure probability at most $2t^{-(l-k)}+2e^{-u^2/2}$. Note also that $\|\mathbf{F}\|=\|\boldsymbol{\Phi}_1^\dagger\|$. Combining the above deductions with Lemma 6.5, we obtain the assertion (note that for $\|\mathbf{A}-P_{(\mathbf{Z}\mathbf{Z}^T)^q\mathbf{A}\boldsymbol{\Omega}}\mathbf{A}\|_F$, we additionally use (6.5)). ∎

6.4. Probabilistic oblique projection error. We first need a lemma.

LEMMA 6.8. *Let $\mathbf{Q},\mathbf{B}$ be generated by Algorithm 2.1 and $\boldsymbol{\Psi}$ be standard Gaussian and independent from $\boldsymbol{\Omega},\boldsymbol{\Phi}$. If $d\geq s+4$, then*

$$\mathbb{P}\left(\|\mathbf{A}-\mathbf{Q}\mathbf{B}\|>\frac{e\sqrt{d}t}{d-s+1}e_2+\left(1+t\sqrt{\frac{3s}{d-s+1}}+u\frac{e\sqrt{d}t}{d-s+1}\right)e_1 \mid E\right)\leq e^{-u^2/2}+2t^{-(d-s)},$$

where $E=\left\{\boldsymbol{\Omega},\boldsymbol{\Phi}\mid \|\mathbf{A}-P_{\mathbf{Q}}\mathbf{A}\|<e_1,\|\mathbf{A}-P_{\mathbf{Q}}\mathbf{A}\|_F<e_2\right\}$.

Proof. According to (A.1), it suffices to estimate $\|\mathbf{Q}^T\mathbf{A}-\mathbf{B}\|$, which is equivalent to $\|\boldsymbol{\Psi}_2^\dagger\boldsymbol{\Psi}_1\mathbf{Q}_\perp^T\mathbf{A}\|$; here $\boldsymbol{\Psi}_1=\boldsymbol{\Psi}\mathbf{Q}_\perp$ and $\boldsymbol{\Psi}_2=\boldsymbol{\Psi}\mathbf{Q}$ with $\mathbf{Q}_\perp$ the orthogonal complement of $\mathbf{Q}$. Define

$$E_1=\left\{\boldsymbol{\Psi}_2 \mid \|\boldsymbol{\Psi}_2^\dagger\|<e_3:=\frac{e\sqrt{d}}{d-s+1}t,\|\boldsymbol{\Psi}_2^\dagger\|_F<e_4:=\sqrt{\frac{3s}{d-s+1}}t\right\}$$

with some $t>1$. The deviations bounds in Lemmas 5.3 and 5.4 show that $\mathbb{P}(E_1^C)\leq 2t^{-(d-s)}$ for any fixed $\mathbf{Q}$.

We next use Corollary 5.7 to bound $\|\boldsymbol{\Psi}_2^\dagger\boldsymbol{\Psi}_1\mathbf{Q}_\perp^T\mathbf{A}\|$. However, $\mathbf{Q}$ itself randomly depends on $\boldsymbol{\Phi},\boldsymbol{\Omega}$, making the independence between $\boldsymbol{\Psi}_1$ and $\mathbf{Q}_\perp^T\mathbf{A}$ questionable, such that Corollary 5.7 cannot be directly applied. Nevertheless, we can use the law of total probability to resolve it. To this end, fixed a pair $(\boldsymbol{\Omega},\boldsymbol{\Phi})=(\boldsymbol{\Omega}_\gamma,\boldsymbol{\Phi}_\gamma)$; then $\mathbf{Q}$ is also fixed, so $\boldsymbol{\Psi}_1,\boldsymbol{\Psi}_2,\mathbf{Q}_\perp^T\mathbf{A}$ are independent. Using Corollary 5.7 we get

$$\begin{aligned}&\mathbb{P}\left(\|\mathbf{B}-\mathbf{Q}^T\mathbf{A}\|>e_3\|\mathbf{Q}_\perp^T\mathbf{A}\|_F+(e_4+ue_3)\|\mathbf{Q}_\perp^T\mathbf{A}\| \mid (\boldsymbol{\Omega},\boldsymbol{\Phi})=(\boldsymbol{\Omega}_\gamma,\boldsymbol{\Phi}_\gamma)\right)\\ \leq& e^{-u^2/2}+\mathbb{P}(E_1^C)\leq e^{-u^2/2}+2t^{-(d-s)},\end{aligned}$$

which uniformly holds for any fixed pair $(\boldsymbol{\Omega},\boldsymbol{\Phi})=(\boldsymbol{\Omega}_\gamma,\boldsymbol{\Phi}_\gamma)$. This allows us to use the law of total probability to remove the condition:

$$\begin{aligned}&\mathbb{P}\left(\|\mathbf{B}-\mathbf{Q}^T\mathbf{A}\|>e_3\|\mathbf{Q}_\perp^T\mathbf{A}\|_F+(e_4+ue_3)\|\mathbf{Q}_\perp^T\mathbf{A}\|\right)\\ =&\int\mathbb{P}\left(\|\mathbf{B}-\mathbf{Q}^T\mathbf{A}\|>e_3\|\mathbf{Q}_\perp^T\mathbf{A}\|_F+(e_4+ue_3)\|\mathbf{Q}_\perp^T\mathbf{A}\| \mid (\boldsymbol{\Omega},\boldsymbol{\Phi})=(\boldsymbol{\Omega}_\gamma,\boldsymbol{\Phi}_\gamma)\right)d\mathbb{P}_{(\boldsymbol{\Omega},\boldsymbol{\Phi})}(\boldsymbol{\Omega}_\gamma,\boldsymbol{\Phi}_\gamma)\\ \leq&\int(e^{-u^2/2}+2t^{-(d-s)})d\mathbb{P}_{(\boldsymbol{\Omega},\boldsymbol{\Phi})}(\boldsymbol{\Omega}_\gamma,\boldsymbol{\Phi}_\gamma)=e^{-u^2/2}+2t^{-(d-s)}.\end{aligned}$$

Finally conditioned on E gives the required probabilistic bound. □

The previous preparations yield the main bound stated in Sect. 3.2.

Proof of Theorem 3.2. Use Theorem 6.7 and Lemma 6.8. □

7. Related Approaches. We discuss some relavant approaches in this section.

First, [42, Remark 4.2] (see also [2, Sect. 6.8]) suggests a "half" power iteration: If $\mathbf{A}$ is given column-wisely, then one can compute $\mathbf{Y} = \mathbf{A}\mathbf{A}^T\mathbf{\Omega} = (a_1 a_1^T + a_2 a_2^T + \cdots + a_n a_n^T)\mathbf{\Omega}$ with only a single pass of $\mathbf{A}$, with a_i being the i-th column of $\mathbf{A}$.

In contrast, SPI supports any linear update of $\mathbf{A}$ (see [35] for a detail of linear update). Moreover, if $\mathbf{A}$ is indeed given column-wisely, then the idea of [42] can still be enhanced by SPI : sketch $\mathbf{Z} = \mathbf{A}\mathbf{A}^T\mathbf{\Phi}$ and $\mathbf{Y} = \mathbf{A}\mathbf{A}^T\mathbf{\Omega}$ with only a single pass of $\mathbf{A}$, and compute $\hat{\mathbf{Y}} = (\mathbf{Z}\mathbf{Z}^T)^q\mathbf{Y}$ as well.

Second, the power of a sketch has been used to estimate the Schatten-p norm of a matrix [26, 18]. Let $\mathbf{Z} = \mathbf{A}\mathbf{\Phi}$ with $\mathbf{\Phi}$ having isotropic columns and denote $\mathcal{T}(\mathbf{Z}^T\mathbf{Z})$ the strict upper triangle of $\mathbf{Z}^T\mathbf{Z}$. Then $c \cdot \mathrm{trace}(\mathcal{T}(\mathbf{Z}^T\mathbf{Z})^{p-1}\mathbf{Z}^T\mathbf{Z})$ gives an unbias estimation of $\|\mathbf{A}\|_{2p}^{2p}$, where c only depends on p and the sketch size. Following their work, it would be interesting to perform SPI as $\hat{\mathbf{Y}} = \mathbf{Z}\mathcal{T}(\mathbf{Z}^T\mathbf{Z})^{q-1}\mathbf{Z}^T\mathbf{Y}$ to accelerate the computation. On the other hand, the classical estimation introduced in [26, Sect. 5.3] also motivates a possible variant of SPI $\mathbf{Z}_1\mathbf{Z}_1^T\mathbf{Z}_2\mathbf{Z}_2^T\cdots\mathbf{Z}_p\mathbf{Z}_p^T\mathbf{Y}$, whose advantage is that it is an unbias estimation of $(\mathbf{A}\mathbf{A}^T)^p\mathbf{Y}$. [21] estimates $\|\mathbf{A}\|_{2p}^{2p}$ using several bilinear sketches.

Third, SPI has a subtle connection to the block-stochastic power method for streaming PCA [27]. This algorithm takes the following spiked covariance model: Assume that one has samples $x_1, \ldots, x_n$, with $x_i = \mathbf{A}z_i + w_i \in \mathbb{R}^p$, where $z_i \sim N(0, I_k)$, $w_i \sim N(0, \mathbf{I}_p)$, while $\mathbf{A} \in \mathbb{R}^{p\times k}$ is unknown. The goal is to estimate the orthonormal basis of $\mathbf{A}$, given that the samples x_i's come streamingly. Given a block size B, [27] (in a simplified form) takes the following recursion from $j = 1, \ldots, n/B$:

$$\mathbf{S}_j = \mathtt{QR}(\mathbf{X}_j\mathbf{X}_j^T\mathbf{S}_{j-1}, 0),$$

where $\mathbf{S}_0 \in \mathbb{R}^{p\times k}$ is a standard Gaussian matrix, and $\mathbf{X}_j$ takes the j-th block of the samples, i.e., $\mathbf{X}_j = [x_{(j-1)B+1}, \ldots, x_{jB}] \in \mathbb{R}^{p\times B}$. Note that the setting of [27] has some connections to SPI: $\mathbf{A}$ is not visible to the algorithms and the storage is constrained. However, the algorithms are different, and SPI only performs very few steps to enhance the rangefinder, while this algorithm requires several iterations to approximate the orthonormal basis of $\mathbf{A}$; it also requires an additional noise w_i; see [27] for more details.

Fourth, letting $q \to \infty$ in SPI algebraically recovers the subspace spanned by the leading s singular vectors of $\mathbf{Z}$. This case has certain connections to the truncated-SVD-based rangefinder approaches [32, 25, 41], but key distinctions are outlined below:

1) Goals differ. In [32, 25, 41], the truncated-SVD-based rangefinder replaces the QR step and final truncated in TYUC17 (or TYUC17-SPI). Specifically, this means

applying SVD to the rangefinder $\mathbf{Y} \in \mathbb{R}^{m\times s}$ and keeping the leading r singular vectors as an orthonormal basis, where $s - r$ is small (e.g., $8, 12$, and 20 in [41, 32, 25]). By contrast, SPI derives $\hat{\mathbf{Y}} \in \mathbb{R}^{m\times s}$ from a considerably larger sketch $\mathbf{Z} \in \mathbb{R}^{m\times l}$ (with $l - s = d > s$; see Sect. 2.6) with only iterations $q = 1, 2$, mimicking the way of power iteration.

2) Even if one applies truncated SVD directly to $\mathbf{Z}$ (to obtain its leading s left singular subspace to play a similar role as $\hat{\mathbf{Y}}$), key differences remain. First, the SVD-based approach requires storage at least comparable to SPI—potentially larger with high-precision SVD algorithms. Second, in the mixed-precision strategy, one may use a half-double precision model instead of a single-double one to obtain even larger sketches in half-precision (FP16), in which FP16 has become ubiquitous in modern machine learning [31]. However, mainstream libraries (e.g. LAPACK, MKL, Matlab) currently offer no native FP16 QR or SVD routines, making SVD-based approach inapplicable. In contrast, in the same context, SPI uses only FP16 matrix-multiplication kernels, which has been supported by NVIDA's cuBLAS, OpenCL BLAS, Tensorflow, and Matlab. Third, computing the SVD of $\mathbf{Z}$ can be more expensive. Meanwhile, empirical results in Sect. 4 imply that SPI with only $q = 1$ nearly attains the same subspace with substantially less overhead.

Fifth, [6] recently advocates replacing unfolding matrices in power iteration by column-subset based sketches to accelerate their multiple-pass algorithm for tensor approximation. Several essential points differ their work and ours:

1) Objective: Tensor unfoldings are typically very fat, so [6] focuses on *accelerating* power iteration via approximate matrix multiplications, which naturally leads to column sampling for sketch construction [10]. By contrast, SPI is designed to *enhance* approximation accuracy of one-pass algorithms via mimicking power iteration, in which subspace embedding techniques are therefore the appropriate choices. To that end, we also have to consider momery-efficient strategies to reduce memory cost and derive sketch sizes a priori via theoretical guidance, which play a central role in SPI.

2) Sketch: Even though one could use column-subset based sketches in SPI, only uniform sampling is feasible, as others are data-dependent that require multiple passes over the data. Unfortunately, uniform sampling demands the strong incoherence assumption to work well, in which SRHT is more preferred [1].

3) Error bound: [6]'s bound ([6, Corollary 3]) depends explicitly on the unfolding matrix's sizes, rank, and Frobenius norm. In contrast, by leveraging recent advances in subspace embedding techniques [13, 34, 26], our bound depends only on the sketch sizes and tail energy, decoupling accuracy guarantees from the ambient matrix size.

Finally, with storage budget constraints, it is also possible to implement TYUC17 in mixed-precision to enlarge sketches. The case is as follows. $\mathbf{W}$ may indeed be obtained in lower precision to accommodate a larger size. However, $\mathbf{Y}$ must still be sketched and maintained in higher precision: If $\mathbf{Y}$ were stored in, say, single-precision, then $\mathbf{Q}$ would necessarily also be single-precision (since otherwise $\mathbf{Q}$ requires twice the storage of $\mathbf{Y}$, which is infeasible under budget constraints; see Fig. 14). However, a single-precision $\mathbf{Q}$ suffers from a marked loss of orthonormality ($\|\mathbf{Q}^T\mathbf{Q} - \mathbf{I}\| \approx 1e-6$, compared to 10^{-15} for double-precision). If one is willing to accept a single-precision $\mathbf{Q}$, TYUC17-SPI may still achieve improved performance by replacing the single-double

precision model in Sect. 2.5 with a half-single precision model (i.e., using half-precision sketches). We will describe the formal algorithm in Sect. SM4, and compare it with the SPI variant (Sect. 2.7).

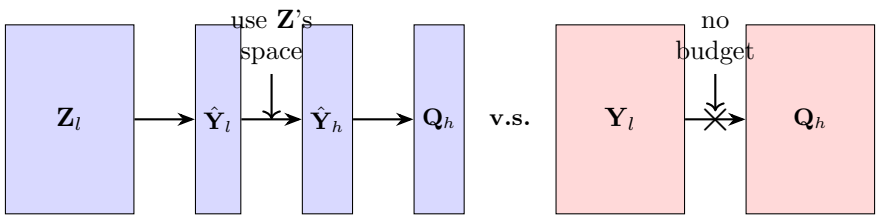


Fig. 14: Comparison of generating $\mathbf{Q}$ by SPI (left) and by ordinary method (right) in mixed-precision. Subscripts "l" and "h" denote lower and higher precision. SPI exploits $\mathbf{Z}$'s storage to convert from low to high precision. The right part shows that directly obtaining a higher precision $\mathbf{Q}$ form a lower precision $\mathbf{Y}$ is impossible due to storage dudget constraint.

8. Conclusions. In this work, we introduced the sketch-power iteration (SPI) as an effective approach to enhance one-pass algorithms for low-rank approximation. By leveraging sketches, SPI substantially improves accuracy without requiring multiple passes over the data matrix. Mixed-precision and two hardware-dependent strategies were proposed to reduce the storage cost. Theoretical analysis established rigorous error bounds, which provide practical guidance on setting sketch sizes a priori for different singular spectrum decay. Numerical results demonstrate the improved accuracy of TYUC17 enhanced by SPI in approximating the dominant singular values and singular vectors. We also observed that the deduced sketch sizes well match empirical optima, enabling efficient parameter selection for better approximation accuracy. Future research may better understand the theoretical behavior of SPI and explore its potential combinations into one-pass algorithms for eigenvalue decompositions and tensor approximations.

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Appendix A. Proofs of Technical Lemmas.

Proof of Corollary 5.5. The first relation comes from the proof of [13, Thm. 10.5]. Here we prove the second relation:

$$\begin{aligned}\mathbb{E}_{\mathbf{G}}\|\mathbf{A}\mathbf{G}_2\mathbf{G}_1^\dagger\|^2 &= \mathbb{E}_{\mathbf{G}_1}\mathbb{E}_{\mathbf{G}_2}\|\mathbf{A}\mathbf{G}_2\mathbf{G}_1^\dagger\|^2 \le \mathbb{E}_{\mathbf{G}_1}\left(\|\mathbf{A}\|\,\|\mathbf{G}_1^\dagger\|_F + \|\mathbf{A}\|_F\,\|\mathbf{G}_1^\dagger\|\right)^2 \\ &\le \|\mathbf{A}\|^2\,\mathbb{E}\|\mathbf{G}_1^\dagger\|_F^2 + \|\mathbf{A}\|_F^2\,\mathbb{E}\|\mathbf{G}_1^\dagger\|^2 + 2\,\|\mathbf{A}\|\,\|\mathbf{A}\|_F\,(\mathbb{E}\|\mathbf{G}_1^\dagger\|_F^2)^{1/2}(\mathbb{E}\|\mathbf{G}_1^\dagger\|^2)^{1/2} \\ &= \left(\|\mathbf{A}\|\,(\mathbb{E}\|\mathbf{G}_1^\dagger\|_F^2)^{1/2} + \|\mathbf{A}\|_F\,(\mathbb{E}\|\mathbf{G}_1^\dagger\|^2)^{1/2}\right)^2,\end{aligned}$$

where the first inequality is from Lemma 5.2 and the second one is due to Hölder inequality. We then use Lemmas 5.3 and 5.4 to complete the proof. ∎

Proof of Corollary 5.7. Conditioned on E, Lemma 5.2 together with Jensen inequality shows that

$$\mathbb{E}_{\mathbf{G}}\left[\|\mathbf{SGT}\| \mid E\right] < \|\mathbf{S}\|_F \|\mathbf{T}\| + \|\mathbf{S}\| \|\mathbf{T}\|_F \leq e_1 e_4 + e_2 e_3,$$

which in connection with Lemma 5.6 gives

$$\mathbb{P}\left(\|\mathbf{SGT}\| > \mathbb{E}_{\mathbf{G}}\left[\|\mathbf{SGT}\| \mid E\right] + \|\mathbf{S}\| \|\mathbf{T}\| u \mid E\right) < e^{-u^2/2},$$

and so $\mathbb{P}\left(\|\mathbf{SGT}\| > e_1 e_4 + e_2 e_3 + e_2 e_4 u \mid E\right) < e^{-u^2/2}$. Use $\mathbb{P}\left(E^C\right)$ to remove the condition and obtain the final result. ∎

Proof of Lemma 5.14. The Frobenius case is similar to that of [35, Thm. 4.3]. For the spectral case, first, we use the triangle inequality to obtain:

$$\|\mathbf{A} - \mathbf{QB}\| \leq \left\|\mathbf{A} - \mathbf{QQ}^T\mathbf{A}\right\| + \left\|\mathbf{QQ}^T\mathbf{A} - \mathbf{QB}\right\| \leq \left\|\mathbf{A} - \mathbf{QQ}^T\mathbf{A}\right\| + \left\|\mathbf{Q}^T\mathbf{A} - \mathbf{B}\right\|. \tag{A.1}$$

[35, Lemma A.4] shows that $\mathbf{B} - \mathbf{Q}^T\mathbf{A} = \mathbf{\Psi}_2^\dagger \mathbf{\Psi}_1 \mathbf{Q}_\perp^T \mathbf{A}$, where $\mathbf{\Psi}_1 = \mathbf{\Psi}\mathbf{Q}_\perp$ and $\mathbf{\Psi}_2 = \mathbf{\Psi}\mathbf{Q}$ are independent and standard Gaussian, with $\mathbf{Q}_\perp$ the orthonormal complement of $\mathbf{Q}$. It then follows from Lemma 5.2 that

$$\mathbb{E}_{\mathbf{\Psi}}\|\mathbf{B} - \mathbf{Q}^T\mathbf{A}\| = \mathbb{E}_{\mathbf{\Psi}}\left\|\mathbf{\Psi}_2^\dagger \mathbf{\Psi}_1 \mathbf{Q}_\perp^T \mathbf{A}\right\| \leq \mathbb{E}\|\mathbf{\Psi}_2^\dagger\|_F \left\|\mathbf{Q}_\perp^T\mathbf{A}\right\| + \mathbb{E}\|\mathbf{\Psi}_2^\dagger\| \left\|\mathbf{Q}_\perp^T\mathbf{A}\right\|_F. \tag{A.2}$$

It follows $\|\mathbf{Q}_\perp^T\mathbf{A}\| = \|\mathbf{Q}_\perp\mathbf{Q}_\perp^T\mathbf{A}\|$ and $\|\mathbf{Q}_\perp^T\mathbf{A}\|_F = \|\mathbf{Q}_\perp\mathbf{Q}_\perp^T\mathbf{A}\|_F$, while $\mathbf{Q}_\perp\mathbf{Q}_\perp^T\mathbf{A} = \mathbf{A} - P_{\mathbf{Q}}\mathbf{A}$. On the other hand, by Lemmas 5.3, 5.4, and Jensen's inequality, we have:

$$\mathbb{E}\|\mathbf{\Psi}_2^\dagger\| \leq \sqrt{\mathbb{E}\|\mathbf{\Psi}_2^\dagger\|^2} \leq \frac{e\sqrt{d+s}}{\sqrt{2}(d-s)}$$

$$\mathbb{E}\|\mathbf{\Psi}_2^\dagger\|_F \leq \sqrt{\mathbb{E}\mathrm{trace}\left(\mathbf{\Psi}_2\mathbf{\Psi}_2^{\mathrm{T}}\right)^{-1}} = \sqrt{\mathrm{trace}\left(\mathbb{E}(\mathbf{\Psi}_2\mathbf{\Psi}_2^{\mathrm{T}})^{-1}\right)} = \sqrt{\frac{s}{d-s-1}},$$

which together with (A.1) and (A.2) yields the result. ∎

Appendix B. Sketch Size Deduction via Theorem 3.1. This section details how to deduce the sketch sizes in Sect. 2.6 under the mixed-precision strategy. Recall that the storage budget is parameterized by $T = (c(l+s)+d)/2$. For simplicity we only consider $m = n$, i.e., $c = 1$ case. To make the derivation go through, we have to simplify the bound in Theorem 3.1. Assume that the last two terms are higher-order ones. For the first term, relax the constant 1 to $2\hat{\xi}^2$ and assume that $\hat{\xi} < C$ for some $C > 0$. The bound can be simplified as

$$\mathbb{E}\left[\|\mathbf{A} - \mathbf{QB}\|_F^2 \,|E_F\right] \lesssim O(1) \cdot \min_{s \leq d, s+d=T/2, r \leq \varrho \leq s} \frac{d}{d-s-1}\frac{l}{l-\varrho-1}\tau_{\varrho+1}^2. \tag{B.1}$$

Recall that Sect. 2.5 implies $l = d + s = T$ when $m = n$; thus we only need to determine s. Assume that $T > 2r$ for convenience in the sequel.

B.1. Flat spectrum. In this case, the tail energy $\tau_{\varrho+1}^2$ is the dominant term, for which $\varrho \approx r$ to minimize the bound. Thus (B.1) remains to minimize

$$\min_{s \leq d, s+d=T, r \leq s} \frac{d}{d-s-1} = \min_{r \leq s} \frac{T-s}{T-2s-1},$$

which is an increasing function of s, whose minimizer is ottained at $s = r$.

B.2. Polynomial spectrum decay. In this case, write $\sigma_i = i^{-\alpha}$ for some $\alpha > 0$. When $\alpha \neq 1/2$, $\tau_{\varrho+1}^2 = \sum_{i=\varrho+1}^n i^{-2\alpha} \approx \int_{\varrho+1}^n x^{-2\alpha}\mathrm{d}x = C_{2\alpha}(n^{1-2\alpha} - (\varrho+1)^{1-2\alpha})$ with $C_{2\alpha} = (1-2\alpha)^{-1}$. Due to the different behavior of $\tau_{\varrho+1}^2$ for different 2α, we consider the argument separately.

When $\alpha < 1/2$, $\tau_{\varrho+1}^2 \approx n^{1-2\alpha}$ for large enough n, and so the problem reduces to $\min_{s+d=T, r\leq\varrho\leq s\leq d} \frac{d}{d-s-1}\frac{l}{l-\varrho-1}n^{1-2\alpha}$. The minimum occurs at $s = \varrho = r$.

When $\alpha > 1/2$, $\tau_{\varrho+1}^2 \approx 1/(\varrho+1)^{2\alpha-1} - 1/n^{2\alpha-1} \approx 1/(\varrho+1)^{2\alpha-1}$, and so the problem reduces to

$$\min_{s+d=T, r\leq\varrho\leq s\leq d} \frac{d}{d-s-1}\frac{l}{(l-\varrho-1)(\varrho+1)^{2\alpha-1}}. \tag{B.2}$$

We first minimize ϱ without constraints, which takes $\varrho = \frac{(2\alpha-1)(l+1)-1}{2\alpha}$. Consider two cases: 1) $s \geq \frac{(2\alpha-1)(l+1)-1}{2\alpha}$: the minimizer of (B.2) with respect to ϱ is $\varrho = \frac{(2\alpha-1)(l+1)-1}{2\alpha}$, and after some calculations, solving (B.2) gives $s = \varrho = \mathrm{P}_{[r,T/2]}(\frac{(2\alpha-1)(l+1)-1}{2\alpha})$; here $\mathrm{P}_{[r,T/2]}(\cdot)$ means the projection onto $[r, T/2]$. 2) $s < \frac{(2\alpha-1)(l+1)-1}{2\alpha}$: the minimizer of (B.2) with respect to ϱ is $\varrho = s$, and (B.2) reduces to

$$\min_{s+d=T, r\leq s\leq \min\{\frac{(2\alpha-1)(l+1)-1}{2\alpha}, \frac{T}{2}\}} \frac{d}{d-s-1}\frac{l}{(l-s-1)(s+1)^{2\alpha-1}} \approx \frac{l}{(T-2s-1)(s+1)^{2\alpha-1}},$$

where the approximation is due to $d = l - s \approx l - s - 1$, whose minimizer occurs at $s = \max\{r, \frac{(2\alpha-1)(T+3)-2}{4\alpha}\}$. After some simple comparisons, this is the minimizer of (B.2).

When $\alpha \approx 1/2$, $\tau_{\varrho+1}^2 \approx \log n - \log(\varrho+1)$, and so the problem reduces to

$$\min_{s\leq d, s+d=T, r\leq\varrho\leq s} \frac{d}{d-s-1}\frac{l}{l-\varrho-1}(\log n - \log(\varrho+1)).$$

We first minimize ϱ without constraints, which satisfies $\log(\frac{n}{\varrho+1}) = \frac{l-\varrho-1}{\varrho+1}$. It has no closed-form solution; however, its solution can be represented as $\varrho = -\frac{l}{W(-l/ne)} - 1$, where $W(\cdot)$ is the Lambert W function. Then, similar to the discussions in the $\alpha > 1/2$ case, we find that $s \approx \mathrm{P}_{[r,T/2]}(-\frac{T+1}{2W(-(T+1)/2ne)} - 1)$.

B.3. Exponential spectrum decay. In this case, write $\sigma_i = e^{-\alpha i}$ for some $\alpha > 0$. Then $\tau_{\varrho+1}^2 \approx \int_{\varrho+1}^n e^{-2\alpha x}\mathrm{d}x \approx C_\alpha(e^{-2\alpha(\varrho+1)} - e^{-2\alpha(n+1)}) \approx C_\alpha e^{-2\alpha(\varrho+1)}$ where $C_\alpha = (1-e^{-2\alpha})^{-1}$. Substituting this into (B.1) yields:

$$\min_{s\leq d, s+d=T, r\leq\varrho\leq s} \frac{d}{d-s-1}\frac{l}{l-\varrho-1}e^{-2\alpha(\varrho+1)}. \tag{B.3}$$

Its minimum with respect to ϱ occurs at $\varrho = l - (2\alpha)^{-1}$. Then consider three cases:

1. $r \leq l - (2\alpha)^{-1} \leq s$: Now minimizing (B.3) with respect to ϱ gives $\varrho = l - (2\alpha)^{-1}$, and so the problem reduces to $\min_{s\leq d, s+d=T, l-(2\alpha)^{-1}\leq s} \frac{d}{d-s-1}$; thus $s = \min\{l-(2\alpha)^{-1}, T/2\} = \min\{T-(2\alpha)^{-1}, T/2\}$.

2. $s \le l - (2\alpha)^{-1}$: Now minimizing (B.3) respect to ϱ takes the minimum $\varrho = s$, and so the problem reduces to

$$\min_{s \le d, s+d=T} \frac{d}{d-s-1} \frac{l}{l-s-1} e^{-2\alpha(s+1)} \approx \frac{l \cdot e^{-2\alpha(s+1)}}{d-s-1},$$

where in the approximation, we used that $d = T - s \approx l - s - 1$. The minimum occurs at $s \approx \max\{T/2 - (2\alpha)^{-1}, r\}$.

3. $r \ge l - (2\alpha)^{-1}$: Now minimizing (B.3) with respect to ϱ takes the minimum $\varrho = r$, and so the problem reduces to $\min_{s \le d, s+d=T} \frac{d}{d-s-1}$; thus $s = r$.

In summary, if α is very small (say, $\alpha \le \frac{1}{2T}$), then $s = r$; Otherwise, $s \approx T/2$.

Finally, we remark that although the bound in Theorem 3.1 is for $q = 1$ case, empirically we observe that the derived parameters work well for $q \ge 1$.

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